PPC LAW OF GRAVITY: MATHEMATICAL DERIVATION OF THE FIRST AND SECOND FRIEDMANN EQUATIONS WITHOUT AN EXPLICIT COSMOLOGICAL CONSTANT AND OBSERVATIONAL TESTING AGAINST MODERN COSMOLOGICAL DATA

Proposed Theory: Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law of Gravity)

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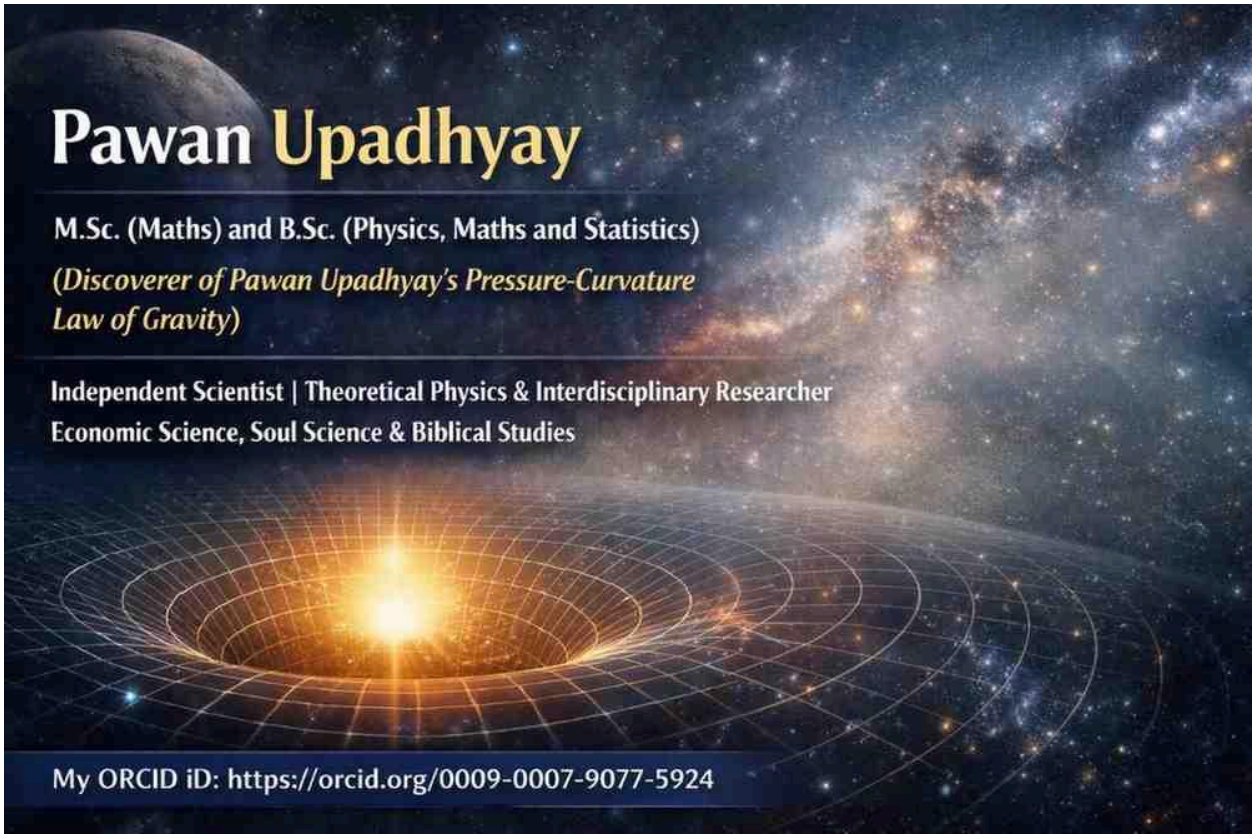
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ABSTRACT

This research paper presents the proposed Pressure–Curvature Law of Gravity (PPC Law) and investigates its application to homogeneous and isotropic cosmology.

The principal objective is to derive the first and second Friedmann equations without introducing an explicit cosmological constant into Einstein's field equations and to establish an observational testing framework using modern cosmological measurements.

The PPC framework proposes a relativistic perfect-fluid energy-momentum tensor:

$T_{\mu\nu}(PPC) = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

The proposed PPC equation of state is:

$P_g(t) = \omega(t)E_d(t)$

Where E_d represents total energy density associated with total mass equivalent density, P_g represents gravitational pressure, and ω represents the dimensionless equation-of-state parameter.

The corresponding mass-equivalent density is defined by:

$\rho = E_d/c^2$

Here, ρ is the mass-equivalent density of the energy represented by the effective PPC fluid, not necessarily the ordinary rest-mass density of its constituent particles.

Using Einstein's field equations without an explicit cosmological constant, the paper derives:

First PPC Friedmann equation

$H^2 = (8\pi G\rho/3) - (kc^2/a^2)$

Second PPC Friedmann equation

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

The paper further investigates the conservation of PPC energy density, time-dependent equations of state, cosmological redshift, luminosity distances, baryon acoustic oscillations, and cosmic microwave background constraints.

For observational testing, it incorporates the published DESI DR2 BAO measurements, Pantheon+ Type Ia Supernova compilation, and Planck 2018 cosmological constraints.

The analysis distinguishes between the mathematical derivation of the PPC Friedmann equations and the separate statistical task of determining whether the proposed framework fits observational data.

The PPC framework is mathematically equivalent to a relativistic perfect-fluid formulation of general relativity at the homogeneous background level. Its independent scientific validity requires a specified physical mechanism for gravitational pressure, a complete perturbation theory, reproducible numerical parameter estimation, and observational predictions distinguishable from existing cosmological models.

Keywords: PPC Law of Gravity, Pressure–Curvature Law, Pawan Upadhyay, Friedmann Equations, Gravitational Pressure, General Relativity, Cosmology, Dark Energy, DESI DR2, Pantheon+, Planck 2018, Cosmic Acceleration.

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1. INTRODUCTION

Modern cosmology describes the expansion of the universe using the Friedmann equations, which follow from Einstein's field equations under the assumptions of spatial homogeneity and isotropy.

The standard cosmological framework, known as Lambda Cold Dark Matter (ΛCDM), incorporates ordinary matter, dark matter, radiation, spatial curvature, and a cosmological constant.

The discovery of the accelerated expansion of the universe through Type Ia supernova observations established an important challenge for theoretical cosmology.

The proposed PPC Law of Gravity investigates whether the expansion and acceleration of the universe can be described through an effective gravitational pressure and total energy density without introducing an explicit cosmological constant into Einstein's field equations.

The central proposition is that gravitational pressure and energy density should be investigated together when describing the cosmological evolution of spacetime.

This research develops the PPC formulation through five principal objectives:

1. Derive the first Friedmann equation using the PPC energy-momentum tensor.
2. Derive the second Friedmann equation using gravitational pressure.
3. Establish the corresponding energy-conservation equation.
4. Investigate possible time-dependent gravitational pressure.
5. Develop a quantitative observational testing framework using modern cosmological datasets.

The research does not assume that the absence of an explicit cosmological constant automatically establishes an alternative explanation of dark energy. Instead, the resulting equations and their predictions must be compared with observational evidence.

2. THE PPC PERFECT-FLUID ENERGY-MOMENTUM TENSOR

The proposed PPC energy-momentum tensor is:

$T_{\mu\nu}(PPC) = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Where:

- E_d = Total energy density of the effective PPC fluid. (Total local energy density with vacuum energy density represented by the PPC fluid.)
- P_g = Gravitational pressure.
- u_μ = Four-velocity of the fluid.
- $g_{\mu\nu}$ = Spacetime metric tensor.

The four-velocity satisfies:

$u_\mu u^\mu = -1$

This convention uses the metric signature (-+++).

In the local rest frame, the energy-momentum tensor takes the diagonal form:

$T^\mu_{\ \nu} =$

[-E_d 0 0 0]

[0 P_g 0 0]

[0 0 P_g 0]

[0 0 0 P_g]

This establishes that the PPC tensor represents a relativistic perfect fluid with isotropic pressure.

The proposed gravitational pressure is therefore a physical quantity within the effective fluid description.

However, to establish a new gravitational theory beyond the standard perfect-fluid formulation, the physical origin and independent dynamical evolution of P_g must be specified.

3. THE PPC EQUATION OF STATE

The proposed PPC equation of state is:

$P_g = \omega E_d$

Consequently:

$\omega = P_g/E_d$

The dimensionless parameter ω determines the relationship between gravitational pressure and energy density.

For homogeneous and isotropic cosmology:

$P_g(t) = \omega(t)E_d(t)$

A spatially varying equation-of-state parameter would require a more general inhomogeneous treatment.

The PPC framework permits the investigation of time-dependent gravitational pressure.

Its physical evolution, however, must be determined by an additional dynamical law or an explicitly specified phenomenological parameterization.

4. TOTAL ENERGY DENSITY AND MASS-EQUIVALENT DENSITY

The PPC framework defines the relationship between total energy density and mass-equivalent density as:

$E_d = \rho c^2$

Therefore:

$\rho = E_d/c^2$

Here, ρ represents the total mass-equivalent density corresponding to the energy density of the effective PPC fluid.

An important distinction must be maintained between mass-equivalent density and ordinary rest-mass density.

For example, radiation carries energy and contributes to gravitational dynamics even though photons have zero rest mass.

Accordingly, if the PPC fluid represents matter, radiation, and other effective energy components together, ρ must be interpreted as the total mass-equivalent density.

For a collection of effective components:

$E_d = \Sigma E_i$

$P_g = \Sigma P_i$

The combined equation-of-state parameter becomes:

$\omega_{eff} = (\Sigma P_i)/(\Sigma E_i)$

This expression assumes that the components can be represented by a common cosmological rest frame and an isotropic effective pressure.

The distinction is important for avoiding double counting of matter, radiation, and other energy contributions during observational analysis.

5. EINSTEIN'S FIELD EQUATION WITHOUT AN EXPLICIT COSMOLOGICAL CONSTANT

The PPC framework adopts Einstein's field equation in the form:

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}(PPC)$

Where:

- $G_{\mu\nu}$ = Einstein tensor.
- G = Newton's gravitational constant.
- c = Speed of light.
- $T_{\mu\nu}(PPC)$ = PPC energy-momentum tensor.

The equation contains no explicit cosmological constant.

The objective is to investigate whether an effective gravitational pressure and total energy density can reproduce the observed cosmological expansion.

The absence of an explicit cosmological constant is a feature of the mathematical formulation. It does not, by itself, demonstrate the absence of vacuum-like energy or an equivalent physical contribution.

6. FLRW SPACETIME AND COSMOLOGICAL EXPANSION

For a homogeneous and isotropic universe, the FLRW metric is:

$ds^2 = -c^2dt^2 + a^2(t)[dr^2/(1 - kr^2) + r^2d\theta^2 + r^2\sin^2\theta\, d\varphi^2]$

Where:

- a(t) = Cosmological scale factor.
- k = Spatial curvature parameter.
- t = Cosmic time.

The Hubble parameter is:

$H = \dot{a}/a$

The three possible spatial geometries are:

- k = 0: Spatially flat universe.
- k > 0: Positively curved universe.
- k < 0: Negatively curved universe.

7. MATHEMATICAL DERIVATION OF THE FIRST PPC FRIEDMANN EQUATION

Starting from Einstein's field equation:

$G_{\mu v} = (8\pi G/c^4)T_{\mu v}(PPC)$

The time-time component of Einstein's field equation for the FLRW metric gives:

$G_{00} = (3/c^2)(H^2 + kc^2/a^2)$

In the comoving rest frame, the energy-momentum component is:

$T_{00} = E_d$

Therefore:

$(3/c^2)(H^2 + kc^2/a^2) = (8\pi G/c^4)E_d$

Multiplying by $c^2/3$:

$H^2 + kc^2/a^2 = (8\pi G/3c^2)E_d$

Rearranging:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2$

Using:

$E_d = \rho c^2$

We obtain:

FIRST PPC FRIEDMANN EQUATION

$H^2 = (8\pi G\rho/3) - (kc^2/a^2)$

This equation establishes the relationship between the expansion rate, total mass-equivalent density, and spatial curvature.

No explicit cosmological constant appears in this equation.

8. MATHEMATICAL DERIVATION OF THE SECOND PPC FRIEDMANN EQUATION

The spatial components of Einstein's field equations give:

$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g)$

Substituting the PPC equation of state:

$P_g = \omega E_d$

We obtain:

$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3\omega E_d)$

Taking E_d as a common factor:

$\ddot{a}/a = -(4\pi G E_d/3c^2)(1 + 3\omega)$

Using:

$E_d = \rho c^2$

The equation becomes:

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

Therefore:

SECOND PPC FRIEDMANN EQUATION

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

For positive total energy density:

Accelerated expansion:

$\omega < -1/3$

Decelerated expansion:

$\omega > -1/3$

Zero acceleration contribution:

$\omega = -1/3$

Thus, negative gravitational pressure can contribute to accelerated cosmological expansion.

9. PPC ENERGY-MOMENTUM CONSERVATION

Einstein's field equations require:

$\nabla_\mu T^{\mu\nu}(PPC) = 0$

For homogeneous and isotropic cosmology, the time component gives:

$\dot{E}_d + 3H(E_d + P_g) = 0$

Substituting:

$P_g = \omega E_d$

We obtain:

$\dot{E}_d + 3HE_d(1 + \omega) = 0$

This is the PPC cosmological energy-conservation equation.

For a constant equation-of-state parameter:

$E_d(a) = E_{d,0}\, a^{[-3(1 + \omega)]}$

For a time-dependent equation of state:

$E_d(a) = E_{d,0} \exp[-3\int_1^a (1 + \omega(x))dx/x]$

The conservation equation is essential for obtaining a mathematically consistent cosmological expansion history.

10. SPECIAL VALUES OF THE PPC EQUATION-OF-STATE PARAMETER

Different values of ω produce different cosmological behaviours.

10.1 PRESSURELESS MATTER

For:

$\omega = 0$

We obtain:

$P_g = 0$

$E_d \propto a^{-3}$

This corresponds to pressureless matter.

10.2 RADIATION

For:

$\omega = 1/3$

We obtain:

$P_g = E_d/3$

$E_d \propto a^{-4}$

This corresponds to relativistic radiation.

10.3 ZERO ACCELERATION CONTRIBUTION

For:

$\omega = -1/3$

We obtain:

$E_d \propto a^{-2}$

The fluid contributes zero to the acceleration equation.

10.4 VACUUM-LIKE ENERGY

For:

$\omega = -1$

We obtain:

$P_g = -E_d$

Energy conservation gives:

$E_d = \text{constant}$

The acceleration equation becomes:

$\ddot{a}/a = 8\pi G E_d/(3c^2)$

For positive energy density, this produces accelerated expansion.

This behaviour is mathematically equivalent at the homogeneous background level to a positive cosmological-constant contribution.

Consequently, a PPC explanation of cosmic acceleration must establish whether gravitational pressure arises from a distinct physical mechanism rather than merely replacing the notation of vacuum energy.

10.5 STIFF FLUID

For:

$\omega = 1$

We obtain:

$P_g = E_d$

$E_d \propto a^{-6}$

This is the stiff-fluid equation of state.

Its application to black-hole event horizons would require an independent gravitational analysis and cannot be inferred from the cosmological equation alone.

11. TIME-DEPENDENT PPC EQUATION OF STATE

A time-dependent equation-of-state parameter allows the PPC framework to investigate evolving gravitational pressure.

A possible phenomenological parameterization is:

$\omega(a) = \omega_0 + \omega_a(1 - a)$

Where:

- ω_0 = Present equation-of-state parameter.
- ω_a = Evolution parameter.

Using the conservation equation, integration gives:

$E_d(a) = E_{d,0}\, a^{[-3(1 + \omega_0 + \omega_a)]} \exp[3\omega_a(a - 1)]$

This parameterization is equivalent to the commonly used Chevallier–Polarski–Linder (CPL) form.

It allows observational investigations of both constant and evolving PPC gravitational pressure.

The parameterization is a modelling choice and must not be interpreted as a uniquely derived consequence of the PPC tensor.

12. PPC COSMOLOGICAL EXPANSION AND REDSHIFT

The relationship between cosmological redshift and scale factor is:

$1 + z = 1/a$

For a present scale factor normalized to unity:

$a = 1/(1 + z)$

The PPC energy-density evolution becomes:

$f(z) = E_d(z)/E_{d,0}$

For the CPL equation of state:

$f(z) = (1 + z)^{[3(1 + \omega_0 + \omega_a)]} \exp[-3\omega_a z/(1 + z)]$

Define:

$\Omega_{d,0} = 8\pi G E_{d,0}/(3c^2H_0^2)$

$\Omega_{k,0} = -kc^2/H_0^2$

The first PPC Friedmann equation becomes:

$H^2(z)/H_0^2 = \Omega_{d,0}\, f(z) + \Omega_{k,0}(1 + z)^2$

For a spatially flat universe:

$\Omega_{k,0} = 0$

Therefore:

$H(z) = H_0\sqrt{[\Omega_{d,0} f(z)]}$

This last expression applies only when the PPC effective fluid contains the entire cosmological energy budget.

For a realistic early-universe calculation, matter, radiation, and their perturbations must be represented consistently. They must either be included in the total PPC fluid or explicitly decomposed into non-overlapping components.

13. OBSERVATIONAL DATASETS FOR PPC COSMOLOGICAL TESTING

A cosmological theory must be tested against independently measured observational quantities.

The PPC framework can be investigated using the following datasets.

13.1 DESI DR2 BARYON ACOUSTIC OSCILLATION OBSERVATIONS

The Dark Energy Spectroscopic Instrument (DESI) provides precision measurements of the baryon acoustic oscillation scale.

DESI DR2 uses three years of survey observations and measurements involving more than 14 million galaxies and quasars.

Its measurements constrain the relationship between cosmological distances and redshift.

The principal observables are:

Transverse comoving distance:

$D_M(z)$

Hubble distance:

$D_H(z) = c/H(z)$

Volume-averaged distance:

$D_V(z) = [zD_M^2(z)D_H(z)]^{(1/3)}$

The BAO measurements are generally expressed relative to the sound horizon at the baryon drag epoch:

r_d

Thus:

$D_M(z)/r_d$

$D_H(z)/r_d$

$D_V(z)/r_d$

The following published DESI DR2 measurements provide an observational benchmark for PPC cosmological calculations.

Redshift 0.295:

$D_V/r_d = 7.942 \pm 0.075$

Redshift 0.510:

$D_M/r_d = 13.588 \pm 0.167$

$D_H/r_d = 21.863 \pm 0.425$

Redshift 0.706:

$D_M/r_d = 17.351 \pm 0.177$

$D_H/r_d = 19.455 \pm 0.330$

Redshift 0.934:

$D_M/r_d = 21.576 \pm 0.152$

$D_H/r_d = 17.641 \pm 0.193$

Redshift 1.321:

$D_M/r_d = 27.601 \pm 0.318$

$D_H/r_d = 14.176 \pm 0.221$

Redshift 1.484:

$D_M/r_d = 30.512 \pm 0.760$

$D_H/r_d = 12.817 \pm 0.516$

Redshift 2.330:

$D_M/r_d = 38.988 \pm 0.531$

$D_H/r_d = 8.632 \pm 0.101$

These are published central values and uncertainties, not PPC predictions or a PPC best fit.

The full covariance matrix, including correlations between observables, must be used in statistical fitting.

Source: DESI DR2 BAO measurements and associated cosmological analysis, published in Physical Review D, 2025.

13.2 PANTHEON+ TYPE IA SUPERNOVA OBSERVATIONS

Type Ia supernovae provide measurements of luminosity distances across cosmological redshifts.

The Pantheon+ compilation contains:

- 1,701 supernova light curves.
- 1,550 unique spectroscopically confirmed Type Ia supernovae.
- Measurements collected from 18 different surveys.

The PPC framework must predict the luminosity distance:

$D_L(z) = (1 + z)D_M(z)$

For a spatially flat universe:

$D_L(z) = (1 + z)c\int_0^z dz'/H(z')$

For non-flat geometry, the transverse comoving distance must include the appropriate curvature dependence.

The theoretical distance modulus is:

$\mu_{th}(z) = 5\log_{10}[D_L(z)/\text{Mpc}] + 25$

The predicted distance modulus must be compared with the observed Pantheon+ distance moduli.

The supernova likelihood should include the published statistical and systematic covariance matrix.

The absolute magnitude and calibration nuisance parameters must be treated consistently.

The Pantheon+ dataset is particularly important because it tests whether PPC gravitational pressure can reproduce the observed distance-redshift relationship.

Source: Scolnic et al. (2022), The Pantheon+ Analysis.

13.3 PLANCK 2018 COSMIC MICROWAVE BACKGROUND OBSERVATIONS

The Cosmic Microwave Background provides constraints on the early universe, the matter-energy composition, spatial geometry, and primordial perturbations.

The Planck 2018 cosmological analysis reports, under its baseline flat ΛCDM model:

$\Omega_c h^2 = 0.120 \pm 0.001$

$\Omega_b h^2 = 0.0224 \pm 0.0001$

$n_s = 0.965 \pm 0.004$

$\tau = 0.054 \pm 0.007$

These values are reference constraints within the Planck baseline model, not direct PPC parameter measurements.

For PPC observational testing, the following must be calculated:

1. CMB temperature-temperature power spectrum.
2. Temperature-polarization power spectrum.
3. Polarization-polarization power spectrum.
4. CMB lensing.
5. Angular acoustic scale.
6. Evolution of gravitational potentials.
7. Recombination-era expansion history.

A background Friedmann equation alone cannot establish agreement with CMB observations.

A complete PPC theory must specify the perturbation equations and calculate the resulting CMB spectra.

Source: Planck Collaboration (2020), Planck 2018 Results VI.

13.4 HUBBLE PARAMETER OBSERVATIONS

The PPC expansion history can also be tested using independent measurements of the Hubble parameter.

The theoretical prediction is:

$H(z) = H_0\sqrt{[\Omega_{d,0} f(z) + \Omega_{k,0}(1 + z)^2]}$

Measurements obtained through cosmic chronometers and other observational methods can be compared with this prediction.

The observational likelihood must account for measurement uncertainties, correlated data, and systematic effects.

13.5 LARGE-SCALE STRUCTURE OBSERVATIONS

The distribution and evolution of galaxies provide an independent test of cosmological gravitational dynamics.

A complete PPC model must reproduce:

- Matter-density perturbation growth.
- Galaxy clustering.
- Redshift-space distortions.
- Matter power spectrum.
- Weak gravitational lensing.
- Structure-growth observables.

These tests are essential because two models may produce similar background expansion histories while predicting different structures.

14. MATHEMATICAL FRAMEWORK FOR OBSERVATIONAL FITTING

The PPC framework must be tested through a statistical likelihood analysis rather than through qualitative comparisons alone.

The proposed procedure is to determine the model parameters by minimizing the chi-square statistic.

14.1 GENERAL CHI-SQUARE EXPRESSION

For observational data:

$$\chi^2 = \Delta D^T C^{-1} \Delta D$$

Where:

- D = Observed data vector.
- D_{th} = Theoretical PPC prediction.
- $\Delta D = D - D_{th}$.
- C = Observational covariance matrix.

The total chi-square can be written as:

$$\chi^2_{total} = \chi^2_{SN} + \chi^2_{BAO} + \chi^2_{CMB} + \chi^2_{H(z)}$$

Only statistically compatible and appropriately calibrated likelihoods should be combined.

14.2 PANTHEON+ LIKELIHOOD

Define:

$\Delta\mu = \mu_{obs} - \mu_{th}$

Then:

$\chi^2_{SN} = \Delta\mu^T C_{SN}^{-1} \Delta\mu$

The covariance must include the published systematic and statistical contributions.

The absolute-magnitude nuisance parameter must be marginalized over or jointly fitted.

14.3 DESI DR2 BAO LIKELIHOOD

The BAO likelihood is:

$\chi^2_{BAO} = \Delta B^T C_{BAO}^{-1} \Delta B$

Where the data vector includes the measured transverse, radial, and volume-averaged BAO quantities.

The DESI DR2 full covariance must be used. Independent treatment of correlated distance measurements would produce an incorrect statistical result.

14.4 CMB LIKELIHOOD

The CMB likelihood requires the theoretical angular power spectra and the relevant Planck likelihood products.

The calculation should include:

$C_l^{TT}, C_l^{TE}, C_l^{EE}$

Together with lensing information and the appropriate nuisance parameters.

This calculation requires a complete perturbation theory and cannot be obtained from the homogeneous Friedmann equations alone.

14.5 PARAMETER ESTIMATION

The principal PPC parameters to investigate are:

$\theta_{PPC} = \{H_0, \Omega_{d,0}, \Omega_{k,0}, \omega_0, \omega_a, r_d\}$

Additional parameters may be required when matter, radiation, baryons, neutrinos, and perturbations are modelled separately.

The best-fitting parameters are defined by:

$\theta_{best} = \arg\min_{\theta} \chi^2_{total}(\theta)$

Parameter uncertainties should be calculated using a reproducible statistical procedure, such as Markov Chain Monte Carlo sampling.

A successful fit must report:

- Best-fitting parameters.
- One-dimensional marginalized uncertainties.
- Two-dimensional parameter confidence regions.
- Minimum chi-square.
- Number of independent data points.
- Number of fitted parameters.
- Reduced chi-square where appropriate.
- Information criteria.
- Comparison against ΛCDM.

No numerical PPC best-fit result is claimed in this paper because the complete likelihood calculations and covariance-weighted optimization have not yet been performed.

15. COMPARISON WITH THE ΛCDM MODEL

The standard ΛCDM model uses:

- Ordinary matter.
- Cold dark matter.
- Radiation.
- Spatial curvature.
- Cosmological constant.

The PPC formulation uses an effective total energy density and gravitational pressure.

Both frameworks can produce accelerated expansion.

However, the mathematical similarity between the two formulations must be considered carefully.

For:

$\omega = -1$

The PPC energy density becomes constant:

$E_d = \text{constant}$

This reproduces the background expansion behaviour of a cosmological constant.

Therefore, the absence of an explicit cosmological constant does not automatically establish a new physical explanation for cosmic acceleration.

A meaningful comparison requires investigating whether the PPC framework produces:

1. A statistically comparable or improved observational fit.
2. A different physical mechanism for cosmic acceleration.
3. Distinct predictions for the evolution of gravitational perturbations.
4. Testable differences in the growth of cosmic structures.
5. A physically motivated evolution law for gravitational pressure.

The appropriate comparison must use the same observational datasets and consistent nuisance-parameter treatment for both models.

16. FINITE-DENSITY PRINCIPLE IN THE PPC FRAMEWORK

An additional proposed principle of the PPC framework is that physically admissible solutions should maintain finite physical energy density, gravitational pressure, and mass-equivalent density.

The proposed conditions are:

$E_d < \infty$

$|P_g| < \infty$

$\rho < \infty$

These conditions represent a finite-density requirement.

However, they are additional physical conditions rather than consequences automatically established by the Friedmann equations.

Their validity must be investigated through the mathematical solutions of the PPC model.

Furthermore, finite energy density does not necessarily imply a finite maximum cosmic time.

The possible future-time limit is:

$0 < t_{max} \leq \infty$

The actual future evolution depends on the dynamical behaviour of the scale factor and equation-of-state parameter.

17. PROPOSED FUTURE-TIME BEHAVIOUR

One possible additional PPC hypothesis is:

$\omega(t) \rightarrow -1$ as $t \rightarrow t_{max}$

If the energy density approaches a finite positive constant:

$E_d(t) \rightarrow E_{d,\infty} > 0$

Then:

$P_g(t) \rightarrow -E_{d,\infty}$

This produces an asymptotically vacuum-like equation of state.

“As cosmic time approaches t_{max}, the PPC equation-of-state parameter approaches to -1 , while its energy density approaches a finite positive constant. Consequently, PPC pressure approaches the negative of its energy density, producing an asymptotically vacuum-like gravitational component.”

Such behaviour can generate accelerated expansion.

However, this future-time behaviour is an additional hypothesis of the PPC framework.

It must be established through a specified evolution equation for $\omega(t)$, together with an investigation of the resulting cosmological solutions.

18. SCIENTIFIC LIMITATIONS AND REQUIREMENTS FOR VALIDATION

The PPC energy-momentum tensor has the mathematical form of a relativistic perfect-fluid tensor.

Consequently, its Friedmann equations reproduce the standard general-relativistic perfect-fluid equations under homogeneous and isotropic assumptions.

The PPC framework's proposed physical interpretation must therefore be distinguished from a demonstrated modification or extension of general relativity.

The following requirements are essential for further scientific development.

1. PHYSICAL ORIGIN OF GRAVITATIONAL PRESSURE

A fundamental mechanism must explain why the PPC gravitational pressure exists.

2. EVOLUTION EQUATION

A dynamical equation must determine the evolution of ω.

3. OBSERVATIONAL PARAMETER ESTIMATION

The PPC model must be fitted to DESI DR2, Pantheon+, and appropriate CMB likelihood data.

4. EARLY-UNIVERSE CONSISTENCY

The model must reproduce the conditions necessary for primordial nucleosynthesis, recombination, and the formation of the BAO sound horizon.

5. PERTURBATION THEORY

The framework must predict the evolution of density and metric perturbations.

6. STRUCTURE FORMATION

Its predictions must be compared with galaxy clustering, weak lensing, and redshift-space distortion measurements.

7. STATISTICAL COMPARISON

Its goodness of fit and model complexity must be compared with ΛCDM and established evolving-dark-energy models.

8. INDEPENDENT PREDICTIONS

The PPC framework must provide at least one reproducible, observationally distinguishable prediction.

9. FINITE-DENSITY PRINCIPLE

The proposed finite-density conditions must be demonstrated for physically relevant solutions.

10. REPRODUCIBILITY

The mathematical derivations, computational procedures, datasets, covariance matrices, and parameter-estimation results must be made available for independent verification.

19. CONCLUSION

This research develops the proposed Pawan Upadhyay Pressure–Curvature Law of Gravity as a perfect-fluid cosmological framework.

The proposed PPC energy-momentum tensor is:

$T_{\mu\nu(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Its proposed equation of state is:

$P_g = \omega E_d$

The relationship between energy density and mass-equivalent density is:

$E_d = \rho c^2$

Using Einstein's field equations without an explicit cosmological constant, the first PPC Friedmann equation is derived as:

FIRST PPC FRIEDMANN EQUATION

$H^2 = (8\pi G\rho/3) - (kc^2/a^2)$

$H^2 = (8\pi GE_d/3c\text{^}2) - (kc^2/a^2)$

The second PPC Friedmann equation is:

SECOND PPC FRIEDMANN EQUATION

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

Where, $\rho = E_d/c^2$

$\ddot{a}/a = -(4\pi GE_d/3c\text{^}2)(1 + 3\omega)$

These equations establish that the expansion and acceleration of a homogeneous and isotropic universe depend on its total mass-equivalent energy density, spatial curvature, and effective equation-of-state parameter.

For positive energy density, accelerated expansion occurs when:

$\omega < -1/3$

The research also establishes a mathematical framework for investigating time-dependent gravitational pressure and comparing PPC predictions with modern cosmological observations.

Three observational resources are particularly important:

1. DESI DR2 BAO measurements.
2. Pantheon+ Type Ia supernova observations.
3. Planck 2018 CMB constraints.

These datasets provide independent observational tests of the predicted expansion history and the physical composition of the universe.

Nevertheless, incorporating observational measurements into a theoretical framework is not equivalent to demonstrating a successful statistical fit.

The PPC model must undergo a complete covariance-weighted likelihood analysis, parameter estimation, perturbation calculation, and comparison against ΛCDM before observational consistency can be claimed.

The proposed PPC Law therefore provides a mathematical starting point for further investigation into gravitational pressure, cosmological expansion, and the relationship between energy density and spacetime curvature.

Its status as an independently predictive and observationally validated gravitational theory remains an open scientific question.

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AUTHOR DECLARATION

Proposed Theory: Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law).

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Research Status: Proposed theoretical framework with mathematically derived background cosmological equations. Quantitative observational validation, parameter estimation, and independent experimental confirmation remain necessary.

RESEARCH NOTE 1

TITLE: ASYMPTOTIC EQUATION-OF-STATE CONDITION IN THE PPC COSMOLOGICAL MODEL: ω → -1

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

The PPC cosmological model considers the equation-of-state parameter (ω) as a quantity that may evolve during cosmic expansion. One proposed asymptotic condition of the model is:

$\omega \rightarrow -1$

This condition represents the possibility that the equation-of-state parameter approaches -1 as the universe evolves towards a limiting future state.

It is important to distinguish this asymptotic condition from the exact special case $\omega = -1$.

2. PPC EQUATION OF STATE

The PPC equation of state is expressed as:

$P_g = \omega E_d$

Where:

P_g = Gravitational pressure, ω = Equation-of-state parameter, E_d = Energy density

Under the asymptotic condition $\omega \rightarrow -1$:

$P_g \rightarrow -E_d$

Therefore, gravitational pressure approaches a negative value equal in magnitude to the energy density.

3. ENERGY-CONSERVATION EQUATION

The energy-conservation equation for the PPC perfect fluid is:

$dE_d/dt + 3H(E_d + P_g) = 0$

Substituting $P_g = \omega E_d$:

$dE_d/dt + 3H E_d(1 + \omega) = 0$

When $\omega \rightarrow -1$:

$1 + \omega \rightarrow 0$

Consequently, under this limiting condition:

$dE_d/dt \rightarrow 0$

This indicates that the rate of change of energy density approaches zero, provided the limiting behaviour is sufficiently regular.

The energy density may therefore approach a constant value.

4. PHYSICAL INTERPRETATION

The asymptotic condition $\omega \to -1$ describes a possible evolutionary behaviour of the PPC equation-of-state parameter.

It does not require ω to be exactly -1 throughout the entire cosmic history.

Instead, it allows the equation-of-state parameter to evolve towards -1 as the universe approaches a specified limiting future state.

If this behaviour is realised, the PPC fluid approaches vacuum-energy-like behaviour.

5. COSMOLOGICAL SIGNIFICANCE

The second Friedmann acceleration equation is:

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

As ω approaches -1:

$1 + 3\omega \to -2$

Therefore:

$\ddot{a}/a \to 8\pi G\rho/3$

For positive energy density, this limiting expression corresponds to accelerated cosmic expansion.

6. THEORETICAL STATUS

The condition $\omega \to -1$ is an asymptotic hypothesis within the proposed PPC framework. It is not automatically derived from the standard perfect-fluid conservation equation.

A complete theoretical demonstration requires a dynamical mechanism or a specified function $\omega(a)$ that produces this limiting behaviour.

The hypothesis should also be tested against cosmological observations, including supernovae, baryon acoustic oscillations, cosmic microwave background measurements and structure-growth data.

7. CONCLUSION

The asymptotic condition $\omega \to -1$ describes a possible evolution of the PPC equation-of-state parameter towards a vacuum-like state. Under sufficiently regular limiting behaviour, gravitational pressure approaches the negative of energy density, while the rate of change of energy density approaches zero.

This condition represents an evolutionary possibility rather than an exact equation-of-state condition at every moment of cosmic history.

RESEARCH NOTE 2

TITLE: EXACT VACUUM-LIKE SPECIAL CASE IN THE PPC COSMOLOGICAL MODEL: ω = -1

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

The PPC cosmological model includes the exact special case:

$\omega = -1$

Unlike the asymptotic condition $\omega \rightarrow -1$, this special case assumes that the equation-of-state parameter is exactly equal to -1 throughout the period being considered.

This condition produces a constant energy density and negative pressure equal in magnitude to the energy density.

2. PPC EQUATION OF STATE

The PPC equation of state is:

$P_g = \omega E_d$

Substituting:

$\omega = -1$

We obtain:

$P_g = -E_d$

Therefore:

$P_g + E_d = 0$

This establishes the exact relationship between pressure and energy density in this special case.

3. ENERGY-CONSERVATION EQUATION

The energy-conservation equation is:

$dE_d/dt + 3H(E_d + P_g) = 0$

Substituting $P_g = -E_d$:

$dE_d/dt + 3H(E_d - E_d) = 0$

Therefore:

$dE_d/dt = 0$

Integrating with respect to time:

$E_d = CONSTANT$

Thus, the exact special case $\omega = -1$ produces constant energy density in the homogeneous expanding-universe description.

4. FIRST FRIEDMANN EQUATION

The first Friedmann equation is:

$H^2 = (8\pi G\rho/3) - (kc^2/a^2)$

Where, $\rho = E_d/c^2$

$H^2 = (8\pi GE_d/3c\text{^}2) - (kc^2/a^2)$

For the special case $\omega = -1$, the energy density remains constant.

Since:

$\rho = E_d/c^2$

The mass-equivalent density also remains constant.

Consequently, the contribution of this PPC fluid to the first Friedmann equation remains constant, apart from the evolution of the spatial-curvature term.

5. SECOND FRIEDMANN EQUATION

The second Friedmann equation is:

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

Substituting $\omega = -1$:

$\ddot{a}/a = -(4\pi G\rho/3)(1 - 3)$

Simplifying:

ä/a = -(4πGρ/3)(-2)

Therefore:

ä/a = 8πGρ/3

ä/a = 8πGE_d/3c^2

For positive energy density:

ä/a > 0

This represents accelerated cosmic expansion.

6. PHYSICAL INTERPRETATION

The exact condition ω = -1 represents a vacuum-energy-like perfect fluid.

Its defining mathematical properties are:

1. The equation-of-state parameter is exactly -1.
2. Gravitational pressure is exactly equal to negative energy density.
3. Energy density remains constant.
4. The fluid produces accelerated expansion for positive energy density.

At the homogeneous cosmological-background level, this special case is mathematically equivalent to a cosmological constant contribution in general relativity.

Therefore, omitting an explicit cosmological-constant symbol does not, by itself, eliminate the physical effect associated with vacuum-like energy.

7. THEORETICAL SIGNIFICANCE

Within the PPC framework, ω = -1 can be studied as an exact special case of the proposed equation of state.

However, the perfect-fluid equations alone do not establish a new physical mechanism for vacuum energy. A distinct PPC theory would require additional physical principles, dynamical equations or observational predictions that distinguish it from the standard cosmological-constant description.

8. CONCLUSION

The exact special case ω = -1 establishes a vacuum-energy-like condition within the PPC perfect-fluid framework.

The mathematical consequences are:

$P_g = -E_d$

$dE_d/dt = 0$

$E_d = CONSTANT$

$ä/a = 8\pi G\rho/3$

$ä/a = 8\pi GE_d/3c^2$

For positive energy density, this condition produces accelerated cosmic expansion.

RESEARCH NOTE 3

TITLE: MATHEMATICAL AND PHYSICAL DISTINCTION BETWEEN THE ASYMPTOTIC CONDITION $\omega \rightarrow -1$ AND THE EXACT SPECIAL CASE $\omega = -1$ IN THE PPC COSMOLOGICAL MODEL

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

The PPC cosmological model distinguishes two mathematically different conditions:

ASymptotic condition:

$\omega \rightarrow -1$

Exact special case:

$\omega = -1$

Although both conditions are associated with vacuum-like behaviour, they do not have identical mathematical meanings.

2. ASYMPTOTIC CONDITION: $\omega \rightarrow -1$

The expression $\omega \rightarrow -1$ means that the equation-of-state parameter approaches -1 as the relevant cosmological evolution proceeds towards a limiting state.

The parameter may vary during cosmic evolution.

The corresponding pressure relation is:

$P_g \to -E_d$

The energy-conservation equation is:

$dE_d/dt + 3H E_d(1 + \omega) = 0$

As ω approaches -1, the factor $(1 + \omega)$ approaches zero.

Under sufficiently regular limiting behaviour:

$dE_d/dt \to 0$

Thus, the energy density may approach a constant value.

3. EXACT SPECIAL CASE: $\omega = -1$

The expression $\omega = -1$ means that the equation-of-state parameter is exactly -1.

The pressure relation becomes:

$P_g = -E_d$

The energy-conservation equation becomes:

$dE_d/dt + 3H E_d(1 - 1) = 0$

Therefore:

$dE_d/dt = 0$

Consequently:

$E_d = \text{CONSTANT}$

Unlike the asymptotic condition, this result is exact for the period in which $\omega = -1$ holds.

4. COMPARISON OF BOTH CONDITIONS

Asymptotic condition:

$\omega \to -1$

Meaning: The parameter approaches -1.

Evolution: The parameter may change with time or scale factor.

Pressure: P_g approaches $-E_d$.

Energy density: Its rate of change approaches zero under regular limiting behaviour.

Interpretation: Possible evolution towards vacuum-like behaviour.

Exact special case:

ω = -1

Meaning: The parameter is exactly -1.

Evolution: The parameter remains constant at -1.

Pressure: P_g = -E_d exactly.

Energy density: It remains constant.

Interpretation: Exact vacuum-energy-like behaviour.

5. MATHEMATICAL RELATIONSHIP

The exact special case ω = -1 can be considered a particular constant solution of the equation-of-state relation.

The asymptotic condition ω → -1 is broader because it allows the parameter to approach -1 without being exactly equal to -1 at earlier times.

Therefore:

ω = -1 does not necessarily describe an evolving approach towards -1.

ω → -1 does not necessarily mean that ω is already exactly -1.

6. COSMOLOGICAL IMPLICATION

Both conditions are associated with the acceleration equation:

ä/a = -(4πGρ/3)(1 + 3ω)

As ω approaches -1, the acceleration contribution approaches:

ä/a → 8πGρ/3

For the exact special case ω = -1:

ä/a = 8πGρ/3

ä/a = 8πGE_d/3c^2

For positive energy density, both limiting and exact cases correspond to accelerated expansion, although their full cosmological histories can differ.

7. CONCLUSION

The PPC cosmological model should maintain a clear distinction between the asymptotic condition $\omega \to -1$ and the exact special case $\omega = -1$.

The first describes a possible evolutionary approach towards vacuum-like behaviour. The second describes an exact vacuum-energy-like equation of state with constant energy density.

This distinction is important for mathematical consistency, theoretical interpretation and observational testing of the PPC cosmological model.

THEORETICAL PRINCIPLE: NEAR-VACUUM-LIKE EQUATION OF STATE IN THE PPC COSMOLOGICAL MODEL

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

The PPC cosmological model considers an equation-of-state parameter approaching, but not necessarily becoming exactly equal to, -1.

The proposed condition is:

$\omega \approx -1$

For mathematical representation, define:

$\omega = -1 + \varepsilon$

where ε is a small, nonzero deviation from -1.

Therefore, the PPC equation of state becomes:

$P_g = (-1 + \varepsilon)E_d$

The corresponding PPC energy-momentum tensor is:

$T_{\mu\nu}(PPC) = E_d[\varepsilon u_\mu u_\nu + (-1 + \varepsilon)g_{\mu\nu}]$

Substitution into Einstein's field equations gives:

$G_{\mu\nu} = (8\pi G/c^4)E_d[\varepsilon u_\mu u_\nu + (-1 + \varepsilon)g_{\mu\nu}]$

For homogeneous cosmological evolution, energy conservation gives:

$dE_d/dt + 3HE_d\varepsilon = 0$

Consequently, the energy density evolves according to:

$E_d(a) = E_{d,0} \exp[-3\int_1^a \varepsilon(x)dx/x]$

For a constant nonzero ε:

$E_d(a) = E_{d,0} a^{(-3\varepsilon)}$

This formulation does not require exactly constant energy density.

The theoretical distinction is that the PPC model approaches vacuum-like behaviour through $\omega \approx -1$ while retaining a possible nonzero deviation from the exact vacuum-like value.

The condition $\omega \approx -1$ is therefore distinct from the exact condition $\omega = -1$.

The proposed near-vacuum-like behaviour requires observational investigation and should not be treated as an established cosmological result.

RESEARCH NOTE 1

TITLE: MATHEMATICAL DERIVATION OF THE PPC ACCELERATION EQUATION USING A NEAR-VACUUM-LIKE EQUATION-OF-STATE PARAMETER

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

ABSTRACT

This research note presents the mathematical derivation of the acceleration equation within the PPC cosmological framework using a variable equation-of-state parameter approaching -1 but not necessarily equal to -1.

The proposed condition is:

$\omega \approx -1$

For mathematical representation, the equation-of-state parameter is expressed as:

$\omega = -1 + \varepsilon$

Here, ε represents a small, nonzero deviation from -1.

The resulting acceleration equation describes accelerated cosmic expansion while allowing the PPC energy density to evolve rather than requiring it to remain exactly constant.

1. INTRODUCTION

The acceleration equation is one of the fundamental equations of cosmology. It describes the acceleration or deceleration of the cosmic scale factor.

Within the PPC framework, the equation of state is:

$P_g = \omega E_d$

Where:

P_g = PPC gravitational pressure, ω = Equation-of-state parameter, E_d = PPC energy density

The proposed near-vacuum-like condition is:

$\omega \approx -1$

Instead of imposing the exact condition $\omega = -1$, this research note considers a small deviation represented by ε.

Therefore:

$\omega = -1 + \varepsilon$

This formulation permits the investigation of a dynamical equation-of-state parameter.

2. ORIGINAL PPC ACCELERATION EQUATION

The acceleration equation for a homogeneous and isotropic universe containing a single PPC perfect fluid is:

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

Where:

$\ddot{a}$ = Second derivative of the cosmic scale factor with respect to cosmic time, a = Cosmic scale factor, G = Newtonian gravitational constant, ρ = PPC mass-equivalent density,ω = Equation-of-state parameter

The relationship between energy density and mass-equivalent density is:

$\rho = E_d/c^2$

Therefore, the acceleration equation can also be expressed as:

$\ddot{a}/a = -(4\pi G E_d/3c^2)(1 + 3\omega)$

3. SUBSTITUTION OF THE NEAR-VACUUM-LIKE CONDITION

The proposed PPC equation-of-state condition is:

$\omega = -1 + \varepsilon$

Substituting this value into the original acceleration equation:

$\ddot{a}/a = -(4\pi G\rho/3)[1 + 3(-1 + \varepsilon)]$

Expanding the expression inside the brackets:

$\ddot{a}/a = -(4\pi G\rho/3)(1 - 3 + 3\varepsilon)$

Simplifying:

$\ddot{a}/a = -(4\pi G\rho/3)(-2 + 3\varepsilon)$

Multiplying the negative signs:

$\ddot{a}/a = (4\pi G\rho/3)(2 - 3\varepsilon)$

4. FINAL PPC ACCELERATION EQUATION

The resulting PPC acceleration equation is:

$\ddot{a}/a = (4\pi G\rho/3)(2 - 3\varepsilon)$

Using:

$\rho = E_d/c^2$

The equivalent energy-density formulation is:

$\ddot{a}/a = (4\pi GE_d/3c^2)(2 - 3\varepsilon)$

Therefore, the final equation is:

$\ddot{a}/a = (8\pi G\rho/3)(1 - 3\varepsilon/2)$

This is the acceleration equation obtained from the proposed near-vacuum-like equation-of-state condition.

5. MATHEMATICAL SIGNIFICANCE

The derived equation contains the parameter ε.

This parameter represents the deviation of the PPC equation-of-state parameter from -1.

When ε is small and positive, the acceleration is lower than the acceleration obtained for the exact value ω = -1, assuming the same energy density.

When ε is small and negative, the acceleration is higher, assuming the same energy density.

The resulting equation therefore allows the acceleration rate to depend on the deviation from the vacuum-like equation-of-state value.

6. CONCLUSION

The PPC acceleration equation under the proposed condition ω = -1 + ε is:

ä/a = (4πGρ/3)(2 - 3ε)

Where, ρ=E_d/c^2

ä/a = (4πGE_d/3c^2)(2 - 3ε)

This equation retains a nonzero deviation from -1 and allows the PPC energy density to evolve with cosmic expansion.

The equation is a mathematical consequence of the standard FLRW acceleration equation applied to the proposed PPC perfect-fluid equation of state.

RESEARCH NOTE 2

TITLE: PHYSICAL CONDITION FOR ACCELERATED COSMIC EXPANSION IN THE PPC MODEL WITH ω = -1 + ε

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

The PPC cosmological model considers the near-vacuum-like equation-of-state condition:

ω = -1 + ε

Where ε is a small, nonzero deviation from -1.

The purpose of this research note is to derive the mathematical condition under which the PPC universe undergoes accelerated expansion.

2. ACCELERATION EQUATION

The PPC acceleration equation is:

$\ddot{a}/a = (4\pi G\rho/3)(2 - 3\varepsilon)$

For positive PPC energy density:

$\rho > 0$

$E_d>0$

The sign of cosmic acceleration depends on the factor:

$2 - 3\varepsilon$

3. CONDITION FOR ACCELERATED EXPANSION

Accelerated expansion requires:

$\ddot{a}/a > 0$

Substituting the PPC acceleration equation:

$(4\pi G\rho/3)(2 - 3\varepsilon) > 0$

Since:

$4\pi G\rho/3 > 0$

The remaining condition is:

$2 - 3\varepsilon > 0$

Therefore:

$3\varepsilon < 2$

Hence:

$\varepsilon < 2/3$

The mathematical condition for accelerated expansion is:

$\varepsilon < 2/3$

4. CONDITION FOR ZERO ACCELERATION

The boundary between accelerated and decelerated expansion occurs when:

$\ddot{a}/a = 0$

Therefore:

$2 - 3\varepsilon = 0$

Solving:

$\varepsilon = 2/3$

At this boundary:

$\omega = -1 + 2/3$

Therefore:

$\omega = -1/3$

This is the equation-of-state value at which the acceleration contribution of a single perfect fluid becomes zero.

5. CONDITION FOR DECELERATED EXPANSION

Decelerated expansion requires:

$\ddot{a}/a < 0$

For positive energy density:

$2 - 3\varepsilon < 0$

Therefore:

$\varepsilon > 2/3$

Equivalently:

$\omega > -1/3$

6. INTERPRETATION FOR SMALL DEVIATIONS

The PPC proposal considers:

$0 < |\varepsilon| << 1$

This means that the deviation is much smaller than unity.

For such small deviations:

$2 - 3\varepsilon > 0$

Therefore, accelerated expansion occurs for positive PPC energy density.

The sign of ε determines whether the acceleration is slightly higher or lower than the exact vacuum-like reference case at the same density.

7. CONCLUSION

For a single PPC perfect fluid with positive energy density, the acceleration conditions are:

Accelerated expansion:

$\varepsilon < 2/3$

Zero acceleration:

$\varepsilon = 2/3$

Decelerated expansion:

$\varepsilon > 2/3$

The proposed near-vacuum-like condition, with a sufficiently small deviation from zero, satisfies the accelerated-expansion condition.

RESEARCH NOTE 3

TITLE: ENERGY-DENSITY EVOLUTION AND COSMIC ACCELERATION IN THE PPC MODEL WITH A VARIABLE DEVIATION PARAMETER

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

The near-vacuum-like PPC equation of state is:

$\omega = -1 + \varepsilon$

If ε is nonzero, the PPC energy density is not generally constant.

This research note derives the relationship between the deviation parameter, energy-density evolution and cosmic expansion.

2. PPC ENERGY-CONSERVATION EQUATION

For a homogeneous and isotropic FLRW universe:

$dE_d/dt + 3H(E_d + P_g) = 0$

The PPC equation of state is:

$P_g = \omega E_d$

Substituting:

$dE_d/dt + 3HE_d(1 + \omega) = 0$

Using:

$\omega = -1 + \varepsilon$

We obtain:

$dE_d/dt + 3HE_d[1 + (-1 + \varepsilon)] = 0$

Simplifying:

$dE_d/dt + 3HE_d\varepsilon = 0$

Therefore:

$dE_d/dt = -3HE_d\varepsilon$

3. SOLUTION FOR CONSTANT ε

For constant ε:

$dE_d/E_d = -3\varepsilon Hdt$

Since:

$H = (1/a)(da/dt)$

We obtain:

$dE_d/E_d = -3\varepsilon(da/a)$

Integrating:

$\ln(E_d/E_{d,0}) = -3\varepsilon \ln(a/a_0)$

Choosing the normalization $a_0 = 1$:

$\ln(E_d/E_{d,0}) = -3\varepsilon \ln(a)$

Therefore:

$E_d(a) = E_{d,0}\, a^{(-3\varepsilon)}$

This is the energy-density evolution equation for constant nonzero ε.

4. SOLUTION FOR VARIABLE ε

If ε varies with the scale factor:

$\varepsilon = \varepsilon(a)$

The energy-density evolution equation becomes:

$dE_d/E_d = -3\varepsilon(a)\, da/a$

Integrating:

$E_d(a) = E_{d,0} \exp[-3\int_1^a \varepsilon(x)dx/x]$

This is the general homogeneous solution for a scale-factor-dependent deviation parameter.

5. RELATIONSHIP WITH THE ACCELERATION EQUATION

The PPC acceleration equation is:

$\ddot{a}/a = (4\pi G\rho/3)(2 - 3\varepsilon)$

Since:

$\rho = E_d/c^2$

We obtain:

$\ddot{a}/a = (4\pi G E_d/3c^2)(2 - 3\varepsilon)$

Substituting the energy-density solution for constant ε:

$\ddot{a}/a = (4\pi G E_{d,0}/3c^2)a^{(-3\varepsilon)}(2 - 3\varepsilon)$

This equation displays the combined dependence of cosmic acceleration on the scale factor and the deviation parameter.

6. PHYSICAL INTERPRETATION

For positive ε:

$E_d(a)$ decreases as the scale factor increases.

For negative ε:

E_d(a) increases as the scale factor increases.

For small nonzero ε:

The PPC energy density evolves slowly relative to a fluid with a substantially different equation-of-state parameter.

The acceleration therefore depends on both the instantaneous energy density and the value of ε.

7. CONCLUSION

The PPC model with ω = -1 + ε gives:

Energy-density evolution:

$E_d(a) = E_{d,0}\, a^{(-3\varepsilon)}$

Acceleration:

$\ddot{a}/a = (4\pi G E_d/3c^2)(2 - 3\varepsilon)$

These equations describe the mathematical relationship between the near-vacuum-like equation of state, energy-density evolution and cosmic acceleration.

RESEARCH NOTE 4

TITLE: TIME-DEPENDENT DEVIATION FROM THE VACUUM-LIKE EQUATION OF STATE IN THE PPC COSMOLOGICAL MODEL

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

The PPC model can be extended by allowing the deviation from -1 to evolve with cosmic time.

The equation-of-state parameter is expressed as:

ω(t) = -1 + ε(t)

Where ε(t) is a small, nonzero time-dependent deviation.

This formulation permits the equation-of-state parameter to evolve towards or away from -1.

2. TIME-DEPENDENT PPC EQUATION OF STATE

The PPC equation of state is:

$P_g(t) = \omega(t)E_d(t)$

Substituting:

$\omega(t) = -1 + \varepsilon(t)$

We obtain:

$P_g(t) = [-1 + \varepsilon(t)]E_d(t)$

3. TIME-DEPENDENT ACCELERATION EQUATION

The original acceleration equation is:

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

Substituting the time-dependent parameter:

$\ddot{a}/a = -(4\pi G\rho/3)[1 + 3(-1 + \varepsilon(t))]$

Simplifying:

$\ddot{a}/a = (4\pi G\rho/3)[2 - 3\varepsilon(t)]$

Therefore:

$\ddot{a}/a = (4\pi GE_d/3c^2)[2 - 3\varepsilon(t)]$

This is the time-dependent PPC acceleration equation.

4. TIME-DEPENDENT ENERGY CONSERVATION

The homogeneous conservation equation is:

$dE_d/dt + 3HE_d(1 + \omega) = 0$

Substituting:

$\omega(t) = -1 + \varepsilon(t)$

We obtain:

$dE_d/dt + 3HE_d\varepsilon(t) = 0$

Therefore:

$dE_d/dt = -3HE_d\varepsilon(t)$

Integrating:

$E_d(t) = E_d(t_0) \exp[-3\int_{t_0}^{t} H(t')\varepsilon(t')dt']$

5. ACCELERATION CONDITION

For positive energy density:

$E_d>0$

$\rho > 0$

Accelerated expansion requires:

$2 - 3\varepsilon(t) > 0$

Therefore:

$\varepsilon(t) < 2/3$

If $\varepsilon(t)$ remains small compared with unity, this condition is satisfied.

6. ASYMPTOTIC BEHAVIOUR

If the proposed PPC evolution satisfies:

$\lim_{t\to t_{max}} \varepsilon(t) = 0$

Then:

$\lim_{t\to t_{max}} \omega(t) = -1$

The acceleration equation approaches:

$\ddot{a}/a \to 8\pi G\rho/3$

provided the energy density has a finite limiting value.

Importantly, the asymptotic condition does not require $\omega(t)$ to equal -1 at earlier times.

7. CONCLUSION

The time-dependent PPC formulation is:

$\omega(t) = -1 + \varepsilon(t)$

$P_g(t) = [-1 + \varepsilon(t)]E_d(t)$

Acceleration:

$\ddot{a}/a = (4\pi G\rho/3)[2 - 3\varepsilon(t)]$

Energy conservation:

$dE_d/dt + 3HE_d\varepsilon(t) = 0$

This formulation provides a mathematical framework for studying a time-dependent equation-of-state parameter that remains close to -1 without imposing the exact value.

CLARIFICATION NOTES ON THE DEFINITION OF ENERGY DENSITY (E_d) AND MASS DENSITY (ρ) IN THE PRESSURE–CURVATURE MODEL

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1. INTRODUCTION

The fundamental Pressure–Curvature (PPC) equation is:

$P_g = \omega E_d$

The energy density used in this equation is defined by:

$E_d = \rho c^2$

Where:

- E_d = Energy density represented by the physical system.
- ρ = Corresponding mass density or mass-equivalent density.
- c = Speed of light.

- P_g = Pressure associated with the PPC relation.
- ω = Dimensionless equation-of-state parameter.

The definition of E_d and ρ depends on the physical system and the type of density being represented.

The symbols E_d and ρ must therefore be interpreted according to the physical composition and scope of the system under consideration.

The purpose of these notes is to establish a consistent definition of energy density and mass density for different physical systems.

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2. GENERAL RELATIONSHIP BETWEEN ENERGY DENSITY AND MASS DENSITY

The fundamental relationship is:

$E_d = \rho c^2$

This relationship expresses the energy density associated with a corresponding mass density.

The dimensions are:

Energy density:

$[E_d] = J/m^3$

Mass density:

$[\rho] = kg/m^3$

Speed of light:

$[c] = m/s$

Therefore:

$[\rho c^2] = (kg/m^3)(m^2/s^2)$

$[\rho c^2] = J/m^3$

Thus, the equation is dimensionally consistent.

However, the physical meaning of ρ depends on which mass density is being represented.

It may represent:

1. Ordinary matter mass density.
2. Total local mass-equivalent density.
3. Total non-vacuum mass-equivalent density.
4. Total mass-equivalent density including vacuum energy.
5. Mass-equivalent density corresponding to the combined energy density of matter, radiation, and vacuum energy.

Each definition represents a different physical system or energy budget.

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3. CASE I: ρ AS THE TOTAL MASS DENSITY OF AN INDIVIDUAL PHYSICAL OBJECT

Suppose ρ represents the total mass density of an individual physical object, such as:

- Earth.
- Sun.
- Neutron star.
- Quark star.
- Other stars.
- Other planets.

In this situation:

ρ = Total mass density of the individual physical object.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total energy density associated with the mass density of that particular physical object.

For example:

A. Earth

If ρ represents the mass density of Earth, then:

$E_{d,Earth} = \rho_{Earth} c^2$

B. Sun

If ρ represents the mass density of the Sun, then:

$E_{d,Sun} = \rho_{Sun} c^2$

C. Neutron Star

If ρ represents the mass density of a neutron star, then:

$E_{d,NS} = \rho_{NS} c^2$

D. Quark Star

If ρ represents the mass density of a quark star, then:

$E_{d,QS} = \rho_{QS} c^2$

In each case, E_d represents the energy density associated with the corresponding object's mass density.

The physical system determines the value and interpretation of ρ and E_d.

Important clarification: In relativistic astrophysics, the appropriate density must be defined carefully. Rest-mass density, total local energy density, and gravitational mass density are not automatically identical. The relation $E_d = \rho c^2$ applies directly when ρ is defined as the corresponding mass-equivalent density.

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4. CASE II: ρ AS THE TOTAL LOCAL MASS-EQUIVALENT DENSITY

Suppose ρ represents the total local mass-equivalent density of a physical system.

In that situation:

ρ = Total local mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total local energy density associated with that total local mass-equivalent density.

Here, ρ is not restricted to the ordinary rest-mass density of individual material objects.

It represents the mass equivalent of the total energy density included in the selected physical system.

The definition must specify which energy contributions are included.

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5. CASE III: ρ AS THE TOTAL NON-VACUUM MASS-EQUIVALENT DENSITY

Suppose ρ represents the total non-vacuum mass-equivalent density.

In this situation:

ρ = Total non-vacuum mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total non-vacuum local energy density associated with that mass-equivalent density.

The physical system may include:

1. Ordinary matter energy.
2. Radiation energy.
3. Dark matter energy.
4. Other specified non-vacuum energy contributions.

The vacuum-energy contribution associated with the cosmological constant is excluded under this definition.

Thus:

E_d = Total non-vacuum local energy density.

And:

E_Λ = Vacuum-energy density associated with the cosmological constant.

These are separately defined contributions.

The combined energy density, when both are included in the same total energy budget, is:

$E_{d(total)} = E_d + E_\Lambda$

The corresponding combined mass-equivalent density is:

$\rho_{total} = E_{d(total)}/c^2$

---

6. CASE IV: ρ AS THE TOTAL MASS-EQUIVALENT DENSITY INCLUDING VACUUM ENERGY

Suppose ρ represents the total mass-equivalent density of a system that includes vacuum energy.

In that situation:

ρ = Total mass-equivalent density including vacuum energy.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total local energy density, including vacuum energy.

It may contain contributions from:

1. Ordinary matter.
2. Radiation.
3. Dark matter.
4. Vacuum energy.
5. Other specified energy contributions.

Under this definition, vacuum energy is included in E_d.

Consequently, if the cosmological constant is also retained explicitly in the geometric part of the Einstein field equations, its vacuum-energy contribution must not be counted a second time.

There are two consistent representations:

Representation A: Explicit cosmological constant

- Exclude E_Λ from E_d.
- Retain Λ explicitly in the Friedmann equations.

Representation B: Vacuum energy included in total fluid density

- Include E_Λ in E_d.
- Remove the corresponding explicit Λ term from the geometric side of the Friedmann equations.

These two representations must not double-count the same vacuum-energy contribution.

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7. CASE V: E_d AS THE TOTAL LOCAL ENERGY DENSITY OF ORDINARY MATTER, RADIATION, AND VACUUM ENERGY

Suppose E_d represents the total local energy density containing:

1. Total ordinary matter energy density.
2. Total radiation energy density.
3. Total vacuum-energy density.
4. Other specified energy contributions.

Then:

$$E_d = E_m + E_r + E_\Lambda + E_{other}$$

Where:

E_m = Total ordinary matter energy density.

E_r = Total radiation energy density.

E_Λ = Total vacuum-energy density.

E_{other} = Other included energy contributions.

The corresponding mass-equivalent density is:

$$\rho = E_d/c^2$$

Therefore:

$$E_d = \rho c^2$$

In this case, ρ represents the total mass-equivalent density associated with the complete energy density E_d.

It is important to understand that photons have zero invariant rest mass but possess energy and momentum. Their radiation energy contributes to the total energy density and therefore to the corresponding mass-equivalent density.

Similarly, vacuum energy contributes to the mass-equivalent density when it is included in E_d.

Thus, ρ in this case is not simply the ordinary rest-mass density of matter. It is the mass-equivalent density corresponding to the complete energy budget.

---

8. RELATIONSHIP BETWEEN THE DIFFERENT DEFINITIONS

The following table summarizes the different interpretations.

Case I: Individual physical object

ρ = Mass density of Earth, Sun, neutron star, quark star, or another individual object.

E_d = Energy density associated with that object's mass density.

Case II: Total local mass-equivalent density

ρ = Total local mass-equivalent density.

E_d = Total local energy density associated with that density.

Case III: Total non-vacuum mass-equivalent density

ρ = Total non-vacuum mass-equivalent density.

E_d = Total non-vacuum local energy density.

Case IV: Total mass-equivalent density including vacuum energy

ρ = Total mass-equivalent density, including vacuum energy.

E_d = Total local energy density, including vacuum energy.

Case V: Combined ordinary matter, radiation, and vacuum energy

ρ = Mass-equivalent density associated with the combined energy budget.

E_d = Total local energy density of ordinary matter, radiation, vacuum energy, and other included contributions.

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9. RELATIONSHIP WITH THE PRESSURE–CURVATURE EQUATION

The PPC equation is:

$P_g = \omega E_d$

Since:

$E_d = \rho c^2$

The PPC equation can also be expressed as:

$P_g = \omega \rho c^2$

This shows that the pressure associated with the PPC relation depends on:

1. The selected physical system.
2. The definition of its mass density ρ.
3. The corresponding energy density E_d.
4. The equation-of-state parameter ω.

Therefore, the PPC equation can be applied to different physical systems, provided the relevant density and pressure are defined consistently.

However, the physical interpretation of ω must be specified for each system.

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10. VACUUM-ENERGY-LIKE BEHAVIOUR WHEN $\omega = -1$

The PPC equation is:

$P_g = \omega E_d$

For:

$\omega = -1$

We obtain:

$P_g = -E_d$

Therefore:

The PPC pressure becomes equal to the negative of its associated energy density.

This produces vacuum-energy-like behaviour.

The standard cosmological-constant vacuum-energy relationship is:

$P_\Lambda = -E_\Lambda$

Both have the same mathematical equation-of-state relationship.

However, the physical interpretation depends on the definition of E_d.

If E_d excludes vacuum energy, the PPC fluid produces a separate vacuum-like contribution.

If E_d includes vacuum energy, the PPC fluid's total energy density already contains that contribution.

Therefore, the definition of E_d must always be specified before interpreting the physical meaning of $\omega = -1$.

---

11. APPLICATION TO THE FRIEDMANN EQUATIONS

For a homogeneous and isotropic universe, when E_d represents the selected total fluid energy density, the first Friedmann equation with an explicit cosmological constant is:

$$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$$

The second Friedmann equation is:

$$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$$

Using:

$$P_g = \omega E_d$$

We obtain:

$$\ddot{a}/a = -(4\pi G/3c^2)E_d(1 + 3\omega) + \Lambda c^2/3$$

Important condition:

These equations with explicit Λ require E_d to exclude the vacuum-energy contribution already represented by Λ.

If E_d includes that vacuum energy, the equivalent Λ contribution must instead be incorporated into the fluid and removed from the explicit geometric term.

The density convention must remain consistent throughout the derivation.

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12. IMPORTANT SCIENTIFIC CLARIFICATIONS

The following principles establish the intended meaning of the PPC variables:

1. E_d is defined through $E_d = \rho c^2$.

2. ρ does not have only one universal physical interpretation. Its definition depends on the physical system.

3. If ρ represents the mass density of an individual object, E_d represents the corresponding energy density of that object.

4. If ρ represents total local mass-equivalent density, E_d represents the corresponding total local energy density.

5. If ρ represents total non-vacuum mass-equivalent density, E_d represents total non-vacuum local energy density.

6. If ρ includes vacuum-energy mass equivalence, E_d also includes the corresponding vacuum-energy density.

7. If E_d includes ordinary matter, radiation, and vacuum energy, ρ represents the mass-equivalent density associated with all those contributions.

8. Vacuum energy must not be counted twice when the cosmological constant is retained explicitly.

9. The physical interpretation of ω depends on the physical system and the energy-density definition.

10. The equation P_g = ωE_d establishes an equation of state; a complete gravitational theory requires additional field equations and conservation laws.

---

13. FINAL CONCLUSION

The fundamental PPC relationship is:

$E_d = \rho c^2$

And:

$P_g = \omega E_d$

The definition of E_d and ρ depends on the physical system they represent.

The mass density may represent an individual physical object, total local mass-equivalent density, total non-vacuum mass-equivalent density, or total mass-equivalent density including vacuum energy.

Correspondingly, E_d may represent the energy density of an individual object, total local energy density, total non-vacuum energy density, or total local energy density including vacuum energy.

Therefore, the PPC variables must always be interpreted according to the selected physical system and its specified energy contributions.

FINAL DEFINING PRINCIPLE

The physical system determines what ρ represents, and the corresponding energy-density definition determines what E_d represents.

The general relationship remains:

$E_d = \rho c^2$

The Pressure–Curvature equation remains:

$P_g = \omega E_d$

The definition of E_d is therefore system-dependent, while the mathematical relationship between energy density and its corresponding mass-equivalent density remains consistent.

Clarification Note: Definition and Physical Interpretation of E_d in the PPC Theory

1. Fundamental Definition

In the Pressure–Curvature (PPC) theory, the energy density E_d is defined as:

$E_d = \rho c^2$

where:

- E_d represents the energy density associated with the mass-equivalent density of the physical system.
- ρ represents the mass-equivalent density of the system under consideration.
- c represents the speed of light in vacuum.

The definition of E_d is general. Its physical composition depends on the particular physical system being studied.

2. System-Dependent Meaning of Mass-Equivalent Density

The quantity ρ does not necessarily represent ordinary matter rest-mass density alone.

Depending on the physical system, it may represent the mass-equivalent density associated with matter like Earth, sun, or neutron stars,ordinary matter, dark matter, radiation energy, vacuum energy, or other energy contributions.

Therefore, E_d is not restricted to any single physical component.

3. Distinction Between Ordinary Matter and Radiation

Ordinary matter has mass density ρ_m and its corresponding energy density is:

$$E_m = \rho_m c^2$$

For ordinary nonrelativistic matter, the pressureless approximation is:

$$P_m \approx 0$$

Radiation is physically different. Photons have zero invariant rest mass, but they carry energy and momentum.

The energy of an individual photon is:

$$E_\gamma = h\nu$$

The radiation energy density is determined by the radiation energy contained per unit volume. For a continuous frequency spectrum:

$$E_r = \int_0^{\nu_{max}} u_\nu \, d\nu$$

Radiation can also be expressed through its mass-equivalent energy density, E_r/c^2, without assigning rest mass to photons.

For isotropic radiation:

$$P_r = E_r/3$$

Thus, ordinary matter and radiation remain physically distinguishable even when both contribute to the total energy density of a system.

4. Avoiding Double Counting of Energy Contributions

When E_d represents the total local energy density of a physical system, the individual energy contributions must be treated as components of that total.

For example, schematically:

$$E_d = E_m + E_{\text{dark matter}} + E_r + E_{\text{vacuum}} + \ldots$$

The exact composition depends on the physical system under consideration.

If E_m and E_r are already included in E_d, they must not be added again as independent contributions to the same total energy density.

However, their individual energy densities and pressure relationships may still be identified separately when formulating the total energy-momentum tensor.

5. Fundamental PPC Equation of State

The fundamental PPC equation remains:

$P_g = \omega E_d$

where:

- P_g represents the gravitational pressure field.
- E_d represents the energy density associated with the mass-equivalent density of the system.
- ω represents the dimensionless equation-of-state parameter.

The value of ω describes the pressure-to-energy-density relationship for the physical system or component being modeled.

6. Important Conceptual Distinction

The definition of E_d does not change from one physical system to another:

$E_d = \rho c^2$

What changes is the physical composition of ρ, depending on the system under consideration.

Mass-equivalent density must not be confused with the invariant rest-mass density of individual particles. In particular, radiation contributes energy and gravitational influence without requiring photons to possess rest mass.

7. Conclusion

In the PPC theory, E_d represents the energy density associated with the mass-equivalent density of the physical system under consideration. Its composition is system-dependent and may include different forms of matter and energy.

The distinction between ordinary matter and radiation is preserved because ordinary matter possesses rest mass, whereas photons have zero invariant rest mass but carry energy and momentum.

This clarification establishes a consistent interpretation of E_d and prevents confusion between total energy density, ordinary matter energy density, radiation energy density, and particle rest-mass density.

2nd RESEARCH PAPER

TITLE: OBSERVATIONAL TESTING OF THE PPC COSMOLOGICAL ACCELERATION EQUATION USING PANTHEON+, DESI DR2 AND PLANCK CMB CONSTRAINTS

Proposed by: Pawan Upadhyay **[Independent Researcher and Independent Scientist]**

ABSTRACT

This research paper develops an observational-testing framework for the PPC cosmological acceleration equation using modern cosmological observations, including Type Ia supernova measurements, baryon acoustic oscillations (BAO), and cosmic microwave background (CMB) constraints.

The proposed PPC equation of state is:

$P_g = \omega E_d$

The near-vacuum-like condition considered in this research is:

$\omega \approx -1$

For mathematical investigation, it is represented as:

$\omega(a) = -1 + \varepsilon(a)$

where $\varepsilon(a)$ is a small, potentially time-dependent and nonzero deviation from -1.

Substituting this condition into the Friedmann acceleration equation produces:

$\ddot{a}/a = (4\pi G\rho/3)(2 - 3\varepsilon)$

$\ddot{a}/a = (4\pi G E_d/3c^2)(2 - 3\varepsilon)$

$\ddot{a}/a = -(4\pi G E_d/3c^2)(-2+3\varepsilon)$

when the PPC fluid is the only cosmic component.

When ordinary matter and radiation are included, the complete acceleration equation becomes:

$\ddot{a}/a = -(4\pi G/3c^2)[E_d(-2+3\varepsilon)+E_m+2E_r]$

ä/a = (4πG/3c^2)[E_d(2 - 3ε)-E_m-2E_r]

ä/a = (4πG/3c^2)[E_d(2 - 3ε) - E_m - 2E_r]

The paper develops the mathematical relationship between the PPC equation of state, energy-density evolution, cosmic acceleration and observational quantities. It also describes the statistical methodology required to test the model against Pantheon+, DESI DR2 and Planck observations.

This paper establishes a theoretical and observational-testing framework. It does not claim that a numerical PPC likelihood analysis or observational validation has already been completed.

KEYWORDS

PPC Cosmological Model; PPC Energy-Momentum Tensor; Near-Vacuum-Like Equation of State; Variable Equation-of-State Parameter; Cosmic Acceleration; Pantheon+; DESI DR2; Planck CMB; Dark Energy; Friedmann Equations.

1. INTRODUCTION

The accelerated expansion of the universe is one of the major problems in modern cosmology.

The standard cosmological model describes the observed acceleration using a cosmological constant or an effective dark-energy component.

The PPC cosmological framework investigates an alternative mathematical formulation based on its proposed perfect-fluid energy-momentum tensor.

The PPC energy-momentum tensor is:

T_μv(PPC) = (E_d + P_g)u_μu_v + P_g g_μv

The PPC equation of state is:

P_g = ωE_d

Rather than imposing the exact condition ω = -1, this research considers a near-vacuum-like condition:

ω ≈ -1

The mathematical representation is:

ω = -1 + ε

where ε is a small, nonzero deviation parameter.

The purpose of this research is to determine how this formulation can be compared with modern observational constraints.

2. THEORETICAL FOUNDATION OF THE PPC MODEL

2.1 PPC Energy-Momentum Tensor

The proposed PPC energy-momentum tensor is:

$$T_{\mu\nu(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

Here:

$T_{\mu\nu(PPC)}$ = PPC energy-momentum tensor, E_d = PPC energy density, P_g = PPC gravitational pressure, u_μ = Fluid four-velocity, $g_{\mu\nu}$ = Spacetime metric tensor

2.2 Variable PPC Equation of State

The PPC equation of state is:

$$P_g = \omega E_d$$

For the near-vacuum-like condition:

$$\omega = -1 + \varepsilon$$

Therefore:

$$P_g = (-1 + \varepsilon)E_d$$

This formulation allows the energy density to evolve with cosmic expansion.

2.3 Variable PPC Energy-Momentum Tensor

Substituting $P_g = (-1 + \varepsilon)E_d$ into the PPC tensor:

$$T_{\mu\nu(PPC)} = [E_d + (-1 + \varepsilon)E_d]u_\mu u_\nu + (-1 + \varepsilon)E_d g_{\mu\nu}$$

Simplifying:

$$T_{\mu\nu(PPC)} = \varepsilon E_d u_\mu u_\nu + (-1 + \varepsilon)E_d g_{\mu\nu}$$

Therefore:

$$T_{\mu\nu(PPC)} = E_d[\varepsilon u_\mu u_\nu + (-1 + \varepsilon)g_{\mu\nu}]$$

This is the PPC energy-momentum tensor under the proposed near-vacuum-like equation of state.

3. PPC GRAVITATIONAL FIELD EQUATION

Einstein's field equation without an explicit cosmological constant is:

$G_{\mu v} = (8\pi G/c^4)T_{\mu v}$

Substituting the PPC energy-momentum tensor:

$G_{\mu v} = (8\pi G/c^4)E_d[\varepsilon u_\mu u_v + (-1 + \varepsilon)g_{\mu v}]$

Therefore:

$G_{\mu v} = (8\pi G/c^4)E_d[\varepsilon u_\mu u_v - (1 - \varepsilon)g_{\mu v}]$

This is the PPC field equation under the proposed near-vacuum-like condition.

The absence of an explicit cosmological-constant term does not itself establish that the model is physically distinct from a cosmological-constant model. Its physical distinction must be demonstrated through additional dynamics or observationally distinguishable predictions.

4. DERIVATION OF THE PPC ACCELERATION EQUATION

The standard acceleration equation for a homogeneous and isotropic universe containing a single perfect fluid is:

$\ddot{a}/a = -(4\pi G\rho/3)(1 + 3\omega)$

The relationship between energy density and mass-equivalent density is:

$\rho = E_d/c^2$

Substituting the proposed PPC equation-of-state parameter:

$\omega = -1 + \varepsilon$

We obtain:

$\ddot{a}/a = -(4\pi G\rho/3)[1 + 3(-1 + \varepsilon)]$

Expanding:

$\ddot{a}/a = -(4\pi G\rho/3)(1 - 3 + 3\varepsilon)$

Simplifying:

$\ddot{a}/a = -(4\pi G\rho/3)(-2 + 3\varepsilon)$

Therefore:

ä/a = (4πGρ/3)(2 - 3ε)

Using ρ = E_d/c²:

ä/a = (4πGE_d/3c²)(2 - 3ε)

THE FINAL PPC ACCELERATION EQUATION IS:

ä/a = (4πGE_d/3c^2)(2 - 3ε)

ä/a = (4πGρ/3)(2 - 3ε)

This equation is the central mathematical expression investigated in this research.

5. COMPLETE ACCELERATION EQUATION INCLUDING MATTER AND RADIATION

For an observational comparison with the real universe, ordinary matter and radiation must be included if PPC represents the dark-energy component rather than the entire cosmic fluid.

E_d = Total Energy Density associated with Total mass equivalent density

E_m = Total Ordinary matter energy density associated with ordinary matter mass density

E_r = Total Radiation Energy Density, There are neither mass nor mass density of Radiation.

The total acceleration equation is:

ä/a = -(4πG/3c^2)[E_m + 2E_r + E_d(1 + 3ω)]

Substituting:

ω = -1 + ε

We obtain:

ä/a = -(4πG/3c^2)[E_m + 2E_r + E_d(1 - 3 + 3ε)]

Simplifying:

ä/a = -(4πG/3c^2)[E_m + 2E_r + E_d(-2 + 3ε)]

Therefore:

THE COMPLETE PPC ACCELERATION EQUATION IS:

ä/a = (4πG/3c^2)[E_d(2 - 3ε) - E_m - 2E_r]

This is the appropriate background acceleration equation for testing a PPC dark-energy component alongside ordinary matter and radiation.

Here:

E_d = PPC Energy density, E_m = Total nonrelativistic matter energy density, E_r = Radiation Energy density

The distinction between the single-fluid equation and the complete cosmic equation is essential for observational consistency.

6. CONDITION FOR ACCELERATED EXPANSION

For the single PPC fluid:

ä/a = (4πGE_d/3c^2)(2 - 3ε)

Assuming:

E_d > 0

Accelerated expansion requires:

ä/a > 0

Therefore:

2 - 3ε > 0

Solving:

ε < 2/3

For the proposed near-vacuum-like condition:

|ε| << 1

the single-fluid acceleration is positive.

For a universe containing matter and radiation, the condition becomes:

E_d(2 - 3ε) > E_m + 2E_r

This is the more appropriate condition for accelerated expansion in a multicomponent universe.

7. PPC ENERGY-DENSITY EVOLUTION

The homogeneous energy-conservation equation is:

$$dE_d/dt + 3HE_d(1 + \omega) = 0$$

Substituting:

$$\omega = -1 + \varepsilon$$

We obtain:

$$dE_d/dt + 3HE_d\varepsilon = 0$$

Therefore:

$$dE_d/dt = -3HE_d\varepsilon$$

Using:

$$H = (1/a)(da/dt)$$

We obtain:

$$dE_d/E_d = -3\varepsilon\, da/a$$

For constant ε, integration gives:

$$E_d(a) = E_{d,0}\, a^{(-3\varepsilon)}$$

For a variable equation-of-state parameter:

$$\varepsilon = \varepsilon(a)$$

the solution is:

$$E_d(a) = E_{d,0} \exp[-3\int_1^a \varepsilon(x)dx/x]$$

This expression must be used when calculating the PPC contribution to the expansion history.

8. TIME-DEPENDENT PPC EQUATION OF STATE

A possible extension of the PPC framework is:

$$\omega = \omega(t)$$

The near-vacuum-like formulation becomes:

$$\omega(t) = -1 + \varepsilon(t)$$

The corresponding pressure is:

$P_g(t) = [-1 + \varepsilon(t)]E_d(t)$

The acceleration equation for a single PPC fluid becomes:

$\ddot{a}/a = (4\pi G E_d/3c^2)[2 - 3\varepsilon(t)]$

For a multicomponent universe:

$\ddot{a}/a = (4\pi G/3c^2)[E_d(2 - 3\varepsilon(t)) - E_m - 2E_r]$

The energy-conservation equation is:

$dE_d/dt + 3HE_d\varepsilon(t) = 0$

Its solution is:

$E_d(t) = E_d(t_0) \exp[-3\int(\text{from } t_0 \text{ to } t) H(t')\varepsilon(t')dt']$

This formulation allows the PPC equation-of-state parameter to evolve with cosmic time.

9. CPL PARAMETRIZATION FOR OBSERVATIONAL TESTING

To compare the PPC model with standard observational constraints on evolving dark energy, a useful parametrization is the Chevallier-Polarski-Linder (CPL) form:

$\omega(a) = \omega_0 + \omega_a(1 - a)$

The corresponding PPC formulation is:

$\omega(a) = -1 + \varepsilon(a)$

Therefore:

$\varepsilon(a) = 1 + \omega_0 + \omega_a(1 - a)$

Define:

$\varepsilon_0 = 1 + \omega_0$

$\varepsilon_a = \omega_a$

Then:

$\varepsilon(a) = \varepsilon_0 + \varepsilon_a(1 - a)$

This gives:

$\omega(a) = -1 + \varepsilon_0 + \varepsilon_a(1 - a)$

The present-day value is:

$\varepsilon(a = 1) = \varepsilon_0$

At redshift z:

$a = 1/(1 + z)$

Therefore:

$\varepsilon(z) = \varepsilon_0 + \varepsilon_a[z/(1 + z)]$

This parametrization provides a direct connection between the PPC deviation parameters and the conventional observational parameters.

10. PPC ENERGY DENSITY IN THE CPL PARAMETRIZATION

The general PPC energy-density evolution is:

$E_d(a) = E_{d,0} \exp[-3\int_1^a \varepsilon(x)dx/x]$

Substituting:

$\varepsilon(a) = \varepsilon_0 + \varepsilon_a(1 - a)$

We obtain:

$E_d(a) = E_{d,0} \exp[-3\int_1^a (\varepsilon_0 + \varepsilon_a - \varepsilon_a x)dx/x]$

Integrating:

$E_d(a) = E_{d,0} \exp[-3(\varepsilon_0 + \varepsilon_a)\ln(a) + 3\varepsilon_a(a - 1)]$

Therefore:

$E_d(a) = E_{d,0}\, a^{[-3(\varepsilon_0 + \varepsilon_a)]} \exp[3\varepsilon_a(a - 1)]$

In redshift notation:

$E_d(z) = E_{d,0}\, (1 + z)^{[3(\varepsilon_0 + \varepsilon_a)]} \exp[-3\varepsilon_a z/(1 + z)]$

This equation provides the PPC energy-density function required for calculating the theoretical expansion history.

11. OBSERVATIONAL DATASETS

11.1 Pantheon+ Type Ia Supernovae

Pantheon+ provides Type Ia supernova observations that can be used to test the predicted luminosity-distance relation.

The published Pantheon+ analysis contains 1,701 light curves from 1,550 distinct Type Ia supernovae.

The analysis covers a broad redshift interval and includes a detailed treatment of observational uncertainties and systematic effects.

For a spatially flat universe:

$$D_L(z) = (1 + z)c\int_0^z dz'/H(z')$$

The theoretical distance modulus is:

$$\mu_{th}(z) = 5\log_{10}[D_L(z)/\mathrm{Mpc}] + 25$$

The predicted distance modulus can be compared with the observed supernova distance moduli.

The comparison must use the released Pantheon+ covariance matrix and the appropriate nuisance-parameter treatment.

11.2 DESI DR2 Baryon Acoustic Oscillations

The Dark Energy Spectroscopic Instrument provides BAO measurements that constrain the expansion history and distances across redshift.

Important BAO observables include:

$$D_M(z)/r_d$$

$$D_H(z)/r_d$$

Where:

$D_M(z)$ = Transverse comoving distance $D_H(z) = c/H(z)$ r_d = Sound horizon at the baryon drag epoch

The PPC expansion history must predict these quantities before comparison with the DESI measurements.

DESI DR2 also provides additional information from the Lyman-alpha forest.

The July 2026 DESI DR2 Results IV analysis reports joint Lyman-alpha AP and BAO measurements at an effective redshift of $z = 2.33$:

$$D_H/r_d = 8.600 \pm 0.066$$

$D_M/r_d = 39.32 \pm 0.33$

These are observational distance-ratio measurements, not PPC predictions.

11.3 Planck Cosmic Microwave Background Observations

Planck CMB observations constrain the early-universe expansion history, matter density, baryon density and other cosmological parameters.

The PPC model must reproduce the relevant CMB observables.

These include:

1. The angular acoustic scale.
2. The sound horizon at recombination.
3. The physical baryon density.
4. The physical matter density.
5. The CMB temperature and polarization power spectra.
6. The effects of dark energy on late-time CMB observables.

A comparison using only the present-day acceleration equation is insufficient to establish agreement with CMB observations.

12. OBSERVATIONAL CONSTRAINTS ON THE EQUATION-OF-STATE PARAMETER

Published cosmological constraints provide useful reference values for the PPC deviation parameter.

12.1 Pantheon+ Combined Constraint

The Pantheon+ analysis reported:

$w_0 = -0.978\ (+0.024, -0.031)$

for a flat w_0CDM model combining supernova observations with CMB and BAO constraints.

Since:

$\varepsilon_0 = 1 + w_0$

The corresponding PPC parameter is:

$\varepsilon_0 = 0.022\ (+0.024, -0.031)$

This result is consistent with a small deviation from -1.

It is also statistically consistent with $\varepsilon_0 = 0$.

12.2 Planck 2018 Combined Constraint

The Planck 2018 analysis, when combined with BAO and Type Ia supernova observations, reported:

$w_0 = -1.03 \pm 0.03$

Therefore:

$\varepsilon_0 = -0.03 \pm 0.03$

This constraint is also consistent with $\varepsilon_0 = 0$.

12.3 Interpretation

These published results demonstrate that cosmological observations permit investigation of small deviations from the exact value -1.

However, the reported parameters were obtained under their respective standard cosmological model assumptions.

They cannot be directly presented as a best-fit PPC measurement.

A dedicated PPC likelihood analysis is required to determine the actual PPC parameter constraints.

13. STATISTICAL TESTING METHODOLOGY

The PPC model must be tested using a joint statistical likelihood.

For a Gaussian data likelihood, the chi-square statistic is:

$\chi^2 = \Delta D^T C^{-1} \Delta D$

Where:

ΔD = Difference between observed and theoretical data vectors C = Observational covariance matrix

The total likelihood may include:

$\chi^2_total = \chi^2_SN + \chi^2_BAO + \chi^2_CMB$

The individual terms must use the correct likelihood functions and covariance information for each dataset.

For the supernova observations:

$\Delta\mu = \mu_{observed} - \mu_{theoretical}$

For BAO:

The theoretical D_M/r_d and D_H/r_d values must be compared with the corresponding observed measurements.

For CMB:

The complete CMB likelihood, including the required early-universe physics, must be evaluated.

14. PPC PARAMETERS FOR NUMERICAL FITTING

A possible parameter vector is:

$\theta = \{H_0, \Omega_m, \Omega_k, \varepsilon_0, \varepsilon_a, r_d\}$

Additional parameters may be required for baryon density, radiation density, neutrino physics, sound-horizon calibration and supernova nuisance parameters.

The model should be fitted under at least two alternatives:

MODEL A: CONSTANT DEVIATION

$\varepsilon(a) = \varepsilon_0$

MODEL B: EVOLVING DEVIATION

$\varepsilon(a) = \varepsilon_0 + \varepsilon_a(1 - a)$

The constant-deviation model tests whether a single small departure from -1 is sufficient.

The evolving-deviation model tests whether the deviation changes with cosmic expansion.

15. MODEL COMPARISON

The PPC model should be compared with the standard ΛCDM model and an appropriate w_0w_aCDM reference model.

The comparison should include:

1. Best-fit parameter values.
2. Parameter uncertainties.
3. Minimum chi-square.
4. Reduced chi-square where appropriate.
5. Bayesian evidence.
6. Information criteria.
7. Posterior parameter correlations.

8. Stability under different observational dataset combinations.

A lower chi-square alone does not establish that PPC is a better physical model.

The number of free parameters and the statistical complexity of the competing models must be taken into account.

16. ADDITIONAL THEORETICAL REQUIREMENTS

A successful background expansion fit is not sufficient to establish a complete cosmological model.

The following issues must also be addressed:

16.1 Perturbation Theory

The PPC model needs a consistent treatment of density perturbations and velocity perturbations.

16.2 Sound Speed

The effective sound speed and pressure perturbations must be specified.

16.3 Structure Formation

The model must be tested against galaxy clustering and the growth of cosmic structures.

16.4 Early-Universe Consistency

The PPC formulation must reproduce the early expansion history required by nucleosynthesis, recombination and CMB observations.

16.5 Stability

The proposed evolution must be checked for physical and mathematical instabilities.

16.6 Covariant Conservation

The PPC energy-momentum tensor must satisfy:

$\nabla_\mu T^{\mu\nu}(PPC) = 0$

This is required for consistency with the Einstein field equations.

17. RESULTS AND CURRENT STATUS

The mathematical derivation establishes that the PPC near-vacuum-like equation of state:

$\omega = -1 + \varepsilon$

leads to the acceleration equation:

$\ddot{a}/a = (4\pi G E_d/3c^2)(2 - 3\varepsilon)$

For a universe containing matter and radiation:

$\ddot{a}/a = (4\pi G/3c^2)[E_d(2 - 3\varepsilon) - E_m - 2E_r]$

Published Pantheon+ and Planck constraints permit investigation of small departures from -1.

The 2026 DESI DR2 analysis provides additional observational motivation for testing evolving dark-energy parametrizations.

However, no PPC-specific numerical optimization, joint likelihood evaluation, parameter posterior analysis or Bayesian model comparison is presented in this paper.

Accordingly, the observational findings should be treated as motivation and reference constraints, not as proof that PPC has passed observational testing.

18. DISCUSSION

The PPC framework provides a mathematical way to express a near-vacuum-like equation of state through a deviation parameter.

The formulation:

$\omega = -1 + \varepsilon$

is useful because it distinguishes the proposed near-vacuum-like behaviour from the exact value -1.

The parameter ε determines the deviation of the pressure from the negative energy density:

$P_g = (-1 + \varepsilon)E_d$

It also affects the evolution of energy density:

$E_d(a) = E_{d,0} \exp[-3\int_1^a \varepsilon(x)dx/x]$

And the acceleration of the universe:

$\ddot{a}/a = (4\pi G E_d/3c^2)(2 - 3\varepsilon)$

These relationships provide a consistent mathematical starting point for an observational investigation.

Nevertheless, the PPC tensor has the standard perfect-fluid mathematical form. To demonstrate a distinct physical theory, the PPC framework must provide additional physical principles or predictions that differ measurably from existing cosmological models.

19. CONCLUSION

This research paper develops the mathematical and observational-testing framework for the PPC cosmological acceleration equation under the proposed near-vacuum-like condition:

$\omega = -1 + \varepsilon$

The resulting single-fluid acceleration equation is:

$\ddot{a}/a = (4\pi G E_d/3c^2)(2 - 3\varepsilon)$

When ordinary matter and radiation are included, the complete equation is:

$\ddot{a}/a = (4\pi G/3c^2)[E_d(2 - 3\varepsilon) - E_m - 2E_r]$

The energy-density evolution for a variable deviation parameter is:

$E_d(a) = E_{d,0} \exp[-3\int_1^a \varepsilon(x)dx/x]$

The observational framework uses Pantheon+ supernova measurements, DESI DR2 BAO observations and Planck CMB constraints.

Published observations provide motivation for testing a small or evolving deviation from -1. However, a dedicated PPC statistical fit is necessary before any claim of observational confirmation can be made.

The next stage of research should therefore be a numerical PPC parameter estimation and comparison with ΛCDM and w_0w_aCDM using released observational likelihoods and covariance matrices.

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SEPARATE RESEARCH NOTE 1

TITLE: OBSERVATIONAL ASSESSMENT OF THE PPC ACCELERATION EQUATION USING PANTHEON+ SUPERNOVA DATA

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. PPC THEORETICAL EQUATION

The proposed PPC equation of state is:

$\omega = -1 + \varepsilon$

The corresponding acceleration equation for a single PPC fluid is:

$\ddot{a}/a = (4\pi G E_d/3c^2)(2 - 3\varepsilon)$

2. OBSERVATIONAL CONSTRAINT

The published Pantheon+ combined analysis reported:

$w_0 = -0.978$ (+0.024, -0.031)

Using:

$\varepsilon_0 = 1 + w_0$

We obtain:

$\varepsilon_0 = 0.022$ (+0.024, -0.031)

3. SUBSTITUTION INTO THE PPC EQUATION

Using the central value $\varepsilon_0 = 0.022$:

ä/a = (4πGE_d/3c^2)(2 - 3 × 0.022)

Therefore:

ä/a = (4πGE_d/3c^2)(1.934)

4. OBSERVATIONAL INTERPRETATION

The Pantheon+ combined constraint permits investigation of a small deviation from the exact equation-of-state value -1.

The result is consistent with $\varepsilon_0 = 0$ within observational uncertainty.

5. CONCLUSION

Pantheon+ observations provide motivation for testing the PPC near-vacuum-like equation of state.

However, this is a conversion of a published w_0 constraint, not an independent PPC parameter fit.

SEPARATE RESEARCH NOTE 2

TITLE: OBSERVATIONAL ASSESSMENT OF THE PPC ACCELERATION EQUATION USING PLANCK CMB CONSTRAINTS

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. PPC EQUATION

The PPC equation of state is:

$\omega = -1 + \varepsilon$

The corresponding acceleration equation is:

ä/a = (4πGE_d/3c^2)(2 - 3ε)

2. PLANCK OBSERVATIONAL CONSTRAINT

The Planck 2018 combined analysis with BAO and Type Ia supernova observations reported:

$w_0 = -1.03 \pm 0.03$

Therefore:

$\varepsilon_0 = 1 + w_0$

$\varepsilon_0 = -0.03 \pm 0.03$

3. SUBSTITUTION INTO THE PPC ACCELERATION EQUATION

Using $\varepsilon_0 = -0.03$:

ä/a = (4πGE_d/3c^2)[2 - 3(-0.03)]

Therefore:

ä/a = (4πGE_d/3c^2)(2.09)

4. OBSERVATIONAL INTERPRETATION

The Planck combined constraint is consistent with $\varepsilon_0 = 0$ within its uncertainty.

It permits investigation of a small negative deviation from -1 but does not establish that such a deviation is physically required.

5. CONCLUSION

Planck observations provide an important reference for testing the PPC near-vacuum-like equation of state.

A dedicated PPC likelihood analysis is necessary to determine whether the PPC model fits the CMB observations.

SEPARATE RESEARCH NOTE 3

TITLE: OBSERVATIONAL ASSESSMENT OF A VARIABLE PPC EQUATION OF STATE USING 2026 SUPERNOVA CONSTRAINTS

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

The PPC model permits a time-dependent equation-of-state parameter:

$\omega(a) = -1 + \varepsilon(a)$

A variable equation of state can be investigated using the Chevallier-Polarski-Linder parametrization.

2. CPL PARAMETRIZATION

The conventional equation is:

$w(a) = w_0 + w_a(1 - a)$

The corresponding PPC formulation is:

$\varepsilon(a) = 1 + w_0 + w_a(1 - a)$

3. PUBLISHED 2026 OBSERVATIONAL CONSTRAINT

One 2026 combined supernova, CMB and BAO analysis reported:

$w_0 = -0.861$ (+0.044, -0.042)

$w_a = -0.60$ (+0.17, -0.19)

Therefore:

$\varepsilon_0 = 1 + w_0$

$\varepsilon_0 = 0.139$ (+0.044, -0.042)

4. SUBSTITUTION INTO THE PPC ACCELERATION EQUATION

At the present epoch, a = 1:

ä/a = (4πGE_d/3c^2)(2 - 3ε_0)

Using $\varepsilon_0 = 0.139$:

ä/a = (4πGE_d/3c^2)(2 - 0.417)

Therefore:

ä/a = (4πGE_d/3c^2)(1.583)

5. VARIABLE PPC DEVIATION

The corresponding deviation function is:

$\varepsilon(a) = 0.139 - 0.60(1 - a)$

This demonstrates how an evolving equation-of-state parameter can be investigated within the PPC framework.

6. OBSERVATIONAL INTERPRETATION

The cited analysis reports a preference for evolving dark energy under its particular dataset combination and statistical assumptions.

Its parameter constraints are not direct PPC measurements.

A PPC-specific numerical fit is required to determine whether the same observations favour the PPC formulation.

SEPARATE RESEARCH NOTE 4

TITLE: COMPLETE PPC ACCELERATION EQUATION FOR OBSERVATIONAL TESTING WITH MATTER AND RADIATION

Proposed by: Pawan Upadhyay Independent Researcher and Independent Scientist

1. INTRODUCTION

For observational comparison with the real universe, ordinary matter and radiation must be included if PPC represents the dark-energy component.

2. GENERAL ACCELERATION EQUATION

The total cosmic acceleration equation is:

$\ddot{a}/a = -(4\pi G/3c^2)[E_m + 2E_r + E_d(1 + 3\omega)]$

3. PPC SUBSTITUTION

Using:

$\omega = -1 + \varepsilon$

We obtain:

$\ddot{a}/a = -(4\pi G/3c^2)[E_m + 2E_r + E_d(1 - 3 + 3\varepsilon)]$

Simplifying:

$\ddot{a}/a = (4\pi G/3c^2)[E_d(2 - 3\varepsilon) - E_m - 2E_r]$

4. FINAL MULTICOMPONENT PPC ACCELERATION EQUATION

$\ddot{a}/a = (4\pi G/3c^2)[E_d(2 - 3\varepsilon) - E_m - 2E_r]$

5. CONDITION FOR ACCELERATED EXPANSION

For accelerated expansion:

ä/a > 0

Therefore:

E_d(2 - 3ε) > E_m + 2E_r

6. OBSERVATIONAL SIGNIFICANCE

This equation is more appropriate for comparison with Pantheon+, DESI BAO and Planck CMB observations when PPC is treated as a dark-energy component alongside ordinary matter and radiation.

The expansion history must be calculated using all relevant components.

7. CONCLUSION

The complete PPC acceleration equation provides a mathematical basis for testing the proposed near-vacuum-like equation of state against cosmological observations.

A numerical joint likelihood analysis is still required before observational confirmation can be claimed.

OVERALL OBSERVATIONAL CONCLUSION

Published cosmological analyses permit investigation of a small or evolving deviation from the equation-of-state value -1.

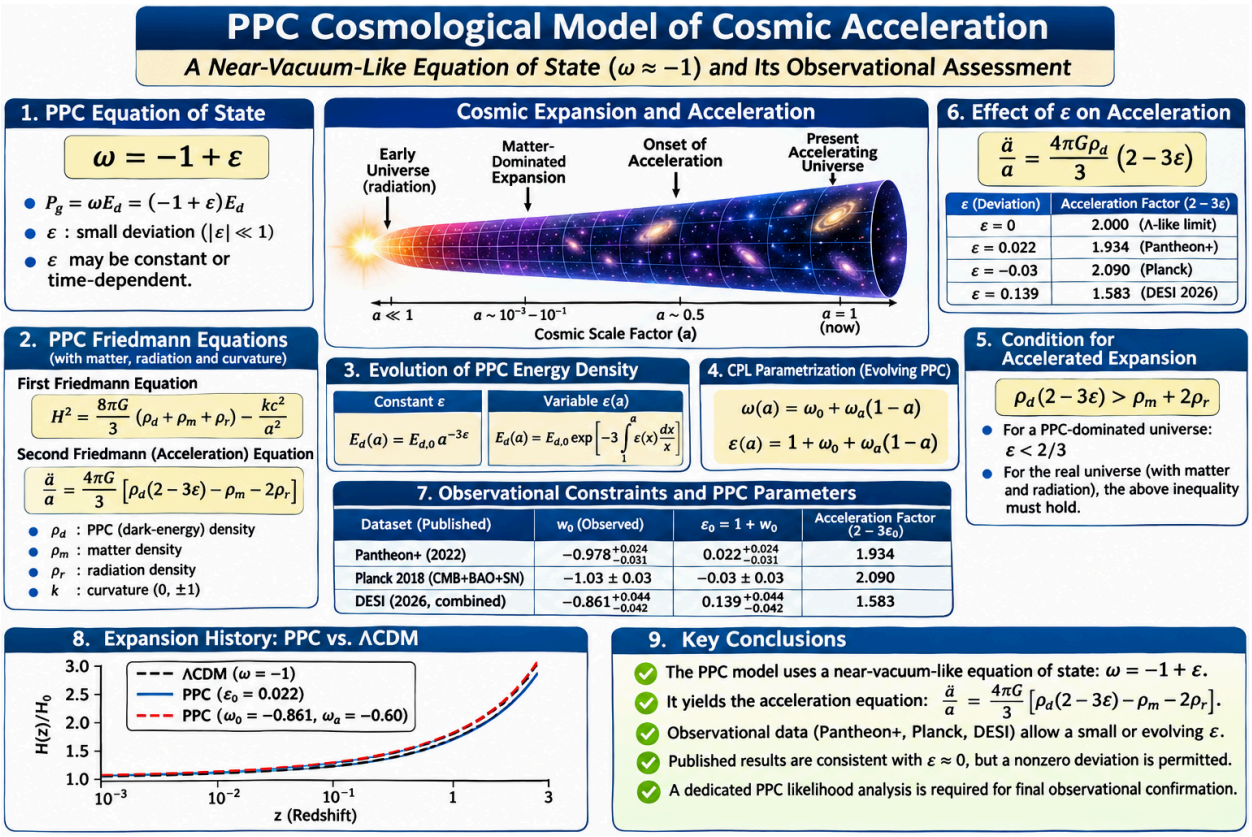
The PPC formulation:

ω = -1 + ε

provides a mathematical framework for this investigation.

However, Pantheon+, Planck and DESI constraints were obtained using their respective cosmological model assumptions. Their published values cannot be described as direct PPC best-fit results.

The PPC model requires its own numerical parameter estimation, covariance-weighted likelihood analysis and comparison with ΛCDM and evolving-dark-energy models.



RESEARCH NOTE

TITLE: PHYSICAL MEANING OF ENERGY DENSITY (E_d) AND ITS RELATIONSHIP WITH MASS-EQUIVALENT DENSITY (ρ) IN THE PPC COSMOLOGICAL MODEL

Author: Pawan Upadhyay

Designation: Independent Researcher and Independent Scientist

1. INTRODUCTION

In the proposed PPC cosmological framework, energy density (E_d) is associated with mass-equivalent density (ρ).

The fundamental relationship is:

$E_d = \rho c^2$

Where:

E_d = Energy density.

ρ = Associated mass-equivalent density.

c = Speed of light in vacuum.

This relationship follows from Einstein's mass-energy equivalence principle.

2. PHYSICAL MEANING OF MASS-EQUIVALENT DENSITY

Mass-equivalent density expresses energy density in units of mass per unit volume.

The dimensional relationship is:

$\rho = kg/m^3$

$E_d = J/m^3$

Consequently:

$\rho = E_d/c^2$

This definition allows energy density to be expressed through its equivalent mass density.

It is important to distinguish rest-mass density from general mass-equivalent density. When kinetic, thermal or other forms of energy are included, their energy also contributes to mass-equivalent density.

3. DISTINCTION BETWEEN MASSIVE MATTER AND LIGHT

Massive particles possess nonzero rest mass and corresponding rest energy:

$E_0 = mc^2$

Photons, which constitute light, have zero rest mass.

However, photons carry energy:

$E_\gamma = hv$

They also carry momentum:

$p_\gamma = E_\gamma/c$

Therefore, zero rest mass does not mean zero energy or zero gravitational influence.

In cosmology, radiation energy density must be included in the total gravitational energy-momentum source.

4. DISTINCTION BETWEEN PPC ENERGY DENSITY, ORDINARY MATTER AND RADIATION

For a cosmological model containing PPC, ordinary matter and radiation, the total energy density can be represented as:

$E_{total} = E_d + E_m + E_r$

Where:

E_d = PPC-associated energy density.

E_m = Ordinary matter energy density, including baryonic matter and dark matter.

E_r = Radiation energy density, including photons.

These components must be defined without counting the same physical energy twice.

5. PPC EQUATION OF STATE

The proposed PPC equation of state is:

$P_g = \omega E_d$

Where:

P_g = PPC gravitational pressure.

ω = PPC equation-of-state parameter.

Three cases are considered:

CASE I: NEAR-VACUUM-LIKE CONDITION

$\omega \approx -1$

CASE II: LIMITING CONDITION

$\omega \rightarrow -1$

CASE III: EXACT CONDITION

$\omega = -1$

These cases represent different mathematical possibilities and must be tested separately against cosmological observations.

6. ENERGY CONSERVATION

If the PPC fluid is separately conserved in a homogeneous and isotropic universe:

$\nabla_\mu T_PPC^{(\mu\nu)} = 0$

Its continuity equation becomes:

$dE_d/dt + 3H(E_d + P_g) = 0$

Using the PPC equation of state:

$P_g = \omega E_d$

We obtain:

$dE_d/dt + 3HE_d(1 + \omega) = 0$

For constant ω:

$E_d(a) = E_d,0 \times a^{[-3(1+\omega)]}$

For the exact condition $\omega = -1$:

$dE_d/dt = 0$

Therefore:

$E_d(a) = E_d,0$

Under the separate conservation assumption, PPC energy density remains constant during cosmic expansion in this exact case.

7. CONCLUSION

The fundamental PPC relationship:

$E_d = \rho c^2$

expresses the relationship between energy density and its mass-equivalent density.

However, the gravitational contribution of the universe cannot be described by rest-mass equivalence alone. Radiation, pressure and other energy-momentum contributions must also be considered.

The PPC framework therefore requires a precise physical definition of E_d, its relationship with ρ, and the physical origin of its effective pressure P_g.

A complete observational investigation must establish whether PPC is an additional cosmic fluid or a different physical description of an existing energy component.

RESEARCH OBJECTIVE

To investigate the cosmological consequences of the relationship between PPC-associated energy density, mass-equivalent density and the equation of state under the framework of General Relativity.

Note: Ordinary matter is pressureless and Light is massless.

RESEARCH NOTE

TITLE: WHY RADIATION ENERGY DENSITY AND ORDINARY MATTER ENERGY DENSITY ARE DIFFERENT FROM PPC TOTAL ENERGY DENSITY

Author: Pawan Upadhyay

Designation: Independent Researcher and Independent Scientist

1. INTRODUCTION

In the proposed PPC cosmological framework, energy density E_d is associated with its mass-equivalent density ρ.

The fundamental relationship is:

$$E_d = \rho c^2$$

Where:

E_d = PPC total energy density.

ρ = Total mass-equivalent density associated with PPC.

c = Speed of light in vacuum.

The PPC equation of state is:

$$P_g = \omega E_d$$

PPC energy density, ordinary matter energy density and radiation energy density must be distinguished because they represent different physical components with different pressure relationships.

2. WHY ORDINARY MATTER ENERGY DENSITY IS DIFFERENT

Ordinary matter consists of baryonic matter and dark matter.

Examples include:

- Protons and neutrons.
- Atoms and molecules.
- Stars and planets.
- Gas and dust.
- Dark matter.

Ordinary nonrelativistic matter has approximately zero pressure:

$P_m \approx 0$

Its energy density is:

$E_m = \rho_m c^2$

Its equation-of-state parameter is approximately:

$\omega_m = P_m/E_m \approx 0$

Therefore, ordinary matter has a different pressure relationship from the proposed PPC fluid.

PPC pressure:

$P_g = \omega E_d$

Ordinary matter pressure:

$P_m \approx 0$

This is why ordinary matter energy density must be identified separately from PPC energy density when PPC is treated as a distinct cosmological component.

3. WHY RADIATION ENERGY DENSITY IS DIFFERENT

Radiation includes electromagnetic radiation, particularly photons.

Photons have zero rest mass:

$m_\gamma = 0$

However, photons carry energy:

$E_\gamma = hv$

They also carry momentum:

$p_\gamma = E_\gamma/c$

Therefore, zero rest mass does not mean zero energy.

Radiation energy density is represented by E_r.

Its pressure is:

$P_r = E_r/3$

Its equation-of-state parameter is:

$\omega_r = 1/3$

Radiation therefore has a different physical relationship between energy density and pressure compared with both ordinary matter and the PPC fluid.

4. COMPARISON OF THE THREE ENERGY DENSITIES

PPC energy density:

$E_d = \rho c^2$

PPC pressure:

$P_g = \omega E_d$

Ordinary matter energy density:

$E_m = \rho_m c^2$

Ordinary matter pressure:

$P_m \approx 0$

Radiation energy density:

E_r

Radiation pressure:

$P_r = E_r/3$

The three components have different equations of state:

PPC: $\omega \approx -1$, $\omega \to -1$, or $\omega = -1$.

Ordinary matter: $\omega_m \approx 0$.

Radiation: $\omega_r = 1/3$.

These differences are important when constructing the cosmological energy-momentum tensor.

5. WHY THEIR ENERGY DENSITIES MUST BE DISTINGUISHED

The distinction is based on their physical properties and equations of state.

A. PPC has its proposed near-vacuum-like or vacuum-like pressure.

B. Ordinary matter has approximately zero pressure under nonrelativistic cosmological conditions.

C. Radiation has positive pressure equal to one-third of its energy density.

Consequently, these components cannot be assigned the same equation-of-state parameter.

In General Relativity, their separate energy-momentum tensors contribute to the total energy-momentum tensor:

$T_{total} = T_{PPC} + T_{matter} + T_{radiation}$

This addition does not mean that their physical definitions or pressure relationships are identical.

It means that all distinct gravitational sources must be included in Einstein's field equations.

6. CONSEQUENCES FOR THE PPC FRIEDMANN EQUATIONS

When PPC, ordinary matter and radiation are treated as separate, the first Friedmann equation without an explicit cosmological constant becomes:

$H^2 = (8\pi G/3c^2)(E_d + E_m + E_r) - kc^2/a^2$

The second Friedmann equation becomes:

$\ddot{a}/a = -(4\pi G/3c^2)[E_d(1 + 3\omega) + E_m + 2E_r]$

These equations assume that the three components are separately conserved perfect fluids.

7. ENERGY-DENSITY EVOLUTION

For constant PPC equation-of-state parameter:

$E_d(a) = E_{d,0} \times a^{[-3(1+\omega)]}$

For ordinary nonrelativistic matter:

$E_m(a) = E_{m,0} \times a^{-3}$

For radiation:

$E_r(a) = E_{r,0} \times a^{-4}$

These different evolution laws demonstrate that the three components respond differently to cosmic expansion.

8. CONCLUSION

The proposed PPC cosmological framework distinguishes PPC energy density, ordinary matter energy density and radiation energy density according to their physical properties and pressure relationships.

PPC energy density is associated with its mass-equivalent density through:

$E_d = \rho c^2$

Ordinary matter has approximately zero pressure, while radiation has zero photon rest mass but nonzero energy density and positive pressure.

PPC has its own proposed equation of state:

$P_g = \omega E_d$

Therefore, the three components require separate physical definitions and separate equations of state.

Their gravitational contributions must nevertheless be combined through their respective energy-momentum tensors when constructing the total source of Einstein's field equations.

RESEARCH OBJECTIVE

To establish the physical distinction between PPC energy density, ordinary matter energy density and radiation energy density, and to investigate their different contributions to cosmic expansion under General Relativity.

SCIENTIFIC CLARIFICATION

Ordinary matter and radiation can both have mass-equivalent energy densities when their total energies are divided by c^2.

Their separate treatment in this research note is justified by their different physical components and equations of state, not simply by saying that photons have zero rest mass or that ordinary matter has zero pressure.

3rd Research Paper

EINSTEIN'S ENERGY-MOMENTUM TENSORS FOR ORDINARY MATTER AND RADIATION: A THEORETICAL STUDY AND THEIR SEPARATE REPRESENTATION IN THE PPC COSMOLOGICAL FRAMEWORK

Author: Pawan Upadhyay
Research Area: General Relativity, Theoretical Cosmology and Energy-Momentum Tensor Analysis

ABSTRACT

Einstein's General Theory of Relativity describes the gravitational influence of matter and energy through the energy-momentum tensor. Ordinary matter and radiation are physically distinct forms of energy and therefore have different equations of state and different energy-momentum tensor representations.

This paper examines the energy-momentum tensors of pressureless ordinary matter and radiation within the framework of General Relativity. It derives their mathematical forms, explains their physical differences, and establishes how both components contribute to the total energy-momentum tensor.

The paper subsequently considers their separate representation alongside the proposed PPC cosmological component, defined by the energy density relation $E_d = \rho c^2$ and the pressure relation $P_g = \omega E_d$.

The central conclusion is that ordinary matter, radiation, and PPC can be represented as distinct physical components, while their energy-momentum tensors are combined to determine the total gravitational source in Einstein's field equations.

Keywords: Einstein's General Relativity, Energy-Momentum Tensor, Ordinary Matter, Radiation, Perfect Fluid, PPC Cosmological Model, Equation of State, Friedmann Equations.

1. INTRODUCTION

Einstein's General Theory of Relativity establishes a relationship between spacetime geometry and the distribution of energy and momentum.

The mathematical foundation is Einstein's field equation:

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$

Here:

$G_{\mu\nu}$ = Einstein tensor
G = Newton's gravitational constant
c = Speed of light
$T_{\mu\nu}$ = Energy-momentum tensor

The energy-momentum tensor describes energy density, momentum density, energy flux, and stresses, including pressure.

Different physical components can have different energy-momentum tensors. Ordinary matter and radiation are two important examples.

Ordinary matter is commonly approximated as pressureless dust in cosmological models, whereas radiation has a pressure equal to one-third of its energy density.

This distinction is important when constructing a cosmological model containing multiple physical components.

2. MATHEMATICAL CONVENTIONS

Throughout this paper, the metric signature is:

(-+++)

The four-velocity is normalized as:

$u_\mu u^\mu = -1$

Energy density and pressure are both expressed in energy-per-unit-volume units.

The metric tensor is represented by $g_{\mu\nu}$.

These conventions give the perfect-fluid energy-momentum tensor:

$T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

Where:

E_d = Energy density
P_g = Gravitational Pressure
u_μ = Four-velocity

The same four-velocity convention is used for the separate fluid components when they are comoving.

3. EINSTEIN'S ENERGY-MOMENTUM TENSOR FOR ORDINARY MATTER

Ordinary matter consists of massive particles, such as baryonic matter and nonrelativistic matter.

On cosmological scales, ordinary matter is frequently approximated as pressureless dust.

Its energy density is:

$E_m = \rho_m c^2$

Where:

ρ_m = Ordinary matter mass density
c = Speed of light

The general perfect-fluid energy-momentum tensor for ordinary matter is:

$T_{\mu\nu}^{(m)} = (E_m + P_m)u_\mu u_\nu + P_m g_{\mu\nu}$

For pressureless ordinary matter:

$P_m \approx 0$

Therefore:

$T_{\mu\nu}^{(m)} = E_m u_\mu u_\nu$

Substituting $E_m = \rho_m c^2$:

$T_{\mu\nu}^{(m)} = \rho_m c^2 u_\mu u_\nu$

This is the energy-momentum tensor of pressureless ordinary matter under the conventions adopted in this paper.

3.1 Physical interpretation

Ordinary matter possesses energy associated with its mass.

In the pressureless approximation, its pressure contribution is negligible compared with its energy density.

In a local comoving orthonormal frame, its energy-momentum tensor has the form:

T_(hat μ hat v)^(m) =

| E_m 0 0 0 |

| 0 0 0 0 |

| 0 0 0 0 |

| 0 0 0 0 |

The symbol is:

$$\boxed{T^{(m)}_{\hat{\mu}\hat{\nu}}}$$

For pressureless ordinary matter:

$$\boxed{T^{(m)}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} E_m & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}$$

Where $E_m = \rho_m c^2$

The hats indicate components in a **local orthonormal frame**.

The temporal component represents energy density, while the spatial diagonal components represent pressure.

The vanishing spatial components express the pressureless-dust approximation. They do not imply that ordinary matter has no gravitational influence.

4. EINSTEIN'S ENERGY-MOMENTUM TENSOR FOR RADIATION

Radiation is physically different from ordinary pressureless matter.

Photons have zero invariant rest mass, but they possess energy and momentum.

The energy of an individual photon is:

$E_\gamma = hv$

Where:

h = Planck's constant
v = Photon frequency

The photon momentum is:

$p_\gamma = E_\gamma/c$

Consequently, zero photon rest mass does not mean zero energy or zero gravitational influence.

For an isotropic radiation fluid, the equation of state is:

$P_r = E_r/3$

Where:

E_r = Radiation energy density
P_r = Radiation pressure

The perfect-fluid energy-momentum tensor for radiation is:

$T_{\mu\nu}^{(r)} = (E_r + P_r)u_\mu u_\nu + P_r g_{\mu\nu}$

Substituting $P_r = E_r/3$:

$T_{\mu\nu}^{(r)} = (E_r + E_r/3)u_\mu u_\nu + (E_r/3)g_{\mu\nu}$

Therefore:

T_μν^(r) = (4E_r/3)u_μu_ν + (E_r/3)g_μν

4.1 Radiation tensor in its comoving orthonormal frame

For isotropic radiation:

T_(hat μ hat ν)^(r) =

| E_r 0 0 0 |

| 0 E_r/3 0 0 |

| 0 0 E_r/3 0 |

| 0 0 0 E_r/3 |

$$T^{(r)}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} E_r & 0 & 0 & 0 \\ 0 & \frac{E_r}{3} & 0 & 0 \\ 0 & 0 & \frac{E_r}{3} & 0 \\ 0 & 0 & 0 & \frac{E_r}{3} \end{pmatrix}$$

Where:

$$P_r = \frac{E_r}{3}$$

- E_r: Radiation energy density.
- P_r: Radiation pressure.
- The hatted indices indicate a local orthonormal frame.

The temporal component represents radiation energy density.

The three spatial diagonal components represent radiation pressure.

Unlike pressureless ordinary matter, radiation has nonzero pressure.

4.2 Important qualification

The perfect-fluid radiation tensor describes an isotropic radiation distribution.

A general electromagnetic field need not be isotropic and is described by the electromagnetic stress-energy tensor derived from the electromagnetic field tensor. Therefore, the radiation-fluid expression is appropriate for an isotropic cosmological radiation component, not every possible electromagnetic configuration.

5. COMPARISON BETWEEN ORDINARY MATTER AND RADIATION

The two components have different physical properties and equations of state.

Physical property	Ordinary matter	Radiation
Energy density	$E_m = \rho_m c^2$	E_r
Pressure	$P_m \approx 0$	$P_r = E_r/3$
Equation-of-state parameter	$w_m \approx 0$	$w_r = 1/3$
Particle rest mass	Massive constituents	Photons have zero invariant rest mass
Energy	Nonzero	Nonzero

Gravitational contribution	Yes	Yes
Cosmological density evolution	a^{-3}	a^{-4}

The difference between the two components is not simply their rest mass.

Their pressure and energy-momentum distributions are also different.

6. SEPARATE ENERGY-MOMENTUM REPRESENTATION IN THE PPC FRAMEWORK

The proposed PPC framework introduces a separate cosmological component.

Its energy density is defined as:

$E_d = \rho c^2$

Its pressure is defined as:

$P_g = \omega E_d$

Where:

E_d = PPC energy density
ρ = Mass-equivalent density associated with PPC
P_g = PPC pressure
ω = PPC equation-of-state parameter

The PPC energy-momentum tensor is:

$T_{\mu v}^{(PPC)} = (E_d + P_g)u_\mu u_v + P_g g_{\mu v}$

Substituting $P_g = \omega E_d$:

$T_{\mu v}^{(PPC)} = E_d[(1 + \omega)u_\mu u_v + \omega g_{\mu v}]$

This represents the proposed PPC component as a perfect fluid.

6.1 Three separate physical components

The three components can now be represented independently:

A. PPC component

$E_d = \rho c^2$

$P_g = \omega E_d$

$T_{\mu\nu}^{(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

B. Ordinary matter component

$E_m = \rho_m c^2$

$P_m \approx 0$

$T_{\mu\nu}^{(m)} = E_m u_\mu u_\nu$

C. Radiation component

$P_r = E_r/3$

$T_{\mu\nu}^{(r)} = (4E_r/3)u_\mu u_\nu + (E_r/3)g_{\mu\nu}$

These expressions distinguish the physical properties and equations of state of the three components.

7. TOTAL ENERGY-MOMENTUM TENSOR

In a cosmological model containing all three components, the total energy-momentum tensor is obtained by adding their individual tensors:

$T_{\mu\nu}^{(total)} = T_{\mu\nu}^{(PPC)} + T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(r)}$

This equation is fundamental to the proposed three-component PPC formulation.

Substituting the individual expressions:

$T_{\mu\nu}^{(total)} =$

$(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

- $(E_m + P_m)u_\mu u_\nu + P_m g_{\mu\nu}$
- $(E_r + P_r)u_\mu u_\nu + P_r g_{\mu\nu}$

For a common four-velocity and pressureless ordinary matter:

$T_{\mu\nu}^{(total)} = (E_d + P_g + E_m + E_r + P_r)u_\mu u_\nu + (P_g + P_r)g_{\mu\nu}$

Using $P_g = \omega E_d$ and $P_r = E_r/3$:

$T_{\mu\nu}^{(total)} = [E_d(1 + \omega) + E_m + (4E_r/3)]u_\mu u_\nu + [\omega E_d + E_r/3]g_{\mu\nu}$

This simplified expression assumes all three components share the same four-velocity.

If their velocities differ, their tensors must be added using their respective four-velocities rather than combining them into one perfect-fluid expression.

8. EINSTEIN FIELD EQUATION FOR THE THREE-COMPONENT FRAMEWORK

Without an explicit cosmological constant, Einstein's field equation is:

$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}^{(total)}$

Substituting the three-component tensor:

$G_{\mu\nu} = (8\pi G/c^4)[T_{\mu\nu}^{(PPC)} + T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(r)}]$

This formulation retains the individual physical identities of PPC, ordinary matter, and radiation while including all three as gravitational sources.

It is important to distinguish separate definitions from separate gravitational contributions.

The densities are defined separately because they describe different physical components. Their energy-momentum tensors are nevertheless added in Einstein's field equation.

9. COVARIANT CONSERVATION OF ENERGY-MOMENTUM

The contracted Bianchi identity requires consistency with:

$\nabla_\mu G^{\mu\nu} = 0$

Consequently, the total energy-momentum tensor satisfies:

$\nabla_\mu T_{total}^{\mu\nu} = 0$

If the three components are separately conserved and do not exchange energy or momentum, then:

$\nabla_\mu T_{PPC}^{\mu\nu} = 0$

$\nabla_\mu T_m^{\mu\nu} = 0$

$\nabla_\mu T_r^{\mu\nu} = 0$

For an FLRW universe, the separate continuity equations become:

9.1 PPC continuity equation

$dE_d/dt + 3H(E_d + P_g) = 0$

Using $P_g = \omega E_d$:

$dE_d/dt + 3HE_d(1 + \omega) = 0$

For constant ω:

$E_d(a) = E_{d,0}\, a^{[-3(1 + \omega)]}$

9.2 Ordinary matter continuity equation

$dE_m/dt + 3H(E_m + P_m) = 0$

For pressureless matter:

$dE_m/dt + 3HE_m = 0$

Therefore:

$E_m(a) = E_{m,0}\, a^{-3}$

9.3 Radiation continuity equation

$dE_r/dt + 3H(E_r + P_r) = 0$

Using $P_r = E_r/3$:

$dE_r/dt + 4HE_r = 0$

Therefore:

$E_r(a) = E_{r,0}\, a^{-4}$

The different scaling laws demonstrate why ordinary matter, radiation, and PPC should be treated as distinct components when their equations of state differ.

If interactions or energy exchange between components are introduced, the individual conservation equations must be modified, while total conservation remains required.

10. COSMOLOGICAL IMPLICATIONS

The separate energy-momentum tensors have important consequences for cosmological expansion.

Ordinary matter dilutes according to the expansion of physical volume:

$E_m \propto a^{-3}$

Radiation dilutes more rapidly:

$E_r \propto a^{-4}$

The additional factor arises from the cosmological redshift of photon energy.

For a constant PPC equation-of-state parameter:

E_d ∝ a^[-3(1 + ω)]

Three cases are of particular interest:

Case 1: Near-vacuum-like PPC

$\omega = -1 + \varepsilon$

Where $|\varepsilon|$ is much smaller than 1.

Then:

$E_d(a) = E_{d,0}\, a^{-3\varepsilon}$

Case 2: Limit approaching vacuum-like behavior

$\omega \rightarrow -1$

The PPC energy density approaches a constant as its equation-of-state parameter approaches -1.

Case 3: Exact vacuum-like behavior

$\omega = -1$

Then:

$P_g = -E_d$

And:

$dE_d/dt = 0$

Therefore:

$E_d(a) = E_{d,0}$

These are mathematical consequences of the assumed perfect-fluid equation of state and separate conservation. They do not by themselves establish that PPC is a physically realized cosmological component.

11. DISCUSSION

This analysis establishes several important distinctions.

First, ordinary matter and radiation have different energy-momentum tensors because their pressure and energy distributions differ.

Second, ordinary matter having approximately zero pressure does not mean it has zero energy density.

Third, photons having zero invariant rest mass does not mean radiation has zero energy density or zero gravitational influence.

Fourth, the PPC density $E_d = \rho c^2$ is a definition for the proposed PPC component. It should not automatically be identified with the total energy density of the universe.

Fifth, separate physical definitions do not imply that the gravitational contributions are excluded from one another. Einstein's field equation requires the total energy-momentum source.

Finally, the PPC formulation requires further investigation of its physical origin, perturbations, stability, interactions, and observational consistency before it can be regarded as an established cosmological model.

12. CONCLUSION

Einstein's General Theory of Relativity provides a unified mathematical framework for describing the gravitational influence of energy and momentum.

However, different physical components require appropriate energy-momentum tensors.

For pressureless ordinary matter:

$T_{\mu\nu}^{(m)} = E_m u_\mu u_\nu$

For isotropic radiation:

$T_{\mu\nu}^{(r)} = (4E_r/3)u_\mu u_\nu + (E_r/3)g_{\mu\nu}$

For the proposed PPC component:

$T_{\mu\nu}^{(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$

The total gravitational source is:

$T_{\mu\nu}^{(total)} = T_{\mu\nu}^{(PPC)} + T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(r)}$

The central theoretical result is that PPC, ordinary matter, and radiation can be represented as three separately defined components with distinct energy densities and equations of state.

Their individual energy-momentum tensors must be combined to obtain the total source in Einstein's field equation.

This provides a mathematically consistent starting point for further development and testing of the three-component PPC cosmological framework, subject to the assumptions and limitations described in this paper.

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4th Research Paper

FRIEDMANN EQUATIONS WITH THE COSMOLOGICAL CONSTANT: DERIVATION, THREE PPC EQUATION-OF-STATE CASES, AND COMPARISON WITH MODERN COSMOLOGICAL OBSERVATIONS

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Research Area: General Relativity, Theoretical Cosmology and PPC Cosmological Framework

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ABSTRACT

This research examines the two Friedmann equations in the presence of a cosmological constant and develops their mathematical formulation for a cosmological framework containing three separately defined physical components: PPC, ordinary matter and radiation, together with the cosmological constant.

The PPC energy density is defined as $E_d = \rho c^2$, and its pressure is $P_g = \omega E_d$. Ordinary matter is represented by $E_m = \rho_m c^2$ with approximately zero pressure. Radiation is represented independently by its energy density E_r and pressure $P_r = E_r/3$. The cosmological constant is represented by its equivalent energy density $E_\Lambda = \Lambda c^4/(8\pi G)$.

Both Friedmann equations are derived, including the pressure contributions of radiation and the negative pressure associated with the cosmological constant.

Three PPC equation-of-state cases are investigated:

1. ω approximately equal to -1.
2. ω approaches -1.
3. ω exactly equal to -1.

The resulting equations are compared with modern observational constraints from Planck, DESI DR2, baryon acoustic oscillations, supernovae and cosmic chronometers.

The analysis identifies an important degeneracy between exact vacuum-like PPC and the cosmological constant. It also establishes the requirements for a genuine numerical observational fit of the PPC framework.

15.1 Vacuum Energy and the Cosmological Constant

Einstein's field equations with a cosmological constant are

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The cosmological constant term can be rewritten as an effective energy–momentum tensor:

$$T^{(\Lambda)}_{\mu\nu} = -\frac{\Lambda c^4}{8\pi G} g_{\mu\nu}$$

Comparing this with the perfect fluid energy–momentum tensor

$$T_{\mu\nu} = (E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu}$$

we identify the vacuum energy density

$$E_\Lambda = \frac{\Lambda c^4}{8\pi G}$$

and the corresponding pressure

$$P_\Lambda = -E_\Lambda$$

Thus

$$\omega_\Lambda = \frac{P_\Lambda}{E_\Lambda} = -1$$

which confirms that the cosmological constant behaves as a perfect fluid with equation-of-state parameter $\omega = -1$.

KEYWORDS

Friedmann Equations, Einstein Field Equations, PPC Cosmological Model, Energy Density, Radiation, Ordinary Matter, Cosmological Constant, Dark Energy, Equation of State, DESI DR2, Planck CMB, Cosmological Observations.

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1. INTRODUCTION

The Friedmann equations describe the expansion and acceleration of the universe within the homogeneous and isotropic Friedmann-Lemaître-Robertson-Walker (FLRW) cosmological framework.

They are derived from Einstein's General Theory of Relativity.

Einstein's field equation with a cosmological constant is:

$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$

Here:

$G_{\mu\nu}$ = Einstein tensor

Λ = Cosmological constant

$g_{\mu\nu}$ = Metric tensor

G = Newton's gravitational constant

c = Speed of light

$T_{\mu\nu}$ = Energy-momentum tensor

The present research considers three separate physical components:

1. PPC.
2. Ordinary matter.
3. Radiation.

The cosmological constant is included as a separate contribution.

The objective is to derive the two Friedmann equations and investigate their behaviour for three different PPC equation-of-state cases.

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2. DEFINITIONS OF THE FOUR COSMOLOGICAL CONTRIBUTIONS

Before deriving the Friedmann equations, the energy densities must be defined separately.

2.1 PPC energy density

The PPC energy density is:

$E_d = \rho c^2$

Here:

E_d = PPC energy density. (Total Energy Density associated with Total mass equivalent density)

ρ = Mass-equivalent density associated with PPC.

c = Speed of light.

The PPC pressure is:

$P_g = \omega E_d$

Therefore, the PPC equation-of-state parameter is:

$\omega = P_g/E_d$

2.2 Ordinary matter energy density

Ordinary matter energy density is:

$E_m = \rho_m c^2$

Here:

ρ_m = Ordinary matter mass density.

For nonrelativistic pressureless matter:

$P_m \approx 0$

2.3 Radiation energy density

Radiation is represented by its energy density:

E_r = Radiation energy density.

Its pressure is:

$P_r = E_r/3$

Radiation does not have photon rest mass or photon rest-mass density.

The radiation component is described by energy density and pressure, not by a photon rest-mass density.

2.4 Cosmological constant energy density

The cosmological constant can be represented as an equivalent vacuum energy density:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Its pressure is:

$P_\Lambda = -E_\Lambda$

This is the vacuum equation of state.

The four contributions must remain separately identified.

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3. FIRST FRIEDMANN EQUATION WITH COSMOLOGICAL CONSTANT

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)E_{d(total)} - kc^2/a^2 + \Lambda c^2/3$

Where:

$H = \dot{a}/a$

a = Scale factor.

k = Spatial curvature parameter.

$E_{d(total)}$ = Total energy density of the physical matter and energy components.

For the present framework:

$E_{d(total)} = E_d + E_m + E_r$

Substituting:

$H^2 = (8\pi G/3c^2)(E_d + E_m + E_r) - kc^2/a^2 + \Lambda c^2/3$

This is the first Friedmann equation with PPC, ordinary matter, radiation and an explicit cosmological constant.

3.1 Introducing cosmological constant energy density

The cosmological constant energy density is defined as:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Rearranging:

$\Lambda c^2/3 = (8\pi G/3c^2)E_\Lambda$

Substituting into the first Friedmann equation:

$H^2 = (8\pi G/3c^2)(E_d + E_m + E_r) - kc^2/a^2 + (8\pi G/3c^2)E_\Lambda$

Therefore, the final first Friedmann equation becomes:

$H^2 = (8\pi G/3c^2)(E_d + E_m + E_r + E_\Lambda) - kc^2/a^2$

FINAL FIRST FRIEDMANN EQUATION

$H^2 = (8\pi G/3c^2)(E_d + E_m + E_r + E_\Lambda) - kc^2/a^2$

---

4. SECOND FRIEDMANN EQUATION WITH COSMOLOGICAL CONSTANT

The general acceleration equation in General Relativity is:

$\ddot{a}/a = -(4\pi G/3c^2)\Sigma(E_i + 3P_i) + \Lambda c^2/3$

For the three physical components:

$\ddot{a}/a = -(4\pi G/3c^2)[(E_d + 3P_g) + (E_m + 3P_m) + (E_r + 3P_r)] + \Lambda c^2/3$

Now introduce their respective pressure relations.

For PPC:

$P_g = \omega E_d$

For ordinary matter:

$P_m \approx 0$

For radiation:

$P_r = E_r/3$

Substituting these relations:

$\ddot{a}/a = -(4\pi G/3c^2)[E_d + 3\omega E_d + E_m + E_r + 3(E_r/3)] + \Lambda c^2/3$

Simplifying:

$\ddot{a}/a = -(4\pi G/3c^2)[E_d(1 + 3\omega) + E_m + 2E_r] + \Lambda c^2/3$

4.1 Introducing cosmological constant energy density

We have:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Therefore:

$\Lambda c^2/3 = (8\pi G/3c^2)E_\Lambda$

Rewriting this contribution with the common acceleration-equation coefficient:

$(8\pi G/3c^2)E_\Lambda = (4\pi G/3c^2)(2E_\Lambda)$

Thus:

$\ddot{a}/a = -(4\pi G/3c^2)[E_d(1 + 3\omega) + E_m + 2E_r - 2E_\Lambda]$

FINAL SECOND FRIEDMANN EQUATION

$\ddot{a}/a = -(4\pi G/3c^2)[E_d(1 + 3\omega) + E_m + 2E_r - 2E_\Lambda]$

Equivalently:

$\ddot{a}/a = (4\pi G/3c^2)[-E_d(1 + 3\omega) - E_m - 2E_r + 2E_\Lambda]$

---

5. WHY RADIATION HAS THE FACTOR 2E_r

The radiation pressure equation is:

$P_r = E_r/3$

In the acceleration equation, each component contributes through:

$E + 3P$

For radiation:

$E_r + 3P_r$

Substituting:

$E_r + 3(E_r/3)$

Therefore:

$E_r + 3P_r = 2E_r$

Consequently, the radiation contribution to the acceleration equation is:

$-2E_r$

The factor 2 arises from the combination of radiation energy density and radiation pressure.

It does not arise from photon rest mass.

---

6. WHY THE COSMOLOGICAL CONSTANT HAS THE FACTOR 2E_Λ

The equivalent cosmological constant energy density is:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Its pressure is:

$P_\Lambda = -E_\Lambda$

Therefore:

$E_\Lambda + 3P_\Lambda$

$= E_\Lambda - 3E_\Lambda$

$= -2E_\Lambda$

Its acceleration contribution is therefore positive:

$+2E_\Lambda$

The positive sign arises because the cosmological constant has negative pressure.

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7. INTRODUCING THE PPC PARAMETER ε

For the PPC framework, define:

$\omega = -1 + \varepsilon$

Therefore:

$1 + 3\omega = 1 + 3(-1 + \varepsilon)$

Expanding:

$1 + 3\omega = 1 - 3 + 3\varepsilon$

Therefore:

$1 + 3\omega = -2 + 3\varepsilon$

Substituting this result into the second Friedmann equation:

$\ddot{a}/a = -(4\pi G/3c^2)[E_d(-2 + 3\varepsilon) + E_m + 2E_r - 2E_\Lambda]$

Distributing the negative sign:

FINAL PPC ACCELERATION EQUATION

$\ddot{a}/a = (4\pi G/3c^2)[E_d(2 - 3\varepsilon) - E_m - 2E_r + 2E_\Lambda]$

This is the PPC acceleration equation for the three physical components with an explicit cosmological constant.

---

8. THE THREE PPC EQUATION-OF-STATE CASES

Three different mathematical cases must be examined.

CASE A: ω APPROXIMATELY EQUAL TO -1

In this case:

$\omega \approx -1$

Define:

$\omega = -1 + \varepsilon$

Where ε is small but not necessarily zero.

The PPC energy density is:

$E_d(a) = E_{d,0}\, a^{(-3\varepsilon)}$

The first Friedmann equation becomes:

$H^2 = (8\pi G/3c^2)[E_{d,0}\, a^{(-3\varepsilon)} + E_m + E_r + E_\Lambda] - kc^2/a^2$

The second Friedmann equation becomes:

ä/a = (4πG/3c²)[E_d(2 - 3ε) - E_m - 2E_r + 2E_Λ]

Interpretation:

PPC energy density changes slowly with the scale factor when ε is small.

For positive ε, the PPC energy density decreases as the universe expands.

For negative ε, it increases as the universe expands.

This case permits a departure from exact vacuum-like behaviour.

CASE B: ω APPROACHES -1

In this case:

ω → -1

Equivalently:

ε → 0

The PPC density is:

E_d(a) = E_d,0 a^(-3ε)

Taking the limit:

lim(ε→0) E_d(a) = E_d,0

Therefore, PPC approaches a constant energy density.

The first Friedmann equation in the limit becomes:

H² = (8πG/3c²)(E_d,0 + E_m + E_r + E_Λ) - kc²/a²

The second Friedmann equation becomes:

ä/a = (4πG/3c²)[2E_d,0 - E_m - 2E_r + 2E_Λ]

Interpretation:

The limiting PPC behaviour approaches the vacuum-like case.

CASE C: ω EXACTLY EQUAL TO -1

In this case:

$\omega = -1$

Therefore:

$\varepsilon = 0$

The PPC pressure is:

$P_g = -E_d$

The PPC continuity equation is:

$dE_d/dt + 3H(E_d + P_g) = 0$

Substituting:

$dE_d/dt + 3H(E_d - E_d) = 0$

Therefore:

$dE_d/dt = 0$

Consequently:

$E_d(a) = E_{d,0}$

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)(E_{d,0} + E_m + E_r + E_\Lambda) - kc^2/a^2$

The second Friedmann equation is:

$\ddot{a}/a = (4\pi G/3c^2)[2E_{d,0} - E_m - 2E_r + 2E_\Lambda]$

Interpretation:

PPC behaves as a constant vacuum-like component in the homogeneous background expansion.

---

9. COMPARISON OF THE THREE CASES

PPC case	Equation of state	PPC density behaviour

Near vacuum-like	ω ≈ -1	Slowly evolving
Limiting vacuum-like	ω → -1	Approaches a constant
Exact vacuum-like	ω = -1	Exactly constant

The acceleration equation for all three cases is:

ä/a = (4πG/3c²)[E_d(2 - 3ε) - E_m - 2E_r + 2E_Λ]

For the limiting and exact cases, ε approaches or equals zero.

Their limiting acceleration equations are therefore identical.

The distinction between the limiting case and exact case concerns whether ω reaches -1 or only approaches it.

---

10. ENERGY-DENSITY EVOLUTION EQUATIONS

For a separately conserved perfect-fluid component in an FLRW universe:

dE_d/dt + 3H(E_d + P_g) = 0

10.1 PPC density evolution

For constant ω:

E_d(a) = E_d,0 a^[-3(1 + ω)]

Since ω = -1 + ε:

E_d(a) = E_d,0 a^(-3ε)

10.2 Ordinary matter density evolution

For pressureless ordinary matter:

P_m = 0

Therefore:

$dE_m/dt + 3HE_m = 0$

Integrating:

$E_m(a) = E_{m,0}\, a^{-3}$

10.3 Radiation energy-density evolution

For radiation:

$P_r = E_r/3$

Therefore:

$dE_r/dt + 3H(E_r + E_r/3) = 0$

Simplifying:

$dE_r/dt + 4HE_r = 0$

Integrating:

$E_r(a) = E_{r,0}\, a^{-4}$

Radiation energy density decreases as a^{-4} because the expansion reduces the number density of freely propagating photons and cosmological redshift reduces the energy of each photon.

10.4 Cosmological constant energy density

For the cosmological constant:

$P_\Lambda = -E_\Lambda$

Its energy density remains constant:

$E_\Lambda(a) = E_{\Lambda,0}$

---

11. SPATIALLY FLAT PPC COSMOLOGICAL MODEL

For a spatially flat universe:

$k = 0$

The first Friedmann equation becomes:

$H^2 = (8\pi G/3c^2)(E_d + E_m + E_r + E_\Lambda)$

For constant ε, substituting the density evolution equations gives:

$H^2(a) = (8\pi G/3c^2)[E_{d,0}\, a^{(-3\varepsilon)} + E_{m,0}\, a^{-3} + E_{r,0}\, a^{-4} + E_{\Lambda,0}]$

At the present epoch:

$a = 1$

$H = H_0$

Therefore:

$H_0^2 = (8\pi G/3c^2)(E_{d,0} + E_{m,0} + E_{r,0} + E_{\Lambda,0})$

Define the density parameters:

$\Omega_i = 8\pi G\, E_{i,0}/(3c^2 H_0^2)$

For a spatially flat universe:

$\Omega_{d,0} + \Omega_{m,0} + \Omega_{r,0} + \Omega_{\Lambda,0} = 1$

Consequently:

$H^2(a)/H_0^2 = \Omega_{d,0}\, a^{(-3\varepsilon)} + \Omega_{m,0}\, a^{-3} + \Omega_{r,0}\, a^{-4} + \Omega_{\Lambda,0}$

This is the background expansion equation for the spatially flat, constant-ε PPC framework.

---

12. COMPARISON WITH MODERN COSMOLOGICAL OBSERVATIONS

Modern cosmological observations include:

1. Planck cosmic microwave background observations.
2. DESI baryon acoustic oscillation observations.
3. Type Ia supernova distance measurements.
4. Cosmic chronometer measurements.
5. Large-scale structure and gravitational lensing observations.

These datasets constrain the expansion history and the properties of cosmic matter and dark energy.

12.1 Planck 2018 observational benchmark

The Planck 2018 base-ΛCDM analysis reported approximately:

$H_0 = 67.36 \pm 0.54$ km/s/Mpc

$\Omega_m = 0.315 \pm 0.007$

$\Omega_\Lambda = 0.6847 \pm 0.0073$

These values are reference constraints within the standard spatially flat ΛCDM model. They are not PPC-specific fitted values.

12.2 DESI DR2 observations

DESI DR2 released BAO measurements based on its first three years of observations.

The analysis uses measurements from more than 14 million galaxies and quasars.

Its results provide more precise information about the expansion history of the universe.

The DESI collaboration reported that combinations of DESI BAO with CMB and supernova observations show interesting preferences for time-evolving dark energy in some parameterizations.

However, the significance depends on the datasets and model assumptions.

12.3 Recent combined observational analysis

A 2026 analysis combining DESI DR2, DES Year 6 BAO, Planck CMB, supernovae and cosmic chronometers reported the following results for its ΛCDM model:

$H_0 = 68.41 \pm 0.30$ km/s/Mpc

$\Omega_m = 0.301 \pm 0.004$

$r_{drag} = 147.55 \pm 0.20$ Mpc

For its constant-w model, it reported:

$w = -0.992 \pm 0.028$

Converting to the PPC notation:

$\omega = -1 + \varepsilon$

$\varepsilon = 1 + \omega$

Therefore:

$\varepsilon = 0.008 \pm 0.028$

This result is compatible with $\varepsilon = 0$ within the reported uncertainty.

However, this is a constraint from a conventional constant-w analysis, not a direct observational fit of PPC with an independently fitted cosmological constant.

---

13. OBSERVATIONAL ASSESSMENT OF THE THREE PPC CASES

CASE A: ω APPROXIMATELY EQUAL TO -1

Modern constant-w observational analyses permit values close to -1.

A nonzero ε allows PPC energy density to evolve with the scale factor.

The observational viability of this case depends on whether the predicted H(z), luminosity distances, BAO distances and structure growth agree with measurements.

Assessment:

Near-vacuum-like PPC is worth testing and is not excluded merely because ω differs slightly from -1.

CASE B: ω APPROACHES -1

The limit $\varepsilon \rightarrow 0$ produces constant PPC energy density.

The background expansion approaches vacuum-like behaviour.

Assessment:

This limiting case approaches the standard cosmological-constant expansion behaviour.

CASE C: ω EXACTLY EQUAL TO -1

For exact $\omega = -1$:

$P_g = -E_d$

$E_d(a) = E_{d,0}$

PPC has constant energy density and negative pressure.

Assessment:

This is compatible with the vacuum-like equation of state used in ΛCDM.

However, the PPC component and the independently included cosmological constant then have identical background expansion behaviour.

---

14. IMPORTANT PARAMETER DEGENERACY BETWEEN PPC AND THE COSMOLOGICAL CONSTANT

This is one of the most important mathematical results of the present research.

When:

$\omega = -1$

PPC energy density is constant.

The cosmological constant energy density is also constant.

The first Friedmann equation contains:

$E_{d,0} + E_{\Lambda,0}$

The acceleration equation contains:

$2E_{d,0} + 2E_{\Lambda,0}$

Therefore, background expansion observations constrain their sum:

$E_{d,0} + E_{\Lambda,0}$

They cannot independently determine $E_{d,0}$ and $E_{\Lambda,0}$ when both behave as constant vacuum-energy components.

This is an exact degeneracy at the homogeneous background level.

To distinguish the two components, additional physical predictions are needed, such as different perturbation behaviour, interactions, sound speeds or other independently testable signatures.

When ε is nonzero, PPC evolves with the scale factor and can potentially be distinguished from a constant cosmological constant.

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15. REQUIREMENTS FOR A GENUINE PPC OBSERVATIONAL FIT

A direct PPC-specific numerical fit has not been performed in this paper.

A genuine fit requires:

1. DESI DR2 BAO measurements and their covariance matrix.
2. Planck CMB likelihood or an appropriately justified compressed likelihood.

3. A specified supernova dataset and its covariance matrix.
4. Cosmic chronometer H(z) measurements, if included.
5. A clearly specified parameterization of ε.
6. Priors on Ω_d, Ω_m, Ω_r and Ω_Λ.
7. A consistent treatment of early-universe physics and the sound horizon.
8. A perturbation model for PPC to calculate CMB anisotropies and structure growth.
9. Numerical likelihood optimization or Bayesian parameter estimation.

The analysis should calculate:

Best-fit parameter values.

Confidence intervals or posterior distributions.

Minimum chi-square.

AIC and BIC.

Bayesian evidence where feasible.

The PPC model should be compared with ΛCDM using the same observational datasets and consistent nuisance parameters.

For the exact ω = -1 case, the independent PPC and Λ contributions must be handled carefully because of their background degeneracy.

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16. DISCUSSION

The mathematical derivations demonstrate that radiation and the cosmological constant have different pressure contributions in the acceleration equation.

For radiation:

$P_r = E_r/3$

Therefore:

$E_r + 3P_r = 2E_r$

For the cosmological constant:

$P_\Lambda = -E_\Lambda$

Therefore:

$E_\Lambda + 3P_\Lambda = -2E_\Lambda$

These results explain the respective terms -2E_r and +2E_Λ.

The PPC component has its own pressure relation:

$P_g = \omega E_d$

The ordinary matter component has approximately zero pressure.

These separate definitions are necessary because the physical components have different equations of state.

Their individual energy-momentum tensors are added in Einstein's field equation to obtain the total gravitational source.

The observational analysis also shows that a near-vacuum equation of state is compatible with published constant-w constraints. Nevertheless, compatibility with a conventional dark-energy parameter range does not prove the physical existence of PPC.

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17. CONCLUSION

This research derives both Friedmann equations with the cosmological constant and introduces PPC, ordinary matter and radiation as separate physical components.

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)(E_d + E_m + E_r + E_\Lambda) - kc^2/a^2$

The second Friedmann equation is:

$\ddot{a}/a = (4\pi G/3c^2)[E_d(2 - 3\varepsilon) - E_m - 2E_r + 2E_\Lambda]$

Where:

$E_d = \rho c^2$

$E_m = \rho_m c^2$

$P_r = E_r/3$

$E_\Lambda = \Lambda c^4/(8\pi G)$

$\omega = -1 + \varepsilon$

Three PPC equation-of-state cases were examined:

1. ω approximately equal to -1.

2. ω approaches -1.
3. ω exactly equal to -1.

The near-vacuum-like case permits evolving PPC energy density.

The limiting case approaches constant PPC energy density.

The exact case produces constant PPC energy density.

Modern cosmological observations are consistent with an equation of state close to -1 in conventional constant-w analyses. Some recent DESI combinations also motivate investigating evolving dark energy.

The exact vacuum-like PPC case is mathematically degenerate with the cosmological constant at the background-expansion level when both are included independently.

Therefore, a complete observational evaluation of PPC requires a dedicated numerical likelihood analysis and a consistent perturbation theory.

The equations developed here provide the mathematical foundation for that further investigation.

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PPC LAW OF GRAVITY DOES NOT PREDICT INFINITY

Separate Theoretical Research Notes

1. Fundamental PPC Principle

The proposed PPC Law of Gravity does not predict infinite physical quantities or gravitational singularities. It proposes that gravitational quantities remain finite, subject to the mathematical conditions of the PPC framework.

2. Scale Factor Approaches Zero but Never Becomes Zero

According to the proposed PPC principle, the cosmic scale factor approaches zero but never reaches exactly zero.

Mathematically:

$a(t) > 0$

$\lim a(t) = 0$ as $t \rightarrow t_{min}$

Therefore:

$a \rightarrow 0$, but $a \neq 0$.

The notation $a \sim 0$ indicates that the scale factor is extremely close to zero, not exactly zero.

3. Maximum Finite PPC Energy Density

PPC energy density may increase as the scale factor approaches zero. However, it can approach a maximum finite value, E_{max}, but will never become infinite.

Mathematically:

$0 < E_d(t) \leq E_{max} < \infty$

At the limiting stage:

$E_d(t) \rightarrow E_{max}$

Here:

- $E_d(t)$ represents PPC energy density at time t.
- E_{max} represents the maximum permitted PPC energy density.
- E_{max} is finite and positive.
- PPC energy density does not become infinite.

4. Asymptotically Vacuum-Like PPC Behaviour

The PPC equation of state is:

$P_g(t) = \omega(t) E_d(t)$

As cosmic time approaches t_{max}, the proposed limiting condition is:

$\omega(t) \rightarrow -1$ as $t \rightarrow t_{max}$

If PPC energy density approaches a finite positive limiting value:

$E_d(t) \rightarrow E_{max} > 0$

then:

$P_g(t) \rightarrow -E_{max}$

Consequently:

$P_g(t) / E_d(t) \rightarrow -1$

This represents an asymptotically vacuum-like equation of state for the PPC component.

5. Divergence in Conventional Cosmological Scaling Laws

Under conventional cosmological scaling laws:

$E_m(a) = E_{m,0}\, a^{-3}$

$E_r(a) = E_{r,0}\, a^{-4}$

As $a \rightarrow 0^+$, both expressions diverge mathematically.

However, the fact that a never equals zero does not automatically prevent these quantities from becoming arbitrarily large.

The PPC framework must therefore establish additional dynamics or modified evolution laws that keep ordinary matter density and radiation energy density bounded, if it claims that these quantities also never become infinite.

6. Finite Gravitational Quantities

To establish the claim that the PPC Law of Gravity does not predict gravitational singularities, the theory must demonstrate that the following quantities remain bounded:

- PPC energy density.
- Ordinary matter energy density.
- Radiation energy density.
- Hubble parameter, H.
- Cosmic acceleration, ä/a.
- Ricci scalar, R.
- Kretschmann curvature invariant, Rμνρσ Rμνρσ.

7. Main Mathematical Conditions

The proposed PPC conditions can be summarized as follows:

$a(t) > 0$

$a(t) \rightarrow 0$, but $a(t) \neq 0$

$0 < E_d(t) \leq E_{max} < \infty$

$E_d(t) \rightarrow E_{max}$

$\omega(t) \rightarrow -1$

$P_g(t) \rightarrow -E_{max}$

These conditions express the proposed asymptotic behaviour of the scale factor and PPC energy density.

8. Conclusion

The PPC Law of Gravity proposes that the cosmic scale factor approaches zero without becoming exactly zero. PPC energy density may approach a maximum finite value, E_max, rather than becoming infinite. As the PPC equation-of-state parameter approaches -1, PPC pressure approaches the negative of its energy density, producing an asymptotically vacuum-like component.

The central principle is that the PPC Law of Gravity aims to describe gravitational evolution without physical infinities or gravitational singularities.

Scientific qualification: These are proposed theoretical conditions. Their validity requires derivation from the complete PPC field equations and demonstration that all relevant energy densities and curvature invariants remain bounded throughout the physical evolution.

Mathematical Approximation Principle in the PPC Law of Gravity

1. Fundamental Principle

In the proposed PPC Law of Gravity, mathematical values may represent approximate, estimated, or limiting physical values. They are not necessarily intended to represent exact numerical values in every physical situation.

The values may be approximate, roughly estimated, or very close to the stated mathematical values.

2. Meaning of Approximate Values

For example:

$\omega \approx -1$

means that the PPC equation-of-state parameter is approximately -1 or very close to -1.

Similarly:

$a \approx 0$

means that the cosmic scale factor approaches or remains extremely close to zero, without necessarily becoming exactly zero.

And:

$E_d \approx E_{max}$

means that PPC energy density approaches or remains very close to its maximum permitted finite value.

3. Approximation and Physical Interpretation

The symbols should be interpreted carefully:

- = means mathematical equality.
- ≈ means approximately equal.
- → means approaches a limiting value.
- < or > means a strict inequality.
- ~ may indicate approximate closeness when explicitly defined.

4. Application to PPC Conditions

The proposed PPC conditions may be expressed as:

$a(t) > 0$

$a(t) \approx 0$

$E_d(t) \approx E_{max}$

$\omega(t) \approx -1$

$P_g(t) \approx -E_d(t)$

These expressions describe approximate physical behaviour, subject to the precise mathematical definitions and conditions of the theory.

5. Scientific Note

Approximation does not mean that mathematical consistency can be disregarded. Exact equalities, inequalities, limits, and approximate numerical values must be distinguished.

Whenever the PPC framework uses approximate values, it should specify the permitted approximation, tolerance, or limiting behaviour.

Conclusion

The proposed PPC Law of Gravity uses mathematical approximations to describe physical behaviour. Its numerical values may be rough estimates or very close to theoretical limiting values rather than exact physical measurements. However, the governing equations and their mathematical relationships must remain internally consistent.

SEPARATE RESEARCH NOTES: RADIATION ENERGY DENSITY (E_r) IN THE PPC LAW OF GRAVITY

1. Definition of Radiation Energy Density

In the PPC Law of Gravity, radiation energy density is represented by E_r.

E_r represents the energy of radiation per unit volume.

Radiation is treated as a separate physical component from ordinary matter and PPC-associated energy density.

2. Energy of One Photon

According to Planck's equation:

$E_\gamma = h\nu$

Where:

- E_γ = energy of one photon.
- h = Planck's constant.
- ν = frequency of one photon.

The frequency ν refers to the frequency of an individual photon.

Photons have zero invariant rest mass, but they carry energy and momentum.

3. Energy Density Contribution of One Photon

If one photon occupies a volume V, its energy density contribution is:

$E_r = E_\gamma/V$

Substituting Planck's equation:

$E_r = h\nu/V$

Define frequency density as:

$v_d = v/V$

Therefore:

$E_r = hv_d$

Here, v_d represents the frequency of one photon divided by the volume.

4. Total Radiation Energy Density

For radiation distributed over a continuous frequency spectrum, define u_v as the spectral energy density per unit frequency interval.

The total radiation energy density is:

$E_r = \int_0^{vmax} u_v \, dv$

Where:

- E_r = total radiation energy density.
- u_v = spectral energy density per unit frequency.
- dv = infinitesimal frequency interval.
- v_{max} = maximum frequency considered in the model.

This integral adds the energy contributions across the radiation frequency spectrum.

5. Radiation Pressure

For isotropic radiation, the pressure-energy-density relationship is:

$P_r = E_r/3$

Therefore:

$3P_r = E_r$

Radiation consequently contributes the following term to the cosmological acceleration equation:

$E_r + 3P_r = 2E_r$

6. Separate Physical Identity of Radiation

Radiation and ordinary matter must be distinguished:

Ordinary matter:

$E_m = \rho_m c^2$

$P_m \approx 0$

Radiation:

E_r = radiation energy density

$P_r = E_r/3$

Photons have zero invariant rest mass. Therefore, radiation energy density is not defined through photon rest-mass density.

Radiation nevertheless carries energy and momentum and contributes to gravitational dynamics.

7. Application to the PPC Acceleration Equation

When PPC, ordinary matter, and isotropic radiation are treated as separate, non-overlapping components, the acceleration equation is:

$\ddot{a}/a = -(4\pi G/3c^2)[E_m + 2E_r + E_d(1 + 3\omega)]$

Here:

- E_m = ordinary matter energy density.
- E_r = radiation energy density.
- E_d = PPC-associated energy density.
- ω = PPC equation-of-state parameter.

8. Scientific Note

The expression $E_r = hv_d$ is valid for the energy density contribution of one photon when $v_d = v/V$.

For multiple photons, the individual photon energies must be summed before dividing by the total volume.

For a continuous radiation spectrum, the spectral integral $E_r = \int_0^{v_{max}} u_v \, dv$ is the general energy-density expression.

The single-photon frequency-density definition $v_d = v/V$ should not be confused with the spectral energy density u_v. They describe different quantities.

Conclusion

In the PPC Law of Gravity, radiation energy density E_r is a separate gravitational energy contribution. It originates from the energy carried by radiation rather than photon rest mass.

The energy of an individual photon is E_γ = hv, while the total radiation energy density is obtained by summing radiation energy over the relevant volume or integrating its spectral energy distribution.

SEPARATE RESEARCH NOTES: SPECTRAL ENERGY DISTRIBUTION OF RADIATION

1. Definition

Spectral Energy Distribution (SED) describes how the energy of radiation is distributed across different frequencies or wavelengths.

It explains how much radiation energy is contained within a particular frequency interval.

In the PPC Law of Gravity, spectral energy distribution provides a mathematical description of radiation energy density without assigning rest mass to photons.

2. Mathematical Definition

Let u_v represent the spectral energy density per unit frequency.

The total radiation energy density is:

$E_r = \int_0^{vmax} u_v \, dv$

($E_r = \int_0^{vmax} u_v \, dv$)

Where:

- E_r = total radiation energy density.
- u_v = spectral energy density per unit frequency.
- v = radiation frequency.

- dv = infinitesimal frequency interval.
- v_max = maximum frequency considered in the model.

3. Meaning of Spectral Energy Density

Spectral energy density describes the radiation energy contained in a unit volume within a unit frequency interval.

Mathematically:

$u_v = dE_r/dv$

For a small frequency interval:

$dE_r = u_v\, dv$

Therefore, integrating over the frequency range gives the total radiation energy density:

$E_r = \int_0^{vmax} u_v\, dv$

($E_r = \int_0^{vmax} u_v\, dv$)

4. SI Units and Dimensional Analysis

The SI unit of radiation energy density is:

$[E_r] = J/m^3$

The SI unit of frequency is:

$[v] = Hz = s^{-1}$

Therefore, the unit of spectral energy density is:

$[u_v] = (J/m^3)/Hz$

$[u_v] = J \cdot s/m^3$

Verification:

$[J \cdot s/m^3] \times [s^{-1}] = J/m^3$

Thus, the spectral integral is dimensionally consistent.

5. Relationship with Planck's Photon-Energy Equation

The energy of an individual photon is:

$E_\gamma = h\nu$

Where:

- E_γ = energy of one photon.
- h = Planck's constant.
- ν = frequency of one photon.

If one photon occupies a volume V, its energy-density contribution is:

$E_r = E_\gamma/V$

$E_r = h\nu/V$

Defining:

$\nu_d = \nu/V$

We obtain:

$E_r = h\nu_d$

This expression describes the energy-density contribution of one photon under the stated frequency-density definition.

For multiple photons, their individual energies must be added before dividing by the total volume.

6. Difference Between Photon Energy and Spectral Energy Density

Photon energy:

$E_\gamma = h\nu$

This represents the energy carried by one photon.

Spectral energy density:

$u_\nu = dE_r/d\nu$

This represents radiation energy density per unit frequency interval.

Total radiation energy density:

$E_r = \int_0^{v_{max}} u_v \, dv$

(duplicate)

This represents the total radiation energy per unit volume across the considered frequency spectrum.

These three quantities have different physical meanings.

7. Radiation Pressure

For isotropic radiation:

$P_r = E_r/3$

Therefore:

$E_r + 3P_r = 2E_r$

Radiation pressure contributes independently to the gravitational dynamics of the universe.

8. Application to the PPC Law of Gravity

In the proposed PPC framework, radiation energy density E_r is treated as a separate physical component from ordinary matter energy density E_m and PPC-associated energy density E_d.

Ordinary matter:

$E_m = \rho_m c^2$

$P_m \approx 0$

Radiation:

$E_r = \int_0^{v_{max}} u_v \, dv$

(duplicate)

$P_r = E_r/3$

PPC component:

$E_d = \rho c^2$

$P_g = \omega E_d$

The components must be defined as non-overlapping contributions if they are added separately in the cosmological equations.

9. Scientific Note

Spectral energy density is not photon rest-mass density and does not require assigning rest mass to photons.

Photons have zero invariant rest mass but carry energy and momentum.

The spectral integral is the general description of total radiation energy density for a continuous frequency distribution.

The single-photon expression $E_r = hv_d$ and the spectral integral describe different levels of the radiation model. They should be connected through a clearly defined frequency distribution to avoid double-counting or dimensional ambiguity.

Conclusion

Spectral Energy Distribution provides a mathematical description of how radiation energy is distributed across frequencies.

The total radiation energy density is obtained by integrating the spectral energy density over the frequency range:

$E_r = \int_0^{vmax} u_v \, dv$

($E_r = \int_0^{vmax} u_v \, dv$)

This formulation allows radiation to be included as a separate energy contribution in the PPC Law of Gravity without introducing photon rest-mass density.

Definition: Spectral Energy Distribution (SED)

A **Spectral Energy Distribution** describes how the energy of radiation is distributed across different frequencies or wavelengths.

In simple words, it tells us how much radiation energy is present in each frequency interval.

1. Mathematical definition

For radiation, let u_ν represent the spectral energy density per unit frequency.

The total radiation energy density is:

$$\boxed{E_r = \int_0^{\nu_{\max}} u_\nu \, d\nu}$$

The total radiation energy density is:

$$\boxed{E_r = \int_0^{\nu_{\max}} u_\nu \, d\nu}$$

Where:

- E_r = total radiation energy density.
- u_ν = spectral energy density per unit frequency.
- ν = radiation frequency.
- $d\nu$ = a very small frequency interval.
- $\nu_{\max}$ = maximum frequency considered.

2. Meaning of u_ν

Suppose radiation occupies a frequency interval from ν to $\nu + d\nu$

The energy density contained in that interval is approximately:

$$dE_r = u_\nu \, d\nu$$

Therefore:

$$\boxed{u_\nu = \frac{dE_r}{d\nu}}$$

This is the spectral energy density.

3. SI units

Energy density:

$$[E_r] = J/m^3$$

Frequency:

$$[\nu] = Hz = s^{-1}$$

Therefore:

$$[u_\nu] = \frac{J/m^3}{Hz}$$

$$\boxed{[u_\nu] = J\,s\,m^{-3}}$$

4. Difference between photon energy and spectral energy density

Quantity	Meaning
$E_\gamma = h\nu$	Energy of one photon
E_r	Total radiation energy per unit volume
u_ν	Radiation energy per unit volume per unit frequency

These quantities are related but physically distinct.

$$\boxed{E_r = \int_0^{\nu_{\max}} u_\nu \, d\nu}$$

It does not require assigning rest mass to photons.

For isotropic radiation:

$$\boxed{P_r = \frac{E_r}{3}}$$

Consequently, its contribution to the acceleration equation is:

$$\boxed{E_r + 3P_r = 2E_r}$$

Conclusion: Spectral energy distribution describes how radiation energy density is spread across frequencies. Integrating that distribution over the frequency range gives the total radiation energy density E_r.

CLARIFICATION NOTES ON THE DEFINITION OF ENERGY DENSITY (E_d) AND MASS DENSITY (ρ) IN THE PRESSURE–CURVATURE MODEL

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1. INTRODUCTION

The fundamental Pressure–Curvature (PPC) equation is:

$P_g = \omega E_d$

The energy density used in this equation is defined by:

$E_d = \rho c^2$

Where:

- E_d = Energy density represented by the physical system.
- ρ = Corresponding mass density or mass-equivalent density.
- c = Speed of light.
- P_g = Pressure associated with the PPC relation.
- ω = Dimensionless equation-of-state parameter.

The definition of E_d and ρ depends on the physical system and the type of density being represented.

The symbols E_d and ρ must therefore be interpreted according to the physical composition and scope of the system under consideration.

The purpose of these notes is to establish a consistent definition of energy density and mass density for different physical systems.

---

2. GENERAL RELATIONSHIP BETWEEN ENERGY DENSITY AND MASS DENSITY

The fundamental relationship is:

$E_d = \rho c^2$

This relationship expresses the energy density associated with a corresponding mass density.

The dimensions are:

Energy density:

$[E_d] = J/m^3$

Mass density:

$[\rho] = kg/m^3$

Speed of light:

$[c] = m/s$

Therefore:

$[\rho c^2] = (kg/m^3)(m^2/s^2)$

$[\rho c^2] = J/m^3$

Thus, the equation is dimensionally consistent.

However, the physical meaning of ρ depends on which mass density is being represented.

It may represent:

1. Ordinary matter mass density.
2. Total local mass-equivalent density.
3. Total non-vacuum mass-equivalent density.
4. Total mass-equivalent density including vacuum energy.
5. Mass-equivalent density corresponding to the combined energy density of matter, radiation, and vacuum energy.

Each definition represents a different physical system or energy budget.

---

3. CASE I: ρ AS THE TOTAL MASS DENSITY OF AN INDIVIDUAL PHYSICAL OBJECT

Suppose ρ represents the total mass density of an individual physical object, such as:

- Earth.
- Sun.
- Neutron star.
- Quark star.
- Other stars.
- Other planets.

In this situation:

ρ = Total mass density of the individual physical object.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total energy density associated with the mass density of that particular physical object.

For example:

A. Earth

If ρ represents the mass density of Earth, then:

$E_{d,Earth} = \rho_{Earth} c^2$

B. Sun

If ρ represents the mass density of the Sun, then:

$E_{d,Sun} = \rho_{Sun} c^2$

C. Neutron Star

If ρ represents the mass density of a neutron star, then:

$E_{d,NS} = \rho_{NS} c^2$

D. Quark Star

If ρ represents the mass density of a quark star, then:

$E_{d,QS} = \rho_{QS} c^2$

In each case, E_d represents the energy density associated with the corresponding object's mass density.

The physical system determines the value and interpretation of ρ and E_d.

Important clarification: In relativistic astrophysics, the appropriate density must be defined carefully. Rest-mass density, total local energy density, and gravitational mass density are not automatically identical. The relation $E_d = \rho c^2$ applies directly when ρ is defined as the corresponding mass-equivalent density.

---

4. CASE II: ρ AS THE TOTAL LOCAL MASS-EQUIVALENT DENSITY

Suppose ρ represents the total local mass-equivalent density of a physical system.

In that situation:

ρ = Total local mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total local energy density associated with that total local mass-equivalent density.

Here, ρ is not restricted to the ordinary rest-mass density of individual material objects.

It represents the mass equivalent of the total energy density included in the selected physical system.

The definition must specify which energy contributions are included.

---

5. CASE III: ρ AS THE TOTAL NON-VACUUM MASS-EQUIVALENT DENSITY

Suppose ρ represents the total non-vacuum mass-equivalent density.

In this situation:

ρ = Total non-vacuum mass-equivalent density.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total non-vacuum local energy density associated with that mass-equivalent density.

The physical system may include:

1. Ordinary matter energy.
2. Radiation energy.
3. Dark matter energy.
4. Other specified non-vacuum energy contributions.

The vacuum-energy contribution associated with the cosmological constant is excluded under this definition.

Thus:

E_d = Total non-vacuum local energy density.

And:

E_Λ = Vacuum-energy density associated with the cosmological constant.

These are separately defined contributions.

The combined energy density, when both are included in the same total energy budget, is:

$E_{d(total)} = E_d + E_\Lambda$

The corresponding combined mass-equivalent density is:

$\rho_{total} = E_{d(total)}/c^2$

---

6. CASE IV: ρ AS THE TOTAL MASS-EQUIVALENT DENSITY INCLUDING VACUUM ENERGY

Suppose ρ represents the total mass-equivalent density of a system that includes vacuum energy.

In that situation:

ρ = Total mass-equivalent density including vacuum energy.

The corresponding energy density is:

$E_d = \rho c^2$

Therefore:

E_d represents the total local energy density, including vacuum energy.

It may contain contributions from:

1. Ordinary matter.
2. Radiation.
3. Dark matter.
4. Vacuum energy.
5. Other specified energy contributions.

Under this definition, vacuum energy is included in E_d.

Consequently, if the cosmological constant is also retained explicitly in the geometric part of the Einstein field equations, its vacuum-energy contribution must not be counted a second time.

There are two consistent representations:

Representation A: Explicit cosmological constant

- Exclude E_Λ from E_d.
- Retain Λ explicitly in the Friedmann equations.

Representation B: Vacuum energy included in total fluid density

- Include E_Λ in E_d.
- Remove the corresponding explicit Λ term from the geometric side of the Friedmann equations.

These two representations must not double-count the same vacuum-energy contribution.

---

7. CASE V: E_d AS THE TOTAL LOCAL ENERGY DENSITY OF ORDINARY MATTER, RADIATION, AND VACUUM ENERGY

Suppose E_d represents the total local energy density containing:

1. Total ordinary matter energy density.
2. Total radiation energy density.

3. Total vacuum-energy density.
4. Other specified energy contributions.

Then:

$E_d = E_m + E_r + E_\Lambda + E_{other}$

Where:

E_m = Total ordinary matter energy density.

E_r = Total radiation energy density.

E_Λ = Total vacuum-energy density.

E_{other} = Other included energy contributions.

The corresponding mass-equivalent density is:

$\rho = E_d/c^2$

Therefore:

$E_d = \rho c^2$

In this case, ρ represents the total mass-equivalent density associated with the complete energy density E_d.

It is important to understand that photons have zero invariant rest mass but possess energy and momentum. Their radiation energy contributes to the total energy density and therefore to the corresponding mass-equivalent density.

Similarly, vacuum energy contributes to the mass-equivalent density when it is included in E_d.

Thus, ρ in this case is not simply the ordinary rest-mass density of matter. It is the mass-equivalent density corresponding to the complete energy budget.

---

8. RELATIONSHIP BETWEEN THE DIFFERENT DEFINITIONS

The following table summarizes the different interpretations.

Case I: Individual physical object

ρ = Mass density of Earth, Sun, neutron star, quark star, or another individual object.

E_d = Energy density associated with that object's mass density.

Case II: Total local mass-equivalent density

ρ = Total local mass-equivalent density.

E_d = Total local energy density associated with that density.

Case III: Total non-vacuum mass-equivalent density

ρ = Total non-vacuum mass-equivalent density.

E_d = Total non-vacuum local energy density.

Case IV: Total mass-equivalent density including vacuum energy

ρ = Total mass-equivalent density, including vacuum energy.

E_d = Total local energy density, including vacuum energy.

Case V: Combined ordinary matter, radiation, and vacuum energy

ρ = Mass-equivalent density associated with the combined energy budget.

E_d = Total local energy density of ordinary matter, radiation, vacuum energy, and other included contributions.

---

9. RELATIONSHIP WITH THE PRESSURE–CURVATURE EQUATION

The PPC equation is:

$P_g = \omega E_d$

Since:

$E_d = \rho c^2$

The PPC equation can also be expressed as:

$P_g = \omega \rho c^2$

This shows that the pressure associated with the PPC relation depends on:

1. The selected physical system.
2. The definition of its mass density ρ.
3. The corresponding energy density E_d.
4. The equation-of-state parameter ω.

Therefore, the PPC equation can be applied to different physical systems, provided the relevant density and pressure are defined consistently.

However, the physical interpretation of ω must be specified for each system.

---

10. VACUUM-ENERGY-LIKE BEHAVIOUR WHEN ω = −1

The PPC equation is:

$P_g = \omega E_d$

For:

$\omega = -1$

We obtain:

$P_g = -E_d$

Therefore:

The PPC pressure becomes equal to the negative of its associated energy density.

This produces vacuum-energy-like behaviour.

The standard cosmological-constant vacuum-energy relationship is:

$P_\Lambda = -E_\Lambda$

Both have the same mathematical equation-of-state relationship.

However, the physical interpretation depends on the definition of E_d.

If E_d excludes vacuum energy, the PPC fluid produces a separate vacuum-like contribution.

If E_d includes vacuum energy, the PPC fluid's total energy density already contains that contribution.

Therefore, the definition of E_d must always be specified before interpreting the physical meaning of $\omega = -1$.

---

11. APPLICATION TO THE FRIEDMANN EQUATIONS

For a homogeneous and isotropic universe, when E_d represents the selected total fluid energy density, the first Friedmann equation with an explicit cosmological constant is:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

The second Friedmann equation is:

$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$

Using:

$P_g = \omega E_d$

We obtain:

$\ddot{a}/a = -(4\pi G/3c^2)E_d(1 + 3\omega) + \Lambda c^2/3$

Important condition:

These equations with explicit Λ require E_d to exclude the vacuum-energy contribution already represented by Λ.

If E_d includes that vacuum energy, the equivalent Λ contribution must instead be incorporated into the fluid and removed from the explicit geometric term.

The density convention must remain consistent throughout the derivation.

---

12. IMPORTANT SCIENTIFIC CLARIFICATIONS

The following principles establish the intended meaning of the PPC variables:

1. E_d is defined through $E_d = \rho c^2$.

2. ρ does not have only one universal physical interpretation. Its definition depends on the physical system.

3. If ρ represents the mass density of an individual object, E_d represents the corresponding energy density of that object.

4. If ρ represents total local mass-equivalent density, E_d represents the corresponding total local energy density.

5. If ρ represents total non-vacuum mass-equivalent density, E_d represents total non-vacuum local energy density.

6. If ρ includes vacuum-energy mass equivalence, E_d also includes the corresponding vacuum-energy density.

7. If E_d includes ordinary matter, radiation, and vacuum energy, ρ represents the mass-equivalent density associated with all those contributions.

8. Vacuum energy must not be counted twice when the cosmological constant is retained explicitly.

9. The physical interpretation of ω depends on the physical system and the energy-density definition.

10. The equation $P_g = \omega E_d$ establishes an equation of state; a complete gravitational theory requires additional field equations and conservation laws.

---

13. FINAL CONCLUSION

The fundamental PPC relationship is:

$E_d = \rho c^2$

And:

$P_g = \omega E_d$

The definition of E_d and ρ depends on the physical system they represent.

The mass density may represent an individual physical object, total local mass-equivalent density, total non-vacuum mass-equivalent density, or total mass-equivalent density including vacuum energy.

Correspondingly, E_d may represent the energy density of an individual object, total local energy density, total non-vacuum energy density, or total local energy density including vacuum energy.

Therefore, the PPC variables must always be interpreted according to the selected physical system and its specified energy contributions.

FINAL DEFINING PRINCIPLE

The physical system determines what ρ represents, and the corresponding energy-density definition determines what E_d represents.

The general relationship remains:

$E_d = \rho c^2$

The Pressure–Curvature equation remains:

$P_g = \omega E_d$

The definition of E_d is therefore system-dependent, while the mathematical relationship between energy density and its corresponding mass-equivalent density remains consistent.

5th Research Paper

FRIEDMANN EQUATIONS WITH THE COSMOLOGICAL CONSTANT AND THE PRESSURE–CURVATURE RELATION

A Mathematical Derivation and Comparison with Current Cosmological Observations

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ABSTRACT

This research paper derives both Friedmann equations using the total local energy density, defined as:

$E_d = \rho c^2$

Here, E_d represents total local energy density without vacuum energy density, and ρ represents the corresponding total mass-equivalent density.

The Pressure–Curvature relation is expressed as:

$P_g = \omega E_d$

No separate matter-energy density E_m or radiation-energy density E_r is added because both are already included as contributions to the total energy density E_d.

An explicit cosmological constant, Λ, is retained in the Friedmann equations.

Three cases are investigated:

1. $\omega \approx -1$, expressed as $\omega = -1 + \varepsilon$.
2. The limiting case $\omega \to -1$.
3. The exact case $\omega = -1$.

The mathematical consequences of these three cases are derived and compared with published cosmological observations, including Planck and DESI results.

The comparison indicates that an equation of state close to -1 is consistent with important cosmological observations. However, some observational analyses also indicate the possibility of evolving dark energy.

An important distinction is maintained throughout this paper: observational constraints on the dark-energy equation of state cannot automatically be interpreted as measurements of the equation-of-state parameter of the entire cosmic energy content.

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

The cosmological constant term can be rewritten as an effective energy–momentum tensor:

$$T^{(\Lambda)}_{\mu\nu} = -\frac{\Lambda c^4}{8\pi G} g_{\mu\nu}$$

Comparing this with the perfect fluid energy–momentum tensor

$$T_{\mu\nu} = (E_d + P_g) u_\mu u_\nu + P_g g_{\mu\nu}$$

we identify the vacuum energy density

$$E_\Lambda = \frac{\Lambda c^4}{8\pi G}$$

and the corresponding pressure

$$P_\Lambda = -E_\Lambda$$

Thus

$$\omega_\Lambda = \frac{P_\Lambda}{E_\Lambda} = -1$$

---

1. INTRODUCTION

The Friedmann equations describe the expansion and acceleration of a homogeneous and isotropic universe within general relativity.

This paper examines these equations using the Pressure–Curvature relation:

$P_g = \omega E_d$

Where:

P_g = Effective pressure associated with the proposed relation.

E_d = Total local non-vacuum energy density represented by the PPC fluid.

ω = Dimensionless equation-of-state parameter.

The total local energy density is defined as:

$E_d = \rho c^2$

Where ρ represents the total mass-equivalent density.

The purpose of this research is to derive both Friedmann equations without adding separate matter and radiation density terms, and to investigate the three specified cases of ω.

---

2. DEFINITION OF TOTAL LOCAL ENERGY DENSITY

The fundamental definition is:

$E_d = \rho c^2$

Here, E_d represents the Total local non-vacuum energy density represented by the PPC fluid, while ρ represents the corresponding total mass-equivalent density.

The total energy density may contain contributions from:

1. Ordinary matter.
2. Radiation.
3. Dark matter.
4. Other relevant physical energy contributions.

These components are already represented within E_d. Therefore, separate E_m and E_r terms must not be added again to the Friedmann equations in this formulation.

The SI units are:

$[E_d] = J/m^3$

$[\rho] = kg/m^3$

Pressure has the same physical dimensions as energy density:

$[P_g] = Pa = J/m^3$

Therefore, ω is dimensionless.

---

3. COSMOLOGICAL FRAMEWORK

We assume a homogeneous and isotropic FLRW universe.

The relevant quantities are:

$a(t)$ = Cosmological scale factor.

H = Hubble parameter.

k = Spatial-curvature parameter.

Λ = Cosmological constant.

G = Gravitational constant.

c = Speed of light.

The Hubble parameter is defined as:

$H = \dot{a}/a$

The Einstein field equations with an explicit cosmological constant are:

$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$

For the homogeneous and isotropic effective fluid, the energy density and pressure are represented by E_d and P_g.

Important clarification concerning the cosmological constant:

To avoid double-counting vacuum energy, E_d in the equations below represents the total fluid energy density excluding the vacuum-energy contribution already represented by the explicit cosmological constant Λ.

If E_d is instead defined to include that same vacuum-energy contribution, the equivalent cosmological-constant term must be absorbed into the fluid and removed from the geometric side of the equations.

This distinction is necessary for mathematical consistency.

---

4. FIRST FRIEDMANN EQUATION WITH COSMOLOGICAL CONSTANT

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

Using:

$H = \dot{a}/a$

We obtain:

$(\dot{a}/a)^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

Therefore, the first Friedmann equation is:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

This equation describes the expansion rate of the universe.

The first Friedmann equation does not contain ω explicitly. However, ω influences the evolution of E_d, which can consequently influence the expansion rate.

---

5. SECOND FRIEDMANN EQUATION WITH COSMOLOGICAL CONSTANT

The acceleration Friedmann equation is:

$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$

According to the Pressure–Curvature relation:

$P_g = \omega E_d$

Substituting this expression:

$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3\omega E_d) + \Lambda c^2/3$

Taking E_d as a common factor:

ä/a = −(4πG/3c²)E_d(1 + 3ω) + Λc²/3

Therefore, the second Friedmann equation becomes:

ä/a = −(4πG/3c²)E_d(1 + 3ω) + Λc²/3

This equation directly demonstrates the influence of ω on cosmic acceleration.

---

6. CASE I: ω ≈ −1, OR ω = −1 + ε

Assume ε is a small dimensionless quantity.

Let:

ω = −1 + ε

According to the Pressure–Curvature relation:

P_g = (−1 + ε)E_d

The factor in the acceleration equation becomes:

1 + 3ω = 1 + 3(−1 + ε)

1 + 3ω = 1 − 3 + 3ε

Therefore:

1 + 3ω = −2 + 3ε

Substituting this result into the second Friedmann equation:

ä/a = −(4πG/3c²)E_d(−2 + 3ε) + Λc²/3

Therefore:

ä/a = (4πG/3c²)E_d(2 − 3ε) + Λc²/3

The final equation is:

ä/a = (4πG/3c²)E_d(2 − 3ε) + Λc²/3

For a small ε, this result approaches the exact ω = −1 case.

If Λ = 0 and E_d is positive, the fluid contribution produces acceleration when:

$2 - 3\varepsilon > 0$

Therefore:

$\varepsilon < 2/3$

This is a mathematical condition for acceleration in the specified model. It is not an observational measurement of ε.

---

7. CASE II: THE LIMIT ω → −1

Consider:

$\omega \to -1$

Assume that the energy density approaches a finite positive limiting value:

$E_d \to E_{d,lim} > 0$

Then:

$P_g \to -E_{d,lim}$

The second Friedmann equation becomes:

$\ddot{a}/a \to -(4\pi G/3c^2)E_{d,lim}(1 - 3) + \Lambda c^2/3$

Since:

$1 - 3 = -2$

We obtain:

$\ddot{a}/a \to (8\pi G/3c^2)E_{d,lim} + \Lambda c^2/3$

Therefore:

$\ddot{a}/a \to (8\pi G/3c^2)E_{d,lim} + \Lambda c^2/3$

This represents an asymptotically vacuum-like equation of state.

The assumption that E_d approaches a finite value is an additional physical assumption. It does not follow from $\omega \to -1$ alone.

---

8. CASE III: EXACT ω = −1

For the exact case:

ω = −1

The pressure becomes:

P_g = −E_d

Substituting into the second Friedmann equation:

ä/a = −(4πG/3c²)E_d(1 − 3) + Λc²/3

Since:

1 − 3 = −2

We obtain:

ä/a = (8πG/3c²)E_d + Λc²/3

Therefore:

ä/a = (8πG/3c²)E_d + Λc²/3

For positive E_d and positive Λ, both contributions produce positive acceleration.

Energy-density conservation

If the fluid is separately conserved, its continuity equation is:

Ė_d + 3H(E_d + P_g) = 0

For:

P_g = −E_d

We obtain:

Ė_d + 3H(E_d − E_d) = 0

Therefore:

Ė_d = 0

And:

E_d = Constant

Thus, under the stated conservation assumption, an exact $\omega = -1$ fluid has constant energy density.

---

9. CONTINUITY EQUATION AND EVOLUTION OF ENERGY DENSITY

For a separately conserved homogeneous fluid:

$\dot{E}_d + 3H(E_d + P_g) = 0$

Using:

$P_g = \omega E_d$

We obtain:

$\dot{E}_d + 3HE_d(1 + \omega) = 0$

Therefore:

$\dot{E}_d + 3HE_d(1 + \omega) = 0$

For constant ω, integration gives:

$E_d(a) = E_{d,0}\, a^{[-3(1 + \omega)]}$

Where the present scale factor is normalized to:

$a_0 = 1$

For:

$\omega = -1 + \varepsilon$

The equation becomes:

$E_d(a) = E_{d,0}\, a^{(-3\varepsilon)}$

As ε approaches zero, the energy density approaches constant behaviour.

For exact:

$\omega = -1$

We obtain:

$E_d(a) = E_{d,0}$

These results assume separate conservation and constant ω. If the effective fluid exchanges energy with other components, the continuity equation must include the corresponding interaction term.

---

10. COMPARISON WITH CURRENT COSMOLOGICAL OBSERVATIONS

The observational comparison must distinguish two different quantities:

1. The equation-of-state parameter of the entire cosmic fluid.
2. The equation-of-state parameter of the dark-energy component.

Observational constraints commonly quoted as $w = -1$ refer to dark energy in a specified cosmological model.

They do not automatically measure the equation-of-state parameter of the entire cosmic energy content.

This distinction is important because ordinary matter and radiation contribute to total energy density but have different pressure properties.

10.1. Planck Observations

The Planck 2018 cosmological-parameter analysis, combined with baryon acoustic oscillation and Type Ia supernova data, reported:

$w_0 = -1.03 \pm 0.03$

at 68% confidence for the stated data combination.

This result is consistent with:

$w = -1$

It therefore supports the observational viability of a cosmological-constant-like dark-energy component.

The Planck analysis does not directly measure the Pressure–Curvature equation-of-state parameter. It constrains parameters within standard cosmological models.

10.2. DESI DR2 Observations

The DESI DR2 cosmological analysis published in 2025 reported that combinations of DESI BAO, cosmic microwave background data, and supernova observations can favour a time-evolving dark-energy equation of state.

For DESI BAO and CMB, the reported preference for an evolving w_0w_a model over ΛCDM was approximately 3.1σ.

With supernova data included, the reported preference ranged from approximately 2.8σ to 4.2σ, depending on the supernova sample.

The favoured region included:

$w_0 > -1$

$w_a < 0$

This is a statistical preference in a particular time-dependent parameterization. It is not proof that the cosmological constant is incorrect.

10.3. Further 2026 Observational Developments

A DESI DR2 Lyman-alpha full-shape analysis released on 30 July 2026 added information from the distribution of intergalactic hydrogen gas.

The DESI collaboration reported that the new measurements shifted towards the standard ΛCDM prediction and could weaken some earlier indications of evolving dark energy.

A separate 2026 analysis combining DESI DR2, DES Year 6 BAO, Planck CMB data, supernovae, and cosmic chronometers reported:

$w = -0.992 \pm 0.028$

for a constant-w model.

This result is consistent with:

$w = -1$

For a time-varying w_0w_a model, that analysis reported:

$w_0 = -0.740 \pm 0.082$

$w_a = -0.893 \pm 0.285$

It found an approximately 3σ preference for evolution in that particular model and data combination.

The authors also cautioned that this preference did not constitute decisive Bayesian evidence.

These results demonstrate that the observational picture is still developing. The inferred result depends on the datasets, model assumptions, and statistical methods used.

---

11. OBSERVATIONAL ASSESSMENT OF THE THREE CASES

CASE A: ω = −1 + ε

This case is compatible with the general possibility of a dark-energy equation of state close to −1.

The Planck result:

$w_0 = -1.03 \pm 0.03$

is consistent with a value close to −1.

Some DESI-based analyses, however, favour a time-dependent equation of state.

Current observations do not establish a unique numerical value of ε for the Pressure–Curvature model.

CASE B: ω → −1

This case represents an equation of state approaching vacuum-like behaviour.

It is compatible with the constant-w observational results.

However, the assumption that E_d approaches a finite limiting value remains a separate PPC hypothesis requiring its own theoretical and observational test.

CASE C: ω = −1

This case corresponds mathematically to:

$P_g = -E_d$

It is the exact equation of state of a cosmological-constant-like fluid.

Planck and the 2026 constant-w analysis are consistent with this value.

Some DESI DR2 combinations nevertheless favour evolving dark energy.

Therefore, exact ω = −1 remains observationally viable, but the possibility of evolution has not been eliminated.

---

12. IMPORTANT SCIENTIFIC DISTINCTION: TOTAL COSMIC DENSITY AND DARK-ENERGY DENSITY

The definition:

$E_d = \rho c^2$

identifies E_d as the total local energy density in the present formulation.

However, the observational value $w \approx -1$ is generally inferred for the dark-energy component, not for the entire cosmic fluid.

In the standard cosmological description, matter and radiation contribute to total energy density but do not individually have an equation-of-state parameter equal to −1.

Consequently, if the PPC parameter ω is intended to describe the entire cosmic fluid, it must be derived from the total pressure divided by the total energy density:

$\omega_{total} = P_{g(total)}/E_{(total)}$

It cannot simply be assigned the observational dark-energy value without specifying how the PPC pressure relates to the pressures of the other components.

Alternatively, if ω is intended to describe a dark-energy-like PPC component, the model must distinguish that component from the other contributions to the total energy density.

This distinction is essential for a valid comparison with observational data.

---

13. SCIENTIFIC LIMITATIONS AND FURTHER REQUIREMENTS

The relation:

$P_g = \omega E_d$

provides an equation of state.

By itself, it does not constitute a complete alternative gravitational field theory.

To establish the PPC model as a predictive theory, further work is required:

1. Specify the complete PPC gravitational field equations.
2. Define whether E_d includes or excludes the vacuum-energy contribution represented by an explicit cosmological constant.
3. Establish the conservation law and any possible energy exchange.

4. Define how the PPC parameter ω relates to the observational dark-energy equation of state.
5. Derive the predicted expansion history H(z).
6. Compare the predictions quantitatively with BAO, CMB, supernova, and structure-growth observations.
7. Estimate the model parameters and uncertainties using observational likelihoods.

The observational values quoted in this paper are published results from other cosmological analyses. They are not a statistical fit of the PPC model.

---

14. CONCLUSION

Both Friedmann equations have been derived using the total local energy density:

$E_d = \rho c^2$

No separate matter-energy density E_m or radiation-energy density E_r has been added.

FIRST FRIEDMANN EQUATION

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

SECOND FRIEDMANN EQUATION

$\ddot{a}/a = -(4\pi G/3c^2)E_d(1 + 3\omega) + \Lambda c^2/3$

FOR $\omega = -1 + \varepsilon$

$\ddot{a}/a = (4\pi G/3c^2)E_d(2 - 3\varepsilon) + \Lambda c^2/3$

FOR $\omega \to -1$

$\ddot{a}/a \to (8\pi G/3c^2)E_{d,lim} + \Lambda c^2/3$

FOR EXACT $\omega = -1$

$P_g = -E_d$

$\ddot{a}/a = (8\pi G/3c^2)E_d + \Lambda c^2/3$

The observational comparison shows that a dark-energy equation of state close to −1 remains consistent with important cosmological measurements.

Some recent DESI-based combinations indicate a preference for evolving dark energy, while other analyses remain consistent with a constant equation of state.

FINAL SCIENTIFIC ASSESSMENT

The cases $\omega \approx -1$ and exact $\omega = -1$ are observationally plausible when interpreted as dark-energy-like equations of state.

However, the PPC model's own ω cannot yet be assigned a measured numerical value from these observations without a complete mapping between the PPC variables and observational cosmological parameters.

The Pressure–Curvature relation therefore provides a mathematical framework for further investigation, but its independent empirical validity remains to be established.

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SEPARATE RESEARCH NOTES

Title: Observational Value and Mathematical Role of the Cosmological Constant (Λ) in the Pressure–Curvature Model

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1. DEFINITION OF THE COSMOLOGICAL CONSTANT

The cosmological constant is represented by:

Λ

Its SI unit is:

$[\Lambda] = m^{-2}$

For the standard spatially flat ΛCDM cosmological model, Planck 2018 reported approximately:

$\Omega_\Lambda = 0.6847 \pm 0.0073$

$H_0 = 67.36 \pm 0.54$ km s^{-1} Mpc^{-1}

The cosmological constant can be calculated using:

$\Lambda = 3\Omega_\Lambda H_0^2/c^2$

Using these observational parameters:

$\Lambda \approx 1.09 \times 10^{-52}$ m^{-2}

This is an inferred value under the standard flat ΛCDM model.

---

2. VACUUM-ENERGY DENSITY ASSOCIATED WITH Λ

In standard general relativity, the cosmological constant can be represented as an equivalent vacuum-energy density:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Using:

$\Lambda \approx 1.09 \times 10^{-52}$ m^{-2}

$c = 2.99792458 \times 10^8$ m/s

$G = 6.67430 \times 10^{-11}$ m^3 kg^{-1} s^{-2}

The equivalent vacuum-energy density is approximately:

$E_\Lambda \approx 5.24 \times 10^{-10}$ J/m^3

Its corresponding mass-equivalent density is:

$\rho_\Lambda = E_\Lambda/c^2$

$\rho_\Lambda \approx 5.83 \times 10^{-27}$ kg/m³

These quantities describe the standard cosmological-constant interpretation. They are not included in E_d in the present PPC formulation.

---

3. EQUATION OF STATE ASSOCIATED WITH THE COSMOLOGICAL CONSTANT

The equivalent pressure associated with a cosmological constant is:

$P_\Lambda = -E_\Lambda$

Its equation-of-state parameter is:

$\omega_\Lambda = P_\Lambda / E_\Lambda$

Therefore:

$\omega_\Lambda = -1$

This is the vacuum-like equation of state associated with the cosmological constant in standard general relativity.

It is important to distinguish this standard Λ equation of state from the PPC parameter ω. Their identification requires a theoretical justification.

---

4. DEFINITION OF E_d IN THE PRESSURE–CURVATURE MODEL

For the present research formulation:

$E_d = \rho c^2$

Here:

- E_d represents the total local energy density included in the PPC fluid.
- ρ represents its corresponding total mass-equivalent density.
- Ordinary matter and radiation may contribute to E_d.
- Other non-vacuum energy contributions may also be included.
- Vacuum energy represented by the explicit cosmological constant Λ is excluded from E_d.

Therefore:

E_d DOES NOT INCLUDE VACUUM ENERGY.

The cosmological constant is treated separately and is not added to E_d.

The equivalent vacuum-energy density E_Λ must not be added to E_d in this formulation.

---

5. FIRST FRIEDMANN EQUATION WITH EXPLICIT COSMOLOGICAL CONSTANT

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

Here:

E_d = Total local non-vacuum energy density included in the PPC fluid.

Λ = Explicit cosmological constant.

No separate E_m, E_r, or E_Λ terms are added because the non-vacuum contributions are already represented in E_d, while the cosmological constant remains a separate geometric term.

---

6. SECOND FRIEDMANN EQUATION WITH EXPLICIT COSMOLOGICAL CONSTANT

The second Friedmann equation is:

$\ddot{a}/a = -(4\pi G/3c^2)(E_d + 3P_g) + \Lambda c^2/3$

Using the Pressure–Curvature relation:

$P_g = \omega E_d$

We obtain:

$\ddot{a}/a = -(4\pi G/3c^2)E_d(1 + 3\omega) + \Lambda c^2/3$

The cosmological constant remains separate from E_d in this equation.

---

7. FINAL NUMERICAL VALUES

Using the Planck 2018 flat ΛCDM parameters:

Cosmological constant:

$\Lambda \approx 1.09 \times 10^{-52}\ m^{-2}$

Equivalent vacuum-energy density in standard general relativity:

$E_\Lambda \approx 5.24 \times 10^{-10}\ J/m^3$

Equivalent mass density:

$\rho_\Lambda \approx 5.83 \times 10^{-27}\ kg/m^3$

Standard cosmological-constant equation-of-state parameter:

$\omega_\Lambda = -1$

The values E_Λ and ρ_Λ are provided only as standard cosmological reference quantities. They are excluded from E_d in the present PPC formulation.

---

8. IMPORTANT SCIENTIFIC CLARIFICATION

The present PPC formulation uses two separate contributions in the Friedmann equations:

1. E_d: Total local non-vacuum energy density represented by the PPC fluid.
2. Λ: Explicit cosmological constant retained separately in the geometric part of the Einstein field equations.

Therefore, the equations must retain the explicit Λ terms while excluding the corresponding vacuum-energy contribution from E_d.

This prevents double-counting.

The numerical value of Λ quoted here is inferred from Planck observations within the standard flat ΛCDM model. It has not yet been independently derived by the Pressure–Curvature theory.

REFERENCE

Planck Collaboration, *Planck 2018 Results. VI. Cosmological Parameters*, Astronomy & Astrophysics 641, A6 (2020).

DOI: 10.1051/0004-6361/201833910.

THE COSMOLOGICAL CONSTANT, VACUUM-ENERGY DENSITY, AND VACUUM-ENERGY-LIKE BEHAVIOUR IN THE PRESSURE–CURVATURE MODEL

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1. PURPOSE OF THESE RESEARCH NOTES

These research notes explain the mathematical relationship between the Pressure–Curvature (PPC) equation, total local energy density, the equation-of-state parameter, vacuum-energy-like behaviour, and the cosmological constant.

The fundamental PPC equation is:

$P_g = \omega E_d$

Three important concepts must be distinguished:

1. PPC energy density, E_d.
2. Vacuum-energy density associated with the cosmological constant, E_Λ.
3. The equation-of-state parameter, ω.

The central objective is to explain why the PPC equation produces vacuum-energy-like behaviour when $\omega = -1$, without automatically identifying E_d with the vacuum-energy density associated with the cosmological constant.

---

2. FUNDAMENTAL PRESSURE–CURVATURE EQUATION

The proposed Pressure–Curvature equation is:

$P_g = \omega E_d$

Where:

P_g = Pressure associated with the Pressure–Curvature relation.

E_d = Total local energy density represented by the PPC fluid.

ω = Dimensionless equation-of-state parameter.

The total local energy density is defined as:

$E_d = \rho c^2$

Where ρ is the corresponding total mass-equivalent density.

In the present formulation, E_d excludes the vacuum-energy contribution represented separately by the explicit cosmological constant Λ.

The PPC energy density may include contributions from ordinary matter, radiation, dark matter, and other non-vacuum physical energy contributions specified by the model.

No separate E_m or E_r terms are added to the Friedmann equations because these contributions are already represented within E_d.

---

3. COSMOLOGICAL CONSTANT AND VACUUM-ENERGY DENSITY

The cosmological constant is represented by:

Λ

Its equivalent vacuum-energy density in standard general relativity is:

$E_\Lambda = \Lambda c^4/(8\pi G)$

The corresponding mass-equivalent density is:

$\rho_\Lambda = E_\Lambda/c^2$

Therefore:

$\rho_\Lambda = \Lambda c^2/(8\pi G)$

The pressure associated with this vacuum energy is:

$P_\Lambda = -E_\Lambda$

Its equation-of-state parameter is:

$\omega_\Lambda = P_\Lambda/E_\Lambda$

Therefore:

$\omega_\Lambda = -1$

This is the standard vacuum-energy relationship associated with the cosmological constant.

Important distinction:

E_Λ is the vacuum-energy density associated with Λ.

It is not included in E_d in the present PPC formulation.

---

4. WHY DOES PPC PRODUCE VACUUM-ENERGY-LIKE BEHAVIOUR WHEN ω = −1?

The PPC equation is:

$P_g = \omega E_d$

Consider the condition:

$\omega = -1$

Substituting this value:

$P_g = (-1)E_d$

Therefore:

$P_g = -E_d$

Now compare this relationship with the standard cosmological-constant relationship:

$P_\Lambda = -E_\Lambda$

Both equations have the same mathematical structure:

$P = -E$

Consequently, when the PPC parameter is exactly −1, the PPC fluid has the same pressure-to-energy-density relationship as cosmological-constant vacuum energy.

This is called VACUUM-ENERGY-LIKE BEHAVIOUR.

However, having the same equation-of-state relationship does not establish that the two energy densities are physically identical.

The correct distinction is:

E_d and E_Λ are separately defined energy-density contributions in the present model.

At the same time:

$\omega_{PPC} = \omega_\Lambda = -1$

can hold mathematically.

The first statement concerns the distinction between the energy-density contributions. The second concerns their equation-of-state behaviour.

---

5. MATHEMATICAL COMPARISON

A. PPC contribution when ω = −1

Energy density:

E_d

Pressure:

$P_g = -E_d$

Equation-of-state parameter:

$\omega_{PPC} = -1$

B. Cosmological-constant vacuum-energy contribution

Energy density:

$E_\Lambda = \Lambda c^4/(8\pi G)$

Pressure:

$P_\Lambda = -E_\Lambda$

Equation-of-state parameter:

$\omega_\Lambda = -1$

Both contributions have vacuum-like pressure relationships.

Nevertheless, their physical origins and definitions can remain separate within the PPC formulation.

---

6. SAME BEHAVIOUR DOES NOT NECESSARILY MEAN THE SAME PHYSICAL ORIGIN

The equation:

$P_g = -E_d$

establishes the mathematical equation of state of the PPC fluid.

It does not, by itself, prove that:

$E_d = E_\Lambda$

Nor does it prove that the PPC fluid is identical to the cosmological constant.

The two contributions can be represented separately in a combined energy budget:

$E_{d(total)} = E_d + E_\Lambda$

This expression is appropriate when both contributions are being counted separately.

If both contributions have an equation-of-state parameter of −1, their combined pressure is:

$P_{g(total)} = -E_d - E_\Lambda$

Therefore:

$P_{g(total)} = -E_{d(total)}$

This result assumes that the PPC contribution itself has $\omega = -1$.

It demonstrates that two distinct contributions can produce the same combined vacuum-like equation-of-state relationship.

---

7. FIRST FRIEDMANN EQUATION WITH EXPLICIT COSMOLOGICAL CONSTANT

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

Where:

H = Hubble parameter.

G = Gravitational constant.

c = Speed of light.

E_d = Total local non-vacuum energy density represented by the PPC fluid.

k = Spatial-curvature parameter.

a = Cosmological scale factor.

Λ = Explicit cosmological constant.

The cosmological constant remains separate from E_d.

The vacuum-energy contribution represented by Λ is not added again as an E_Λ term because the explicit Λ term already accounts for it in this formulation.

---

8. SECOND FRIEDMANN EQUATION WITH EXPLICIT COSMOLOGICAL CONSTANT

The second Friedmann equation is:

ä/a = −(4πG/3c²)(E_d + 3P_g) + Λc²/3

Using the PPC equation:

P_g = ωE_d

We obtain:

ä/a = −(4πG/3c²)(E_d + 3ωE_d) + Λc²/3

Taking E_d as a common factor:

ä/a = −(4πG/3c²)E_d(1 + 3ω) + Λc²/3

Therefore, the second Friedmann equation becomes:

ä/a = −(4πG/3c²)E_d(1 + 3ω) + Λc²/3

The PPC contribution and the cosmological-constant contribution remain separately represented.

---

9. COSMIC ACCELERATION WHEN PPC HAS EXACTLY ω = −1

For:

ω = −1

The PPC pressure becomes:

P_g = −E_d

Substituting into the second Friedmann equation:

ä/a = −(4πG/3c²)E_d(1 − 3) + Λc²/3

Since:

$1 - 3 = -2$

We obtain:

$\ddot{a}/a = (8\pi G/3c^2)E_d + \Lambda c^2/3$

Therefore:

$\ddot{a}/a = (8\pi G/3c^2)E_d + \Lambda c^2/3$

This equation contains two separately represented contributions to cosmic acceleration.

First, the PPC contribution:

$(8\pi G/3c^2)E_d$

Second, the cosmological-constant contribution:

$\Lambda c^2/3$

For positive E_d and positive Λ, both contributions are positive.

Thus, the PPC equation with $\omega = -1$ produces an additional vacuum-like contribution to acceleration alongside the explicit cosmological-constant contribution.

This conclusion follows mathematically from the assumed Friedmann framework and PPC equation of state. It does not independently establish that the PPC contribution exists observationally.

---

10. ENERGY-DENSITY EVOLUTION WHEN $\omega = -1$

For a separately conserved PPC fluid, the continuity equation is:

$\dot{E}_d + 3H(E_d + P_g) = 0$

Using:

$P_g = -E_d$

We obtain:

$\dot{E}_d + 3H(E_d - E_d) = 0$

Therefore:

$\dot{E}_d = 0$

And:

E_d = Constant

Thus, under the separate-conservation assumption, the PPC fluid has constant energy density when $\omega = -1$.

This is another mathematical property shared with cosmological-constant vacuum energy.

However, the constant-density result depends on the separate-conservation assumption. If the PPC fluid exchanges energy with other components, its evolution requires an appropriate interaction term.

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11. OBSERVATIONAL INTERPRETATION

The standard cosmological-constant equation of state is:

$\omega_\Lambda = -1$

Observational studies commonly find that the dark-energy equation of state is consistent with a value close to −1, although some combinations of recent DESI data and other cosmological observations have indicated a possible preference for evolving dark energy.

The PPC equation predicts:

$\omega_{PPC} = -1$

Therefore:

$P_g = -E_d$

This makes the PPC fluid mathematically vacuum-like in this limit.

However, observational measurements of dark-energy ω do not automatically measure the PPC parameter ω.

To compare PPC with observations, it is necessary to establish how the PPC fluid contributes to the expansion history and how its pressure and density map onto the observational cosmological components.

If E_d is intended to include matter and radiation as well as other non-vacuum contributions, assigning $\omega = -1$ to the entire PPC fluid means that its total effective pressure must equal minus its total energy density.

This is a substantive physical assumption, not a property of ordinary matter or radiation individually.

A quantitative comparison must test this assumption against the observed expansion and structure-growth history.

---

12. SCIENTIFIC LIMITATIONS AND FURTHER REQUIREMENTS

The relation:

$P_g = \omega E_d$

establishes an equation of state.

By itself, it does not establish a complete gravitational theory.

Further theoretical work is required to:

1. Specify the complete PPC gravitational field equations.
2. Define the physical origin of the PPC pressure.
3. Establish the conservation law governing E_d.
4. Determine whether E_d remains constant for $\omega = -1$ under the proposed dynamics.
5. Derive the predicted expansion history $H(z)$.
6. Determine how the PPC contribution differs observationally from the cosmological constant.
7. Compare predictions with CMB, BAO, supernova, and structure-growth observations.

In particular, an additional PPC contribution with $\omega = -1$ can be mathematically degenerate with a change in the effective cosmological constant at the level of the homogeneous expansion history.

Independent observational signatures or a distinct theoretical mechanism would be needed to distinguish the two.

---

13. FINAL CONCLUSION

The fundamental Pressure–Curvature equation is:

$P_g = \omega E_d$

For exact:

$\omega = -1$

It becomes:

$P_g = -E_d$

This produces vacuum-energy-like behaviour because it has the same equation-of-state relationship as standard cosmological-constant vacuum energy:

$P_\Lambda = -E_\Lambda$

However, the two energy-density contributions remain conceptually distinct in the present formulation.

The PPC energy density E_d excludes the vacuum-energy contribution represented by the explicit cosmological constant.

The first Friedmann equation is:

$H^2 = (8\pi G/3c^2)E_d - kc^2/a^2 + \Lambda c^2/3$

The second Friedmann equation is:

$\ddot{a}/a = -(4\pi G/3c^2)E_d(1 + 3\omega) + \Lambda c^2/3$

For $\omega = -1$:

$\ddot{a}/a = (8\pi G/3c^2)E_d + \Lambda c^2/3$

FINAL SCIENTIFIC STATEMENT

According to the Pressure–Curvature relation, when $\omega = -1$, the PPC pressure becomes equal to the negative of its associated energy density. This produces vacuum-energy-like behaviour, mathematically corresponding to the equation-of-state relationship of the cosmological constant.

Nevertheless, the PPC energy density E_d and the cosmological-constant vacuum-energy density E_Λ remain separately defined contributions in the present formulation.

Their identical equation-of-state relationships do not, by themselves, establish an identical physical origin.

REFERENCE

Planck Collaboration, *Planck 2018 Results. VI. Cosmological Parameters*, Astronomy & Astrophysics 641, A6 (2020).

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The PPC Law of Gravity

A Research Study on the First and Second Friedmann Equations Without an Explicit Cosmological Constant

Author: Pawan Upadhyay

Abstract

This research presents the proposed PPC Law of Gravity and derives the first and second Friedmann equations without introducing an explicit cosmological constant.

The proposed framework uses:

$$T_{\mu\nu}^{(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

$$P_g = \omega E_d \quad , \quad E_d = \rho c^2$$

Here, P_g is the gravitational pressure, E_d is the total energy density, ρ is the total mass-equivalent density, and ω is the equation-of-state parameter. The resulting model is examined for consistency with modern observational evidence.

Chapter 1: Introduction

- Modern cosmology describes the evolution of the universe through Einstein's theory of relativity and the Friedmann equations.
- The proposed PPC Law of Gravity explores the cosmic expansion using a PPC Perfect Fluid Energy-Momentum Tensor.
- **Objective:** To derive both Friedmann equations without an explicit cosmological constant.
- The PPC framework uses gravitational pressure P_g and total energy density E_d, linked by the parameter ω.

Chapter 2: PPC Energy-Momentum Tensor

The proposed PPC Perfect Fluid Energy-Momentum Tensor is:

$$T_{\mu\nu}^{(PPC)} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

Where:

- $T_{\mu\nu}^{(PPC)}$: PPC energy-momentum tensor
- E_d : total energy density
- P_g : gravitational pressure
- u_μ : four-velocity
- $g_{\mu\nu}$: spacetime metric tensor

The metric signature is $(-+++)$ and

$$u_\mu u^\mu = -1$$

Chapter 3: PPC Equation of State

The proposed equation of state is:

$$P_g(x, y, z, ct) = \omega(x, y, z, ct)E_d(x, y, z, ct)$$

Equivalently:

$$\omega = \frac{P_g}{E_d}$$

- ω is the dimensionless equation-of-state parameter.
- For homogeneous and isotropic cosmology:

$$P_g(t) = \omega(t)E_d(t)$$

Chapter 4: Total Density and Energy Density

- ρ : total mass-equivalent density.
- E_d : total energy density associated with ρ.
- Mass-energy equivalence:

$$E_d = \rho c^2 \quad , \quad \rho = \frac{E_d}{c^2}$$

- For multiple components:

$$E_d = \sum_i E_i \quad , \quad P_g = \sum_i P_i$$

- Effective equation-of-state parameter:

$$\omega = \frac{\sum_i P_i}{\sum_i E_i}$$

Chapter 5: Einstein's Field Equation (without Cosmological Constant)

The PPC formulation uses Einstein's field equation:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{(PPC)}$$

Substituting the PPC tensor:

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left[(E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}\right]$$

- $G_{\mu\nu}$: Einstein tensor.
- G : Newton's gravitational constant.
- No explicit cosmological constant is introduced.

Chapter 6: FLRW Metric

For a homogeneous and isotropic universe, the FLRW metric is:

$$ds^2 = -c^2dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2d\theta^2 + r^2\sin^2\theta d\phi^2\right]$$

Where:

- $a(t)$: scale factor
- k : spatial curvature
- t : cosmic time

The Hubble parameter is:

$$H = \frac{\dot{a}}{a}$$

Chapter 7: First Friedmann Equation

From the time-time component:

$$H^2 = \frac{8\pi G E_d}{3c^2} - \frac{kc^2}{a^2}$$

Using $E_d = \rho c^2$:

$$H^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2}$$

First PPC Friedmann Equation

$$H^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2}$$

Chapter 8: Second Friedmann Equation

From the spatial components:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}(E_d + 3P_g)$$

Using $P_g = \omega E_d$ and $E_d = \rho c^2$:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G\rho}{3}(1 + 3\omega)$$

Second PPC Friedmann Equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G\rho}{3}(1 + 3\omega)$$

- Accelerated expansion: $\omega < -1/3$
- Decelerated expansion: $\omega > -1/3$
- Limiting condition: $\omega = -1/3$

Chapter 9: Energy-Momentum Conservation

Conservation law:

$$\nabla_\mu T_{\mu\nu}^{(PPC)} = 0$$

For a homogeneous and isotropic universe:

$$\dot{E}_d + 3H(E_d + P_g) = 0$$

Using $P_g = \omega E_d$:

$$\dot{E}_d + 3HE_d(1 + \omega) = 0$$

- For constant ω: $E_d(a) = E_{d,0}a^{-3(1+\omega)}$
- For time-dependent ω:

$$E_d(a) = E_{d,0}\exp\left[-3\int_1^a \frac{1+\omega(x)}{x}dx\right]$$

Chapter 10: Special Values of ω

Case	ω	P_g	Energy Density (E_d)	Behaviour
Pressureless fluid	0	0	$E_d \propto a^{-3}$	Matter-like
Radiation-like	$\frac{1}{3}$	$\frac{E_d}{3}$	$E_d \propto a^{-4}$	Radiation
Zero acceleration	$-1/3$	$-\frac{E_d}{3}$	$E_d \propto a^{-2}$	No acceleration contribution
Vacuum-like	-1	$-E_d$	E_d = constant	Accelerated expansion
Stiff fluid	1	E_d	$E_d \propto a^{-6}$	Stiff behaviour

Chapter 11: Time-Dependent Eq. of State

- One possible parameterization:

$$\omega(a) = \omega_0 + \omega_a(1 - a)$$

- Corresponding energy density:

$$E_d(a) = E_{d,0}a^{-3(1+\omega_0+\omega_a)}\exp[3\omega_a(a-1)]$$

- ω_0 : present equation-of-state parameter.
- ω_a : parameter describing its evolution.
- A possible method to investigate an evolving PPC equation of state.

Chapter 12: Redshift and Cosmic Expansion

The redshift–scale factor relation:

$$1 + z = \frac{1}{a}$$

For $a_0 = 1$, define $f(z) = \frac{E_d(z)}{E_{d,0}}$. The first Friedmann equation becomes:

$$\frac{H^2(z)}{H_0^2} = \Omega_{d,0}f(z) + \Omega_{k,0}(1+z)^2$$

Where: $\Omega_{d,0} = \frac{8\pi G E_{d,0}}{3c^2H_0^3}$, $\Omega_{k,0} = -\frac{kc^2}{H_0^3}$

Chapter 13: Consistency with Observations

- **Type Ia Supernovae:** Reproduce observed redshift–luminosity distance: $D_L(z) = (1+z)c\int_0^z \frac{dz'}{H(z)}$.
- **Baryon Acoustic Oscillations:** Match BAO distance measurements and expansion rates.
- **Cosmic Microwave Background:** Reproduce CMB temperature and polarization spectra.
- **Hubble Parameter:** Compare predicted $H(z)$ with observations.
- **Large-Scale Structure:** Explain growth of cosmic structures and galaxy distribution.

Chapter 14: Comparison with ΛCDM Model

- ΛCDM includes matter, radiation, curvature, and a cosmological constant.
- PPC describes the total effective energy density through E_d and ω.
- No explicit cosmological constant is used.
- If $\omega = -1$ and E_d = constant, the PPC fluid produces the same expansion behaviour as a vacuum-energy contribution.
- A scientific comparison must examine predicted observations, not merely the notation.

Chapter 15: Finite-Density Principle

Proposed principle: Physically admissible solutions should have finite values:

$$E_d < \infty \quad , \quad |P_g| < \infty \quad , \quad \rho < \infty$$

- These are proposed physical requirements.
- They are not automatically guaranteed by the Friedmann equations.
- Possible upper limit of cosmic time:

$$0 < t_{tmax} \leq \infty$$

Chapter 16: Future-Time Behaviour

One possible PPC hypothesis:

$$\omega \to -1 \quad \text{as} \quad t \to t_{tmax}$$

- If E_d approaches a finite positive constant, the PPC fluid approaches vacuum-like behaviour.
- This could produce accelerated expansion.
- It requires a specified evolution law for ω and mathematical analysis.

Chapter 17: Limitations and Validation

1. What determines ω?
2. What equation governs the evolution of ω?
3. Can it explain accelerated expansion without Λ?
4. Can it reproduce CMB spectra?
5. Can it reproduce BAO and supernova distances?
6. Can it predict growth of large-scale structures?
7. Does it provide testable predictions distinct from ΛCDM?
8. Can the finite-density principle be demonstrated?

Chapter 18: Conclusion

- Derived both Friedmann equations without an explicit cosmological constant.
- Uses $T_{\mu\nu}^{(PPC)}$, $P_g = \omega E_d$, and $E_d = \rho c^2$.
- Expansion and acceleration depend on ρ, k and ω.
- Accelerated expansion for $\omega < -1/3$.
- Must be tested quantitatively with observations.
- Provides a mathematical basis for further study.

References

[1] Einstein (1915) [2] Friedmann (1922) [3] Lemaître (1927) [4] Planck Collaboration (2020)
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